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Weight-based Sentiment Analysis for Prediction

Aso Hamzehei

A thesis in fulfillment of the requirements for the degree of

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Social networks are more than just connecting people together. It is every individual's broadcasting station. The social networks and microblogs features have brought a unique opportunity to capture public's opinion, sentiment, and mood on variety of topics. It is claimed that computational indicators of public sentiment may have predictive power for socio-political and socio-economic phenomena. This predictive power is mainly for incidents that are public opinion driven. This dissertation proposes an innovative approach for topic-based sentiment analysis that is weighted by authors topic-based social influence to predict events such as elections. This dissertation consists of two major components: 1. Sentiment Analysis 2. social influence measurement and prediction. First, the thesis introduces a scalable sentiment analysis approach, a Semantic Scoring Sentiment Analysis (SSSA) approach, which matches the demand of scalability and efficiency of a sentiment analysis system for a large volume of microblogs. Then, the thesis studies social influence and introduces Topic-based Social Influence Measurement (TSIM). TSIM proposes an approach that incorporates network structure, user-generated content, and user's interactions in the network for topic-based influence measurement. Next, the thesis proposes a collaborative topic-based social influence model, SICTR, that incorporates both network structure and user-generated content for topic-based influence prediction. SICTR predicts topic-based user influence on unobserved topics, based on observed topic-based user influence through their generated contents and activities in social networks. Finally, the thesis concludes by proposing Weight-based Sentiment Analysis (WSA) approach, where the author's influence is included in sentiment analysis measurement. WSA aims to propose a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. For evaluating the proposed approaches, multiple datasets are collected from social networks and used. The experiments demonstrate that the proposed models and approaches outperform the state-of-the-art benchmarks. In summary, this thesis proposes a collective and aggregated sentiment analysis approach. The aggregated sentiment analysis can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets.

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Abstract

The fast growth of the internet availability and smart phones has made social networks available for everyone and everywhere. Social networks are more than just connecting people together. It is every individual's broadcasting station. The social networks and microblogs features such as information sharing, users communication and interaction, real-time distribution of information have brought a unique opportunity to capture public's opinion, sentiment, and mood on variety of topics. A growing body of literature is available that have shown correlation between public mood and sentiment measured from social media of financial market and stock price changes, election results, commercial sales, and influenza epidemics. It is claimed that computational indicators of public sentiment may have predictive power for socio-political and socio-economic phenomena. This predictive power is mainly for incidents that are public opinion driven. The traditional sentiment analysis approaches purely consider the content only and ignore the author of the text. In this dissertation, an innovative approach is proposed for topic-based sentiment analysis that is weighted by authors topic-based social influence. This dissertation consists of two major components: 1. Sentiment Analysis 2. Social Influence Measurement and Prediction.

First, the thesis introduces a scalable sentiment analysis approach. Scalability is one of the main challenges of social media analysis and sentiment analysis. Microblogs as emerging opinion sharing platforms require new approaches that are more scalable and

accurate. The thesis propose, implement, and evaluate SSSA, a Semantic Scoring Sentiment Analysis approach, which matches the demand of scalability and efficiency of a sentiment analysis system for a large volume of microblogs. First, an automated tweet tagging system is built to alleviate the cost of building a dataset for training. SSSA then calculates semantic scores of words in tweets. These scores are the baseline for the proposed subjectivity classification and feature selection steps. Also a polarity classifier is built to estimate the polarity of subjective tweets on top of them. The experiments are performed on two different datasets, mobile phone and 2016 US presidential election. In the first experiment, the scalability and accuracy of SSSA are tested. In the second experiment, the accuracy of the approach is compared with benchmarks. The first experiment shows that compared to conventional approaches that are based on emoticon, hashtag, or semantic scoring, SSSA has improved the scalability and accuracy of microblog sentiment analysis and also outperforms emoticon and hashtag based approaches. The second experiment demonstrated that SSSA outperforms the state-of-the-art benchmarks on the large dataset of 2016 US presidential election dataset.

Then, the thesis studies social influence and introduces Topic-based Social Influence Measurement (TSIM). Users of social networks spread their opinions, get involved in discussions, and consequently, influence each other. However, the level of influence of different users is not the same. It varies not only among users, but also for one user across different topics. The structure of social networks and user-generated content can reveal immense information about users and their topic-based influence. Although many studies have considered measuring global user influence, measuring and estimating topic-based user influence has been under-explored. TSIM proposes an approach that incorporates network structure, user-generated content, and user's interactions in the network for topic-based influence measurement. The experimental analysis performed on Twitter data and it is shown that the proposed TSIM can effectively measures topic-based user influence.

Next, the thesis proposes a collaborative topic-based social influence model (SICTR) that

incorporates both network structure and user-generated content for topic-based influence prediction. SICTR predicts topic-based user influence on unobserved topics, based on observed topic-based user influence through their generated contents and activities in social networks. The experimental analysis performed on Twitter data shows that SICTR outperforms state-of-the-art approaches on recall, accuracy, precision, and F-score for predicting topic-based user influence.

Finally, the thesis is concluded by proposing Weight-based Sentiment Analysis (WSA), where the author's influence is included in sentiment analysis measurement. WSA aims to propose a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. Such predictions traditionally is conducted through polling systems which are highly expensive with limited data points. In the last main chapter, the focus is on the impact of influential users in prediction through sentiment analysis. In particular, the goal is to observe whether influence-based aggregated sentiment analysis is outperforming generic aggregated sentiment analysis in prediction. For evaluating WSA, a dataset collected from an online social network for the US 2016 presidential election. WSA predicts the election results and performs with higher accuracy.

In summary, in this dissertation, a weight-based sentiment analysis approach is proposed where its components are developed in earlier chapters of the thesis and it is tested as a whole in Chapter 5. We apply more advanced sentiment analysis technique that go beyond the simple and discrete classification of positive and negative polarity. Collective and aggregated sentiment analysis can be considered as practical use of sentiment analysis. The aggregated sentiment analysis can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets.

Related Publications

1. **Aso Hamzehei**, Mohamad Ebrahimi, Elaheh Shafiee, Raymond K. Wong and Fang Chen, Scalable Sentiment Analysis for Microblogs Based on Semantic Scoring, *IEEE International Conference on Services Computing (IEEE-SCC)*, New York, NY, 2015, pp. 271-278.

2. **Aso Hamzehei**, Shanqing Jiang, Danai Koutra, Raymond K. Wong, Fang Chen. TSIM: Topic-based Social Influence Measurement for Social Networks. *In Proceeding of the Fourteenth Australasian Data Mining Conference (AusDM)*, Canberra, Australia. 6-8 December 2016. Vol. 170.

3. **Aso Hamzehei**, Shanqing Jiang, Danai Koutra, Raymond Wong, Fang Chen. Topic-based Social Influence Measurement for Social Networks. *Australasian Journal of Information Systems (AJIS)*, November 2017, Vol. 21.

4. **Aso Hamzehei**, Raymond Wong, Danai Koutra, Fang Chen. Collaborative Topic Regression for Predicting Topic-Based Social Influence. *Machine Learning (Special issue of ACML18 Journal Track)*. 2019 (DOI: 10.1007/s10994-018-05776-w).

5. **Aso Hamzehei**, Raymond Wong, Fang Chen. Weight-based Sentiment Analysis: A new Approach for Prediction. (Under Submission)

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Chapter 1

Introduction

1.1 Introduction

”Man is by nature a social animal. Society is something that precedes the individual. Anyone who either cannot lead the common life or is so self-sufficient as not to need to, and therefore does not partake of society, is either a beast or a god.” Aristotle. Similar understanding of human’s social nature has been discussed since Aristotle’s era or perhaps earlier. Due to social interaction, it is important to people that what others think, feel, or do. In personal lives, one seeks for advice from her connected people for better decision making. A student would ask friends of what book to read. A tourist would ask for recommendations on places to visit. A customer would ask others’ opinion on a product to buy. Speaking of seeking information from other members of society, we realise that this phenomenon is a repetitive part of daily life.

Discovering other’s opinion is not limited to personal lives. Businesses are eagerly willing to understand what their customers think about their products or their competitors. In

economics, consumer confidence is about indicating the degree of consumers' optimism and feeling on the economy and their personal financial situation. Also, public opinions and sentiments play an important role in politics. Governments are interested in public's opinion on their strategies and plans. Voting itself, in more formal way, is an act of determining a group of people's opinion regarding the voting topic. Understanding publics' sentiment and opinion is extremely important for any entity with any size in this functioning society.

Sentiment analysis is about determining emotion, feeling, and opinion. It traditionally is measured through questionnaires and polls in the last few decades. However, with the advancement of information technology and computer science, wider range of opinions, in larger scales, more cost effective manner, and faster are accessible to be measured. The term "Sentiment Analysis" refers to a series of techniques, methods, and approaches in text analysis, natural language processing , and machine learning to identify author's mood and opinion. In general terms, sentiment analysis aims to identify polarity of a given text whether it is positive or negative regarding a particular topic. Extracting someone's mood and feeling from text is a very challenging task and there is big room for error and natural language can be complicated. Sarcasm is an example of that complication.

The rapid and intensive growth of the Internet text data creates immense opportunities and challenges for applying techniques to learn "Big Data", e.g., extracting meaningful knowledge and discovering patterns from big corpuses, as well as dealing with collecting, organising and analysing voluminous data. One of the sources of such massive amount of data comes from microblog service providers such as Twitter. Twitter is a platform of sharing information and opinion and users publish tweets on wide variety of topics everyday and has attracted hundreds of millions of users to share their personal activities and public messages (Kwak et al. 2010). Given the huge amount of posts everyday, Twitter becomes an attractive source for conducting large-scale data analytics for many research and commercial projects. In the era of "Big Data", with emphasis on the 5 V's (volume,

velocity, variety, value and veracity), a scalable, accurate and fast algorithm is the centre of challenges in the recent works in this domain.

Social media contents, e.g. tweets, regardless of their content and intention, often carry information about their author's sentiment or opinion (Bollen, Mao & Pepe 2011). It has created great opportunities and considerable promises to capture, measure, and extract public sentiment, mood state, and opinion by analysing them in a more rapid, accurate and cost-effective way. In fact, data obtained from social networks can be mined and analysed extensively in a wide range of applications and disciplines such as marketing research companies, public opinion organisations, and other text mining entities (Mostafa 2013). Also, a growing body of literature is available that have shown correlation between public mood and sentiment measured from social media with financial market and stock market changes(Mao et al. 2011, Bollen & Mao 2011, Ruiz et al. 2012), election results (Bermingham & Smeaton 2011, Miller 1995), commercial sales (Mishne et al. 2006), influenza epidemics (Culotta 2010). It is claimed that computational indicators of public sentiment may have predictive power in regard to socio-economic phenomena (Asur & Huberman 2010). It is assumed that social media can have predictive power mainly for incidents that are public opinion driven.

Collective and aggregated sentiment analysis can be considered as practical use of sentiment analysis. Through aggregated sentiment analysis, businesses can track their customers response and feedback to their services and products. The aggregated sentiment analysis also practically can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets. Such predictions traditionally could be conducted through polling systems which are highly expensive with limited data points. Due to ubiquitous social networks and the advancement of text mining and sentiment analysis, the polling approach can be replaced with more cost-effective, timely, and comprehensive approaches (O'Connor et al. 2010, Asur & Huberman 2010).

Social influence of users can be an identifier of how strongly a user's opinion will impact

on other user's opinions and propagate in the network. Members of a network have different levels of influence in the network. The concept of celebrity is a product of existence of influence phenomena. In marketing, the concept of influence is utilised and influential people are selected for marketing campaigns to maximise a marketing campaign's performance. Given a user's influence and sentiment in a topic, an interesting question could raise that whether we can have a better prediction with aggregated sentiment analysis with utilising a user's influence measure.

Although social influence has been an area of interest for researchers in sociology and more recently in computer science, still there is no agreement on its definition. A very early definition for influential people is "individuals who were likely to influence other persons in their immediate environment" (Katz 1957). Social influence has either been studied to identify influential users (opinion leaders or authorities) or topical and topic-based influential users (Riquelme & González-Cantergiani 2016).

Social science studies, e.g.(Katz & Lazarsfeld 1955), have acknowledged the fact that the social influence of individuals is not identical. Katz (Katz 1957) introduced three main factors that are related to an individual's social influence such as: Who one is, what one knows, and whom one knows. The individual's social influence can be much more easily observed on social media while it is confirmed that the social influence factors are similar in social networks to those in the real society (Libai et al. 2010, Eccleston & Griseri 2008). For example, Eirinaki et al (Eirinaki et al. 2012) introduced two factors (popularity and activity) as factors related to social influence on Online Social Networks (OSN).

One of the main measures studied for influence is information diffusion which measures how important a user is in spreading information in the network. This is equivalent to identify central and hub nodes in the network (Jin & Wang 2013, Hajian & White 2011). Opinion leaders and discussion starters also have been studied as a measure of social influence (Jabeur et al. 2012). A user's position in the network (Jin & Wang 2013), content (Hu, Fang & Godavarthy 2013), and activities (Pal & Counts 2011) have been also studied

as influence measures. Another aspect of studied influence has been the scale of affected users by a post on social network or intensity of emotional and cognitive impact (McNeill & Briggs 2014).

According to (Probst et al. 2013), influential users have different influences on different topics and a very influential user is not necessarily influential on all topics. It is indicated in (Kardara et al. 2015) that topic-based influence measures are more effective and functional than the global ones. One of the differences of topic-related influence studies to network structure analysis is that it takes the posts' (e.g., tweets) content into account. When we consider user influence on topics, no longer the whole network needs to be analyzed, which improves the performance of measures.

In this dissertation, we propose an innovative approach, Weigh-based Sentiment Analysis (WSA), to predict events that are public opinion driven e.g. elections. We weight users' opinions and sentiments according to their influence value in a respective topic. WSA proposes a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. Such predictions, traditionally, is conducted through polling systems which are highly expensive with limited data points. In WSA, the focus is on the impact of influential users in prediction through sentiment analysis. In particular, the goal is to observe whether influence-based aggregated sentiment analysis is outperforming generic aggregated sentiment analysis in prediction. To achieve WSA, we propose a scalable sentiment analysis approach, a social influence measure, and a social influence prediction approach. In other words, our approach has two major components: topic-based influence measurement and sentiment analysis. Both together form WSA, the aggregated weighted topic-based sentiment analysis.

1.2 Integration of sentiment analysis and influence measurement

Usually in the literature, sentiment analysis is referred to document-level sentiment analysis. However, when there is an aspect e.g., a candidate in an election, a product review, or a movie, to measure its sentiment, the document-level sentiments need to be aggregated to measure the aspect-level sentiment also known as collective or aggregated sentiment analysis. To measure a candidate's sentiment in an election, all the sentiment of all the opinions in the election corpus can be measures and then be aggregated to measure collective sentiment analysis for the given aspect.

Collective and aggregated sentiment analysis can be considered as practical use of sentiment analysis. Through aggregated sentiment analysis, businesses can track their customers' response and feedback to their services and products. The aggregated sentiment analysis also practically can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets. Such predictions traditionally could be conducted through polling systems, which are highly expensive with limited data points. Due to ubiquitous social networks and the advancement of text mining and sentiment analysis, the polling approach can be replaced with more cost-effective, timely, and comprehensive approaches (O'Connor et al. 2010, Asur & Huberman 2010).

In collective sentiment, to have a fair collective measurement every user should be counted once in the collective sentiment regardless of the number of documents a particular user could have related to the aspect. It means in a given period of time; a user could have one sentiment value for an aspect. This way the multiple documents of a user would be aggregated in one sentiment value and that user would not be over represented in the collective sentiment.

Moreover, every user could be assigned a coefficient (weight) in the collective sentiment

measurement. A weight equal to one for every user means the users will be equally represented in the collective sentiment value. However, users have different levels of impact and influence on other users in the network. If we could quantify users' aspect-level influence, we could have a weighted collective sentiment analysis for the aspect. Therefore, sentiment analysis and influence measurement are two essential components of weighted collective sentiment analysis, which is the main focus of this dissertation.

1.3 A Brief History of the Related Work

1.3.1 Sentiment Analysis

Sentiment analysis is a technique to measure sentiment of a given textual content. From machine learning point of view, it is mainly considered as a text classification problem. Words are the key features, also called feature vectors, for machine learning algorithms. These features are words and their frequencies, parts of speech (POS) tags, syntax, opinion words and phrases, words positions, negation and etc (Kumar & Teeja 2012). For sentiment analysis on a statement, its polarity needs to be evaluated. A statement or opinion expression is not necessarily subjective and could have neutral polarity.

General Sentiment Analysis

Polarity classification is challenging when it comes to microblogs especially Twitter, due to very short text length (140 characters in the case of Twitter). The challenging tasks of sentiment analysis on tweets are topic and polarity identification on the very short length text and manual tagging of Twitter corpus for supervised polarity classification. To address both challenges, researchers have engaged metadata such as emoticons (Wong et al. 2014) and hashtags for tweets positivity and negativity detection(Wang et al. 2011).

Read (2005) showed the usability of emoticons in enhancing the polarity identification of tweets when using a trained model on a domain and testing it on a different domain.

Another way of determining a document's sentiment is via dictionary-based approaches which depend on existing lexicographical resources (such as WordNet (Miller 1995)). The dictionary-based techniques provide semantic data in regards to individual senses and words (Khan 2011). Several authors have used the advantages of the graph nature of WordNet (Hassan & Radev 2010, Esuli & Sebastiani 2007, Page et al. 1999). SentiWordNet is another lexical resource that is used by the opinion mining community to obtain the semantic value (Denecke 2008, Ogawa et al. 2011). Scalability of semantic-based services is their main drawback due to expensive runtime and low performance when datasets grow.

Social network features are investigated to improve sentiment analysis results. Connected users could possibly have similar opinions. Tan et al. (2011) studied the follower and following networks of user and concluded that connections and network of a user on Twitter can improve the classified sentiment. To measure aggregated sentiment of an aspect, Neves-Silva et al. (2016) looked into influence propagation of opinions to combine with sentiment analysis in there classifier. On a similar note, Hu, Tang, Tang & Liu (2013) utilised social connections to reduce noise in sentiment measurement of social networks short texts. They modeled tweets sentiments relations via graph Laplacian. It means their efforts conducted under the assumption that author's network can explain tweets sentiment in the classification process.

Deep learning as one of the recent emerging techniques of machine learning has been applied in many domains including sentiment analysis. Long Short-Term Memory (LSTM) and its variations have been a promising candidate to measure sentiment in short texts from social media. Wang et al. (2015) introduced an LSTM recurrent network model where they predict tweets sentiment. Naive Bayes classifiers are used in estimating sentiment of documents and microblogs contents. Naive Bayes calculates probability by

measuring frequency of words. Prasad (2010) experimented Naive Bayes to classify sentiment of social media content and concluded that Naive Bayes as positive in terms of performance. Naive Bayes classifier introduced and tested for aspect-based sentiment analysis by Mubarok et al. (2017).

Sentiment Analysis in Prediction

When it comes to applications of sentiment Analysis in prediction, elections have been an interesting event to study. A few studies have tested their sentiment analysis models on elections. Anuta et al. (2017) experimented whether sentiment analysis and social networks can be a replacement for traditional election polls. Bovet et al. (2016) intended to predict 2016 US presidential election in a temporal perspective and concluded that sentiment analysis on twitter can be a predictor for public opinion in a national level. Similar effort is performed in predicting recent French election and they were successful in predicting Macron's victory (Wang & Gan 2017). Paul et al. (2017) proposed a topic sensitive spatio-temporal sentiment analysis framework. This system is able to analyse aspect-based sentiment for a given location and time.

Sentiment analysis is also proposed to be applied in predicting stock market. Alongside the news effect on stock price movement, public and investor sentiment is identified as a more important factor on stocks price volatility. It is validated that market is driven by investor's decisions and investors make decisions based on their sentiment and bullishness (Qian & Rasheed 2007, Rao & Srivastava 2014, Bollen, Mao & Zeng 2011). Recent researches in the area of sentiment analysis have attempted to develop techniques for public sentiment indicators measurement directly through social networks and blogs. (Bollen, Mao & Zeng 2011) showed that Dow Jones Industrial average can be predicted by collective public sentiment on Twitter. There are other studies like (Oh & Sheng 2011) that studied investors blogs such as stocktwits.com and concluded that stock microblogs

content seem to have strong correlation with stock movements. Furthermore, (Liu et al. 2015) studied the social network influence on stock return co-movement. They showed that companies with official twitter account have higher co-movement than companies without official page.

1.3.2 Social Influence Measurement

Sentiment analysis has been mainly about the literal text sentiment and author of the text has been ignored in the process. Studying the text's author can help the process in understanding the author's sentiment. Understanding a document's author and his influence on readers is potentially an important information to consider when the aim is prediction via aggregated sentiment analysis.

Social influence in social networks is generally studied through network structure and user's position and connectivity in the network. The major adopted algorithms for network structure based influence measurement include PageRank Haveliwala (2002) and HITS Kleinberg (1999). Numerous works have applied PageRank algorithm's variations on social network graph to rank user influence according to the network structure. The network structure is relatively static compared to the activities of users in social networks. Some studies have included the social network related meta data (in case of Twitter, the meta data are retweets, mentions, and likes) (Hajian & White 2011).

There are also influence measures where they identify global influencers instead of influencers for specific topics. An example of such works is Barbieri et al. (2012) where they extended the Linear Threshold Model and Independent Cascade Model to be topic-aware, the topics are still obtained based on the network structure, while totally ignoring the valuable content information.

More recently, topic-based influence studies have combined content of user posts with

link-based metrics. Haveliwala Haveliwala (2002) proposed a topic-sensitive extension of PageRank to rank query results in regards to the query topics. The idea of topic-sensitive PageRank was later used and adjusted for social networks such as Twitter for ranking topic-based user influence. Topical authorities were also studied in Pal & Counts (2011). In Welch et al. (2011), they found out that topical relevance is better detectable through the retweet link rather than following links. They used two variations of PageRank algorithm to on retweet and following graphs for that purpose. In a recent work by Katsimpras et al. (2015), they proposed a supervised random walk algorithm for topic sensitive user ranking. As it is obvious from the algorithm name, it needs labeled data which is not very practical in many cases specially with the volume of social networks.

1.4 Dissertation Outline

This thesis is organised into six chapters including introduction. The first section of the thesis is focusing on sentiment analysis (Chapter 2) and the second section focuses on influence measurement and prediction (Chapter 3 and 4). Chapter 2 introduces SSSA, a scalable sentiment analysis approach. Chapter 3 discusses a topic-based influence measurement approach. We introduce a topic-based influence prediction approach in chapter 4 which is based of the influence measured in chapter 3. Chapter 5 is completing the purpose of this thesis by introducing weight-based sentiment analysis, an approach to utilise sentiment analysis and influence of the author for prediction purposes. Chapter 6 is concluding the thesis, summarise the work, explain the limitations and propose future works. In more details, the thesis outline is as follows:

- First, the thesis introduces a scalable sentiment analysis approach. Scalability is one of the main challenges of social media analysis and sentiment analysis. Microblogs as emerging opinion sharing platforms require new approaches that are

more scalable and accurate. The thesis propose, implement, and evaluate SSSA, a Semantic Scoring Sentiment Analysis approach, which matches the demand of scalability and efficiency of a sentiment analysis system for a large volume of microblogs. First, an automated tweet tagging system is built to alleviate the cost of building a dataset for training. SSSA, then calculates semantic scores of words in tweets. These scores are the baseline for the proposed subjectivity classification and feature selection steps. Also, a polarity classifier is built to estimate the polarity of subjective tweets on top of them. The experiments are performed on two different datasets, mobile phone and 2016 US presidential election. In the first experiment, the scalability and accuracy of SSSA are tested. In the second experiment, the accuracy of the approach is compared with benchmarks. The first experiment shows that compared to conventional approaches that are based on emoticon, hashtag, or semantic scoring, SSSA has improved the scalability and accuracy of microblog sentiment analysis and also outperforms emoticon and hashtag based approaches. The second experiment demonstrated that SSSA outperforms the state-of-the-art benchmarks on the large dataset of 2016 US presidential election dataset.

- Then, the thesis studies social influence and introduces Topic-based Social Influence Measurement (TSIM). Users of social networks spread their opinions, get involved in discussions, and consequently, influence each other. However, the level of influence of different users is not the same. It varies not only among users, but also for one user across different topics. The structure of social networks and user-generated content can reveal immense information about users and their topic-based influence. Although many studies have considered measuring global user influence, measuring and estimating topic-based user influence has been under-explored. TSIM proposes an approach that incorporates network structure, user-generated content, and user's interactions in the network for topic-based influence measurement. The experimental analysis performed on Twitter data and it is shown that the proposed TSIM can effectively measure topic-based user influence.

- Next, the thesis proposes a collaborative topic-based social influence model (SICTR) that incorporates both network structure and user-generated content for topic-based influence prediction. SICTR predicts topic-based user influence on unobserved topics, based on observed topic-based user influence through their generated contents and activities in social networks. The experimental analysis performed on Twitter data shows that SICTR outperforms state-of-the-art approaches on recall, accuracy, precision, and F-score for predicting topic-based user influence.

- Finally, the thesis is concluded by proposing Weight-based Sentiment Analysis (WSA), where the author's influence is included in sentiment analysis measurement. WSA aims to propose a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. Such predictions traditionally are conducted through polling systems which are highly expensive with limited data points. In the last main chapter, the focus is on the impact of influential users in prediction through sentiment analysis. In particular, the goal is to observe whether influence-based aggregated sentiment analysis is outperforming generic aggregated sentiment analysis in prediction. For evaluating SICTR, a dataset collected from an online social network for the US 2016 presidential election. WSA predicted the election results and performed with higher accuracy.

Chapter 2

A Sentiment Analysis Approach for Microblogs Based on Semantic Scoring

2.1 Introduction

[1]The rapid and intensive growth of Internet text data creates immense opportunities and challenges for applying techniques to learn "Big Data", e.g., extracting meaningful knowledge and discovering patterns from big corpuses, deals with collecting, organising and analysing voluminous data. One of the sources of such massive amount of data comes from microblog service providers such as Twitter. Twitter is a platform of sharing information and opinion and users publish tweets on wide variety of topics everyday and has attracted hundreds of millions of users to share personal activities and public messages (Kwak et al. 2010). Given the huge amount of posts everyday, Twitter becomes an attractive source for conducting large-scale data analytics for many research and commercial

[1]**Aso Hamzehei** et al., Scalable Sentiment Analysis for Microblogs Based on Semantic Scoring, *IEEE International Conference on Services Computing (IEEE-SCC)*, New York, NY, 2015, pp. 271-278. (Hamzehei et al. 2015)

projects. In the era of ”Big Data”, with emphasis on the 5 V’s (volume, velocity, variety, value and veracity), a scalable, accurate and fast algorithm is the centre of challenges in the recent works in this domain.

In general terms, sentiment analysis aims to identify polarity of a given text whether it is positive or negative regarding to a particular topic. Machine Learning (ML), one of the text sentiment classification approaches, is usually based on supervised binary classification, which classifies a text into two classes of positive and negative. These methods need labelled datasets for training classifiers (Pang et al. 2002). An advantage of supervised learning approaches is their capacity to create context aware models, while their shortcoming is the requirement of labelled datasets. Another category for sentiment analysis consists of the works that use Natural Language Processing (NLP) techniques. Lexical-based approaches engage a word-list that each word’s sentiment is predefined. NLP related techniques focus on the word and feature levels of semantic text analytics, which drill down into the details of the context-aware meaning of the lexicons.

Polarity classification is more challenging when it comes to microblogs especially Twitter, due to their very short text length (Wilson et al. 2009). The challenging tasks of sentiment analysis on tweets are topic and polarity identification on the very short text and manual tagging of Twitter corpus for supervised polarity classification. To address both challenges, researchers have employed metadata such as emoticons (Wong et al. 2014) and hashtags for tweets positivity and negativity detection(Wang et al. 2011). Although this kind of meta-information can reduce expenses of tweets tagging, their accuracy of labelling the correct sense of a tweet is controversial.

Another major challenge for handling big text datasets specially using NLP is reducing the computational time and memory requirement. Due to the computational cost of NLP, there is not enough efforts in the direction of improving the NLP for handling big data. As a result, other branches of ML rather than NLP have received more attention from ML research community as the main research track for development algorithms for handling

big text data. The precision and accuracy of NLP-based models are widely recognised, however they are not well accepted for the big data world. Scalability is a missing component in the semantic-based products due to expensive runtime and low performance when datasets grow.

To address those issues, a novel scalable sentiment analysis approach is presented in this dissertation to improve performance and scalability of sentiment analysis task on big text corpuses of very short texts (tweets). Our approach addresses the big corpus analytics limitation, which has less computational cost and higher performance while not sacrificing the polarity classification accuracy. The proposed approach is evaluated in two experiments. First, it is compared with conventional and common used emoticon-based and hashtag-based sentiment analysis methods in terms of classification accuracy and performance. The main principal behind our approach is combining the features of NLP with ML to reduce the high costs of semantic-based models to step over the limitations on the conventional NLP and ML techniques and build a scalable sentiment analysis system. We use Word Sense Disambiguation (WSD) in order to utilise relevant meaning of all useful words in the tweets to improve the accuracy of subjectivity and objectivity detection. Due to scalability, our approach can be potentially used for sentiment analysis on large tweet corpuses. Moreover, in the second experiment, we test our approach on a big dataset and evaluated its accuracy along with social network related sentiment analysis approaches.

The proposed semantic scoring based sentiment analysis method (after data cleansing and preparation) calculates semantic scores of words of tweets. Subjectivity classification and feature selection are two next main steps, which use semantic value of tweets. Finally, we develop a polarity classifier to estimate tweets polarity. An automated tweet tagging system has also been developed to automatically label the tweets polarity to build our training dataset. The rest of this chapter is organized as follows. We elaborate the proposed scalable approach in Section 2.3. The results and experiments are shown in Section 2.4 and the last section contains the conclusion and future works. In this chapter, the

first experiment focuses on scalability and performance, and accuracy of the SSSA and the second experiment focuses on performance and accuracy of SSSA comparing to other existing sentiment analysis approaches.

The contribution of this chapter is summarised as below:

- We propose a methodology that has engaged semantic scoring for subjectivity and polarity classification to be applied on a large scale of tweets, which is more scalable and accurate.
- In the first experiment, the proposed approach improves accuracy of sentiment prediction by 10% and 2.2% in compare to conventional hashtag-based and emoticon-based approaches. It also shows 3.42% and 7.26% improvement in regard to SSSA2 and SSSA1 (two similar semantic-based approaches). In the second experiment, our approach outperforms VADER by 4.2% and TextBlob by 20%.
- Our approach reduced the sentiment analysis computational cost to one third in regard to conventional emoticon-based, hashtag-based, and similar semantic-based (SSSA1 and SSSA2) approaches.
- We developed an automatic system for tagging the tweets with high precision for single target tweets.
- Our approach is able to filter out news, ads, and noise, automatically.

2.2 Related Work

Sentiment Analysis is mainly considered as a text classification problem. Words are the key features, also called feature vectors, for machine learning algorithms. These features reported in the literature are words and their frequencies, parts of speech (POS) tags,

syntax, opinion words and phrases, words positions, negation and etc (Kumar & Teeja 2012). For sentiment analysis on a statement, its polarity needs to be evaluated, however, a statement or opinion expression is not necessarily subjective. Thus, subjectivity classification is set as a pre-step to filter out objective tweets from the training corpus for polarity classification (Montoyo et al. 2012).

Polarity classification is more challenging when it comes to microblogs especially Twitter, due to very short text length (140 characters in case of Twitter). The challenging tasks of SA on tweets are topic and polarity identification on the very short length text and manual tagging of Twitter corpus for supervised polarity classification. To address both challenges, researchers have engaged metadata such as emoticons (Wong et al. 2014) and hashtags for tweets positivity and negativity detection(Wang et al. 2011). Read (2005) showed the usability of emoticons in enhancing the polarity identification of tweets when using a trained model on a domain and testing it on a different domain. Emoticons also have been engaged for tweets automatic tagging (Wong et al. 2014, Montejo-Rez et al. 2014). Although meta-information can reduce expenses of tweets tagging, their accuracy of labelling the correct sense of a tweet is controversial. First, emoticons and hashtags do not appear in all tweets. (Wang et al. 2011) reported that nearly 14.6% of tweets contains hashtags and this number is less than 10% for messages with emoticons on Twitter (Park et al. 2013). Second, a smiley face does not always represent a positive subjectivity of the author regarding the tweet topic.

Although most of these approaches need immense annotated training datasets, they can overcome this limitation by utilising dictionary-based approaches which depend on existing lexicographical resources (such as WordNet (Miller 1995)). The dictionary-based techniques provide semantic data in regards to individual senses and words (Khan 2011). Several authors have used the advantages of the graph nature of WordNet. Hassan & Radev (2010) applied a Markov random walk model to WordNet, producing a polarity estimate for any given word. In (Esuli & Sebastiani 2007)the objective of the authors is

to obtain a semantic ranking version of WordNet, using a modified version of the original formula of PageRank (Page et al. 1999) to bias the random walk to positive and negative terms. An alternative method was suggested by (Kamps et al. 2004) for utilising WordNet's synonym relations for tagging words.

SentiWordNet is a lexical resource that is used by the opinion mining community. It is based on WordNet, and provides a domain-independent resource for obtaining certain information about the degree of emotion of its concepts (Denecke 2008, Ogawa et al. 2011). In (Read 2005) the authors use SentiWordNet to obtain the semantic value and develop a microblog sentiment analysis system to discover consumers' opinions. The system performs a subjective classification to distinguish subjective and objective tweets. It measures the polarity of the tweets, which is obtained by an aggregation process of the polarity values of each term of the tweet. Montejo-Rez et al. (2014) uses the method of extracting weighted nodes vector from the WordNet's graph along with features of tweets such as emoticons. They performed classification by comparing methods that are widely studied such as SVM and SentiWordNet Rank WordNet (SWN-RW). The results of comparing the two methods of classification showed the proposed method has precision and recall that they are not much different from SVM, but the proposed method has advantages to avoid the ambiguous meaning of word.

Scalability of semantic-based services is their main drawback due to expensive runtime and low performance when datasets grow. Combining the features of NLP with scalable and intuitive systems can reduce the high costs of semantic-based models to step over the limitations on the conventional NLP techniques. WSD in the long text produce the higher quality while it needs more computational time. However, for short text (e.g. tweets), WSD can find the optimal combinations of senses in the short time because there are not many different combinations of senses. Therefore, it improves the performance of the system, which is intended in our work.

2.3 Semantic Scoring Based Sentiment Analysis

The main principal behind our approach is engaging the strength of NLP to handle sentiment analysis on big tweets corpuses. Our approach addresses the big corpus analytics limitation, which its computational cost and performance improved while not scarifying the accuracy of polarity classification. Our approach contains three main steps. First, given a tweet collection, it applies an extensive data cleansing process to remove noise and prepare tweets for the next phase. Second, tweet corpus goes for semantic scoring. The words in each tweet will be weighted. The weights express the polarity degree of the words in a tweet. The assigned weights will be used for scoring subjectivity and polarity of the tweet. In the next phase, an automated tweet tagging system is built to label the tweets. The tagging system uses the calculated weights to detect subjectivity and polarity (in case the subjectivity is approved) of the tweets. In the last phase, we build the subjectivity and polarity classifiers.

2.3.1 Tweets Preprocessing and Cleaning

Removing noise and cleaning text, plays an important role in the whole process of sentiment analysis. Tweets are considered as a very noisy short text that is written in an informal way. When the objective is mining opinion of users on a specific topic, it is important to exclude non-important tweets such as news, advertisements and objective tweets (objective tweets will be addressed in the subjectivity detection section though). Having unrelated tweets filtered, we remove urls, and usernames mentioned in tweets. Moreover, we check the spelling and correct misspelt words. Figure 2.1 shows the steps of our approach (SSSA).

Tweet: daddy got me a galaxy s4. i'm the happiest girl alive #grinning				
Word	**Root**	**POS**	**Sense ID**	**SentiWN Score**
daddy		n	09988063	0
got	get	v	02210855	0.125
me		-	-	-
a		-	-	-
galaxy		n	08270938	0
s4		n	-	-
i		-	-	-
ḿ		v	-	-
the		-	-	-
happiest	happy	a	01148283	0.875
girl		n	10129825	0
alive		a	00094448	0.125
grinning		n	06878071	0.125
			Subjectivity Score	0.312
			Polarity Score	0.312

Table 2.1: Tweets sentiment, subjectivity, and polarity scores through three examples of positive, negative, and neutral tweets

2.3.2 Word Sense Disambiguation (WSD)

Since a tweet may include some ambiguous words with different senses, identifying the most related sense of each word based on the context is necessary to extract the polarity of that tweet. As a formal definition, if we ignore the punctuation, we can view a text (T) as a sequence of words (W), and each word has some senses (S) in Dictionary (D). Therefore, WSD can be formally described as the task of mapping a sequence of words to a sequence of appropriate senses:

$$\begin{aligned} T &: SeqW \\ D &: W \rightarrow PS \\ WSD &: T \rightarrow SeqS \end{aligned} \tag{2.1}$$

For example, if $T = \{W_1, W_2, ..., W_n\}$ and each word W_i has a set of senses in dictionary $D(W_i) = \{S_{i1}, S_{i2}, ..., S_{im}\}$, the output of WSD algorithm will be a sequence of senses such that each sense is an appropriate sense for a correspondent word in the input text $WSD(T) = \{S_1, S_2, ..., S_n\}$. In other words, if we consider WSD as an optimisation problem, the task is finding a sequence of senses, which maximises the overall relatedness value among the words within a certain context window of length n. The objective function can be defined as:

$$argmax_{i=1}^{m} \sum_{j=1}^{n} max\{rel(S_{ti}, S_{j1}), \ldots, rel(S_{ti}, S_{jm})\} \tag{2.2}$$

where W_t is the target word to be disambiguated and rel is the relatedness value between two senses. The overall relatedness is calculated for each sequence and finally the sequence that resulted in the best relatedness is considered. Considering the above definition, our WSD process consists of the following steps:

First of all, we tag a given tweet by the POS tagger, which assigns a part of speech (such as noun, verb, and adjective) to each word (and other tokens in the text). Knowing the POS would dramatically decrease the number of necessary senses and simplifies the problem. In this regard, a POS tagger from Stanford University is used in this step because of its performance and accuracy.

In the next step, we extract the root (lemma) of a given word using a lemmatization algorithm developed by Princeton University. Having the root of each word we can obtain the synsets of the roots from the WordNet.

We employ the adapted lesk algorithm for WSD. This method expands the number of words to be compared by looking up their glosses found through several semantic similarities (extracted from WordNet-based resources). These semantic similarities are: synonyms, hypernyms, hyponyms, holonyms, meronyms, troponyms and attributes.

2.3.3 Semantic Scoring Using SentiWordNet

SentiWordNet (Baccianella et al. 2010), which uses WordNet as a lexical resource, is engaged and returns a synset as set of scores to represent the notions of neutrality, positivity, and negativity. A synset is the basic element of information in WordNet, which represents a ”concept” that is unambiguous. There are relations among synsets over the lexical graph. These relations can be one of the hyperonymy, synonymy, homonymy and etc. Words in the graph are weighted in regards to its subjective and polar sense. The newest version of SentiWordNet (3.0) is build from manual annotations of previous versions and making the complete lexical graph by applying a random walk algorithm. SentiWordNet (3.0) provides a domain-independent resource for achieving certain information about the level of emotion of its concepts.

When we find the appropriate senses of the ambiguous words in a tweet, we send them along with their POS to the SentiWordNet and it returns the score of each sense. An example of this process is shown in Table 2.2.

2.3.4 Automated Tweet Tagging System

The cost for building a practical, manually annotated training dataset is always expensive. It is almost impossible in big corpuses to prepare a dataset with a reasonable size. Some researchers have used emoticons or hashtags to annotate training datasets automatically. However, as discussed in the related works, a small percentage of tweets contain those metadata and a majority of tweets will be ignored with those tagging systems. We use the calculated semantic scores of tweets in the previous step to build an automated tagging system. A tweet is considered as subjective when its subjectivity score is greater than θ, where θ is the summation of absolute values of semantic scores of the words in each

Positive Tweet (TPos):
daddy got me a galaxy s4. i'm the happiest girl alive #grinning
Negative Tweet (TNeg):
i hate having this outdated android phone i need an upgrade #iphone
Neutral Tweet (TNeu):
as i walked around the kitchen with one hand holding my android to my cheek and the other holding pepper

TPos	Score	TNeg	Score	TNeu	Score
daddy	0	i	-	as	-
got	0.125	hate	-0.75	i	-
me	-	having	0.25	walked	0
a	-	this	-	around	0
galaxy	0	outdated	-0.75	the	-
s4	-	android	0	kitchen	0
i	-	phone	0	with	-
ḿ	-	i	-	one	0
the	-	need	-0.25	hand	0
happiest	0.875	an	-	holding	0
girl	0	upgrade	0.125	my	0
alive	0.125	iphone	0	android	0
grinning	0.125			to	-
				my	-
				cheek	0
				the	-
				other	0
				holding	0
				pepper	0
Subjectivity	0.312		0.425		0
Polarity	0.312		-0.575		0

Table 2.2: Tweets sentiment, subjectivity, and polarity scores through three examples of positive, negative, and neutral tweets

tweet.

$$subjectiveweights > \theta \tag{2.3}$$

The chosen subjective tweets are considered as tweets that hold a negative or positive polarity. A tweet is positive if it satisfies the condition in the following two equations where δ is the polarity threshold.

$$polarityweights > \delta => positive \tag{2.4}$$

$$polarityweights < -\delta => negative \tag{2.5}$$

The value of threshold parameters are specified through running the system over a small manually annotated tweets, which contains positive, negative, and neutral tweets from the main corpus.

2.3.5 Feature Selection

Feature selection is usually either lexicon-based that requires manual annotation or automatic. Our proposed feature selection method is a mixture of both, in a way that combines the advantages of both sides. The words semantic values that are calculated in Section III.C are the baseline for feature selection in our approach. Words with zero semantic score are those that do not have sentiment value. We assume that these words will not be able to help the classifier in sentiment detection and removing them improves the model performance. Our assumption is that by excluding these words and keeping the words with sentiment value greater than zero, we can reduce the dimensionality of the corpus while keeping the sentiment prediction accuracy in a reasonable level.

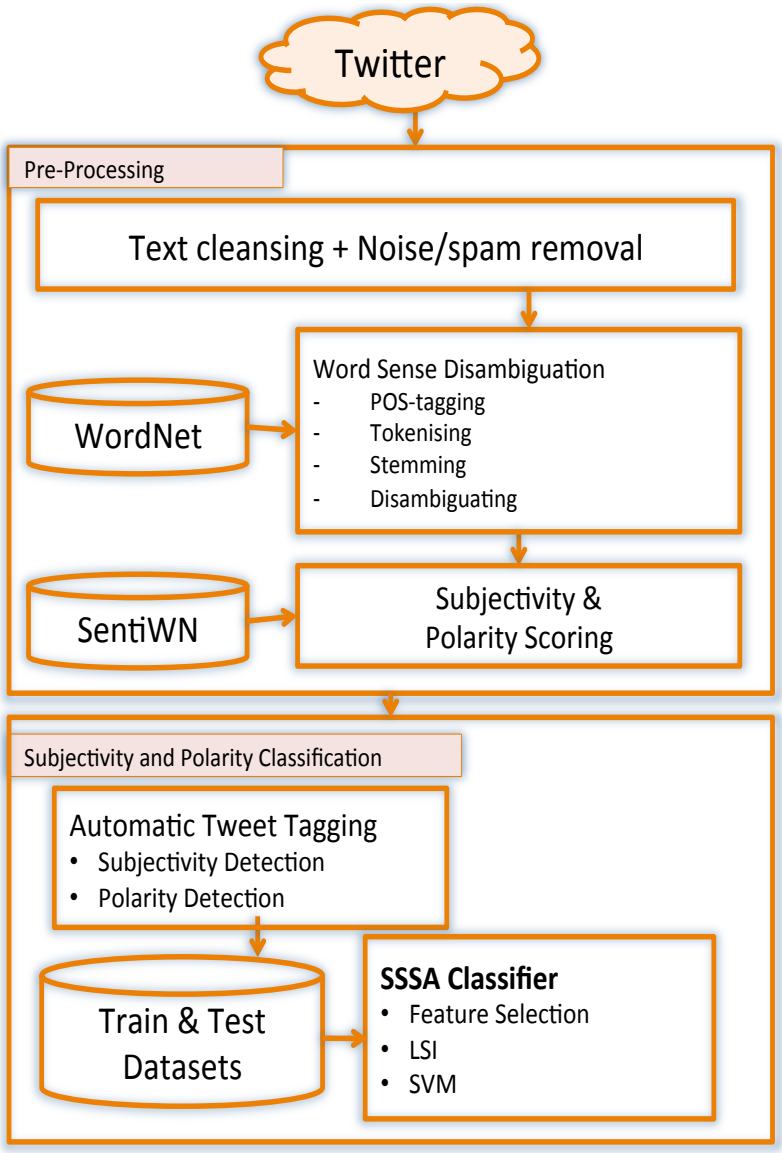


Figure 2.1: Semantic Scoring Sentiment Analysis methodology.

2.3.6 Latent Semantic Indexing (LSI)

LSI is a word-level embedding technique that can capture semantic distance between a pair of words. It has specific advantages in detecting the latent semantics in short text and also in reducing the text dimensionality by applying Singular Value Decomposition (SVD) to term-document co-occurrence matrix. This enables efficient computation of the semantic similarity between tweets and words through a low-dimensional representation of them. There are other methods inspired by LSI like probabilistic LSI and Latent Drichlet Allocation (LDA). Given a set of n documents denoted by $P = \{P_1, ..., P_n\}$, the LSI generates a set of k topics denoted by $T = \{T_1, ..., T_k\}$. Each topic is related to a weighted representation over m words denoted by $Ti = \{W_1, ..., W_m\}$, where W_j is the weight representing the contribution of word j to topic i.

2.3.7 Support Vector Machine (SVM) Classifier

SVM is well suited for text sentiment classification due to the sparse nature of text. E.g. having a set of training tweets $T = \{T_1, T_2, ..., T_n\}$ where each tweet is labeled to a class (pos, neg). The classification model is related to the features in the underlying record to one of the class labels. Then, for a given instance of unknown class, the model is used to predict a class label for it. Alec Go et al. found that SVM works better than Naive Bayes and Maximum Entropy in their experiments (Go et al. 2009). We follow their approach and adopt SVM for building the classifier.

2.4 Experiments and Results

In this section, we evaluate our proposed SSSA on two different datasets and compare with available sentiment analysis approaches for social networks. First, we perform anal-

ysis on mobile phone dataset and compare SSSA with EBSA and HBSA. We also perform comparisons and analysis on big dataset of US 2016 Presidential Election and we compare SSSA with Vader, and TextBlob sentiment analysis approaches.

2.4.1 Experiment 1

Dataset and data pre-processing

For the experiment, we streamed tweets related to mobile phones. Our prepared keywords for stream query contain mobile phone related words such as iphone, samsung, nexus, htc, one, android, etc. We used Twitter API to stream the tweets that contain at least one of the key words in the list of keywords. Among the streamed tweets, we kept the English tweets that are not retweets. As a result, we obtained 3717039 English original tweets.

The train/test dataset preparation process is described as follows:

- We have prepared a sample dataset from the streamed tweets for testing the proposed approach. The dataset contains tweets that have just one target from the list of mobile phones. Our focus in this chapter is on single target tweets. Therefore, the sampled dataset from the main streamed corpus is chosen when only one target from those in our target list appears in a tweet. The comparative tweets are not considered in this work. A comparative tweet has more than one target. For example in this tweet ”I love iphone5 more than HTC one” two targets iphone5 and HTC 1 are comparative. It means positive sentiment over iphone gives negative sentiment to HTC one. It is important to map the determined sentiment to the right target. As comparative tweets are out of the scope of this chapter, they are filtered out and therefore single target tweets are chosen. Further preparation on the corpus is explained in the pre-processing section in the research methodology. After removing

Table 2.3: Accuracy of the proposed approach SSSA3 against SSSA1 and SSSA2 (in %)

Approach	Sample size					Average
	5k	10k	20k	40k	80k	
SSSA1	66.9	72	75.4	78.6	80.1	74.6
SSSA2	72.6	76.2	79.8	81.3	82.3	78.44
SSSA3	79.1	80.4	82	83.1	84.7	81.86

duplicate tweets and affecting the single target condition, total number of 178552 tweets remained.

- The automatic tweet tagging system identified 82768 positive and negative tweets that 80000 (half positive and half negative) of them selected for the experiments. This tagging system is explained in Section 2.3.4. This dataset is used for scalability assessment. Also, it is applied for comparison of our approach with similar semantic-based approaches.

- A dataset consist of 20000 subjective tweets, which derived from the final tweets dataset used for accuracy comparison of our approach with the hashtag-based approach. This dataset contains at least one hashtag.

- A dataset consists of 1062 subjective tweets is prepared and derived from the final tweets dataset. This dataset is used for accuracy comparison of our approach and emoticon-based approach. Tweets in this dataset contain at least an emoticon.

Performance Study

We conducted several experiments to evaluate polarity classifiers, their prediction accuracy, and computational performance. In the first round of experiments, we compare the proposed approach (SSSA3) with two other similar semantic scoring based approaches (SSSA1 and SSSA2). In the second round, we evaluate our approach with conventional

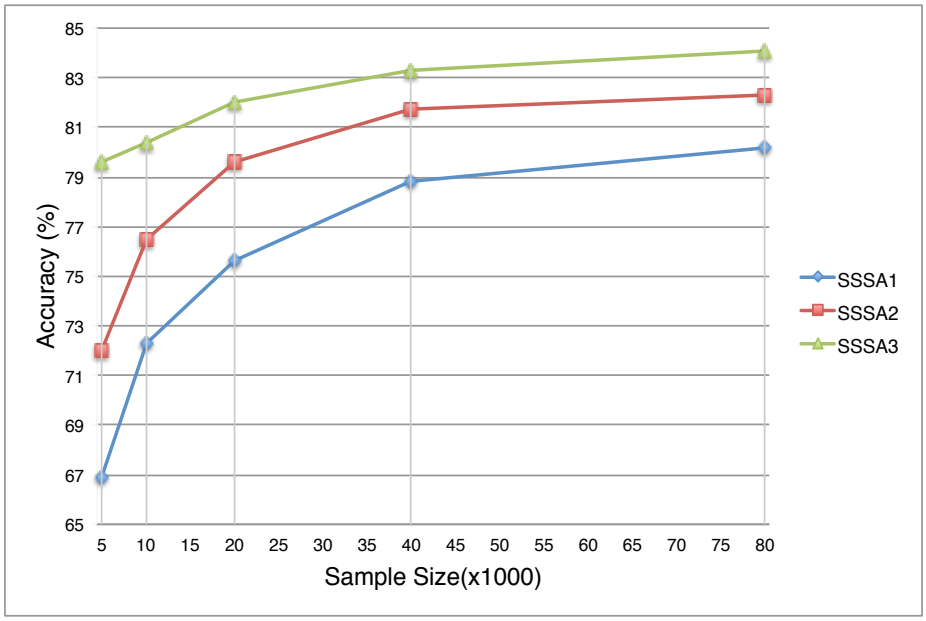


Figure 2.2: Accuracy of SSSA1-3 over 5 different sample size 5k-80k.

hashtag-based approaches. Finally, we compare SSSA3 with conventional emoticon-based approach in terms of prediction accuracy and scalability. To classify the tweets polarity, we build an SVM classifier with 10-fold cross validation. We used gensim (open source vector space and topic modelling toolkit for python) to build LSI models, TF-IDF, and dictionary from the corpus. Also, scikit-learn library for python used to build the SVM classifier. The computer that we used for these experiments has the configuration of processor ”2.6 GHz Intel Core i5” and memory ”8 GB 1600 MHz DDR3”.

1. Comparison with similar semantic-based approaches

   SSSA1, SSSA2, and SSSA3 have similar steps as Figure 2.1 except of their feature selection phase. We run three different sets of experiments that each round has a different set of extracted features as follows:

   - SSSA1 features: For this round of experiments, the remaining words of each tweet after cleaning phase are considered as the features of that tweet.
   - SSSA2 features: the feature space for SSSA2 consists of the words of each

Table 2.4: Prediction accuracy of SSSA3 and HBSA

	SSSA3	HBSA
Accuracy	86%	76%

tweet that are transformed into their root format. For example, all words of go, goes, going, gone, and went are transformed to ”go”.

- SSSA3 features: The features from each tweet in this round are chosen based on the semantic score of the words in each tweet. As it is explained in the feature selection Section 2.3.5, the features that have a semantic value will be the candidate features to be used for training the classifier.

For each of SSSA1, SSSA2, and SSSA3, five different tweet sample sizes (5000, 10000, 20000, 40000, and 80000) are prepared to test and evaluate the accuracy and scalability of the proposed approach over mentioned sample sizes. In each sample size, 75% of the tweets are used for training and 25% for testing of the classifier.

Figure 2.2 shows the accuracy of the polarity classifier. The x-axis represents the sample size, which is from the range of 5000 to 80000 tweets, and y-axis depicts the accuracy of the polarity classifier in each round. There is a similar trend among the accuracy of SSSA1, SSSA2, and SSSA3. The polarity classifier accuracy increases by increasing the sample size. However, SSSA3 outperformed SSSA1 and SSSA2. Table 2.3 shows that SSSA3 outperforms SSSA2 by 3.42% and SSSA1 by 7.26% in average. The average precision of SSSA3 is 81.86% and this percentage is 84.7% for the 80000-sample size.

2. Comparison with hashtag-based approach

We implemented a conventional Hashtag-Based Sentiment Analysis (HBSA) approach from (Tsur & Rappoport 2012) to compare its sentiment prediction accuracy with our approach. A dataset consist of 20000 polarity labelled tweets used in this round of analysis. 75% of this dataset used for training a classifier of our

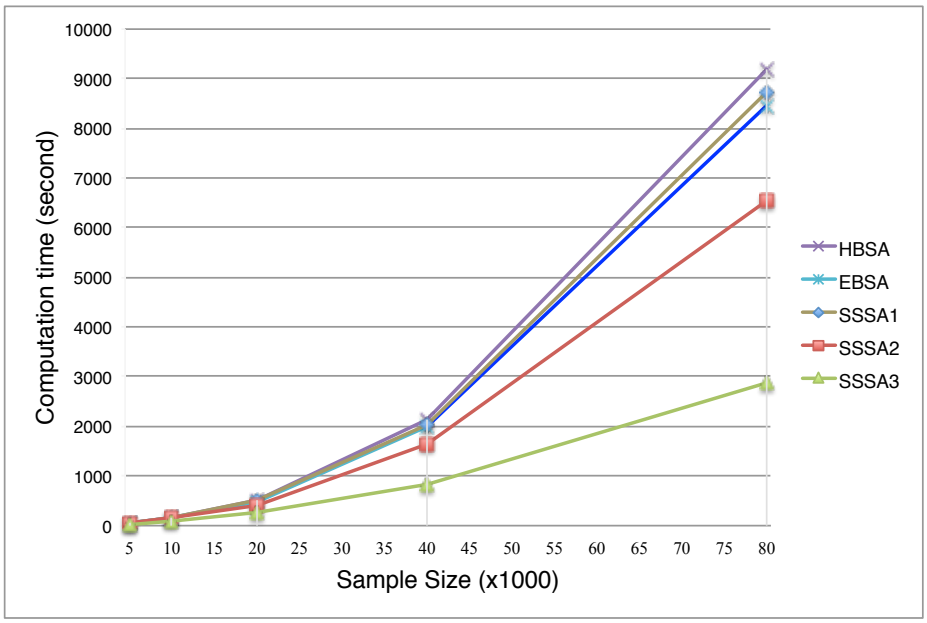


Figure 2.3: Scalability of our approach (SSSA3) against the benchmarks (SSSA1 and SSSA2, HBSA, and EBSA).

Table 2.5: Prediction accuracy of SSSA3 and EBSA

	SSSA3	EBSA
Accuracy	96.9%	94.7%

approach and one from HBSA. Table 2.4 shows the accuracy results of both SSSA3 and HBSA where our approach outperforms the HBSA by 10%.

3. Comparison with emoticon-based approach

   We implemented a conventional Emoticon-Based Sentiment Analysis (EBSA) approach from (Wong et al. 2014) to compare its sentiment prediction accuracy with our approach. A dataset of 1062 polarity labelled tweets was used for training and testing our approach and EBSA. Table 2.5 shows that our approach outperforms EBSA by 2.2%.

Figure 2.3 presents the SSSA1, SSSA2, SSSA3, EBSA, and HBSA computational time and scalability. x-axis is the sample size and y-axis shows the runtime in seconds. SSSA3

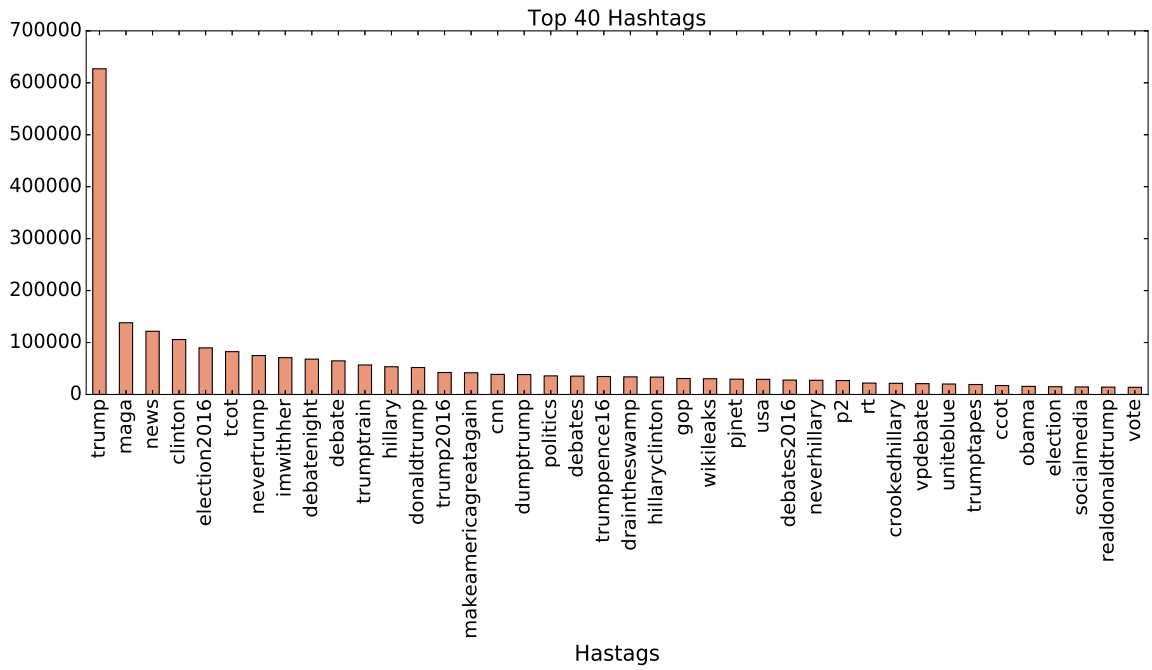


Figure 2.4: The hastgas in Election dataset sorted by the number of tweets they are appeared in.

outperformed the other compared approaches. There is a considerable improvement in the time performance of our proposed approach. The runtime in SSSA3 is one third of EBSA, HBSA, and SSSA1. The scalability diagram depicts the potentiality of our approach for sentiment analysis on big tweet corpuses. The semantic score based feature selection plays an important role in reducing the runtime of our approach. Figure 2.3 also shows that with increasing the sample size, the wisely feature selected approach of SSSA3 significantly reduces the runtime due to correctly reducing the dimensionality of the corpus with keeping the words for SA with a semantic value.

2.4.2 Experiment 2

In this section, we present the experiment outcome of SSSA comparison with Vader and TextBlob sentiment analysis approaches on a big dataset of 2016 US presidential election.

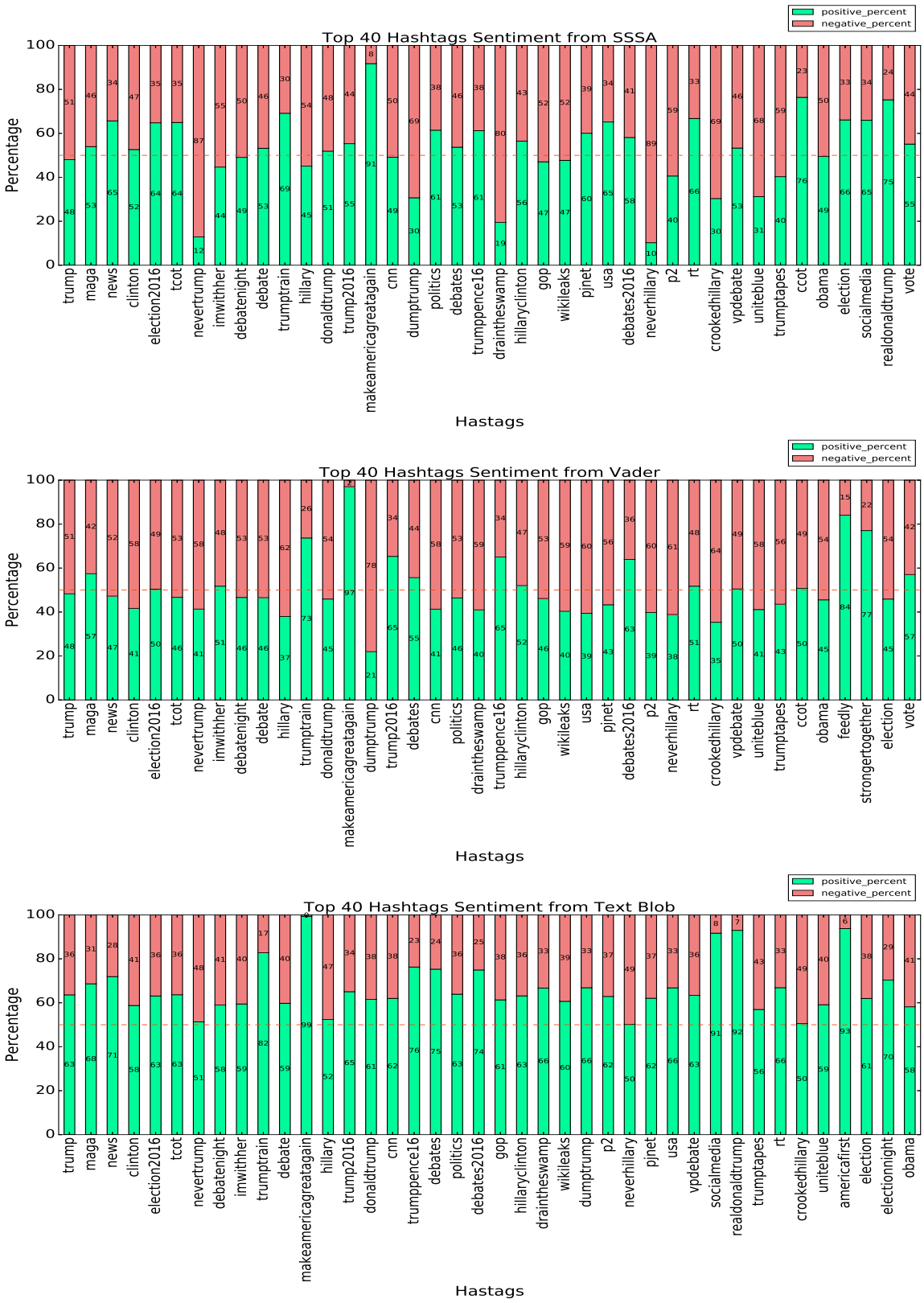


Figure 2.5: Top 40 hashtags and there sentiments measured by SSSA, VADER, and TextBlob

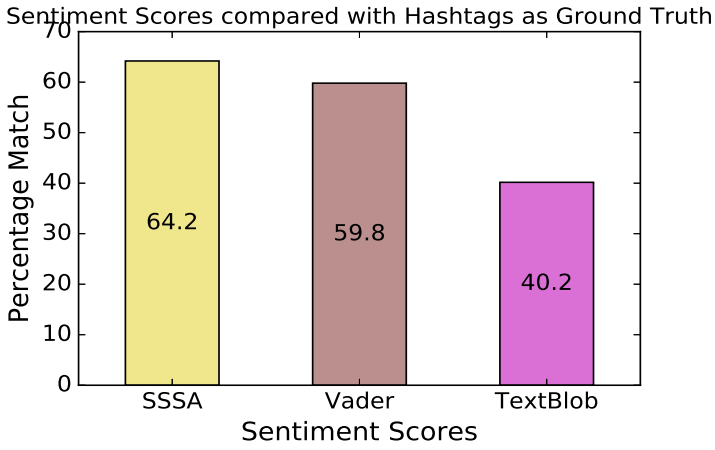


Figure 2.6: Sentiment analysis results by SSSA, VADER, and TextBlob for the entire dataset

Dataset and data pre-processing

In the second experiment, we use the 2016 US presidential election streamed and collected from twitter. The details of this dataset is explained thoroughly in the chapter 3. However, we explain here the subset of the dataset that is used in this study.

The election dataset contains a number of hastags for or against the candidates. The portion of tweets with hashtags is 25.2% of the overall tweets in the dataset. The hashtag dataset that is used in this study contains 296762 unique hashtags with 50 percentile of 1, 75 percentile of 2, mean of 20.99, and max value of 627012. This means that the majority of hashtags are not popular and 75% of the hashtags are appeared in 1 or 2 tweets. The maximum frequency belongs to hashtag ”trump” which is appeared in 627012 tweets. Figure 2.4 shows the high frequency hashtags in the dataset sorted by the number of times they are appeared in the dataset.

However, among all the hashtags, we have chosen those that has obvious negative or positive sentiments related to both major candidates of the 2016 US presidential election. The tweets that contain those hashtags are labeled positive or negative based on the hashtag that is appeared in each tweet. The labelled tweets are considered as ground truth and will

be used to evaluate SSSA and benchmark algorithms in the performance study.

Benchmark algorithms

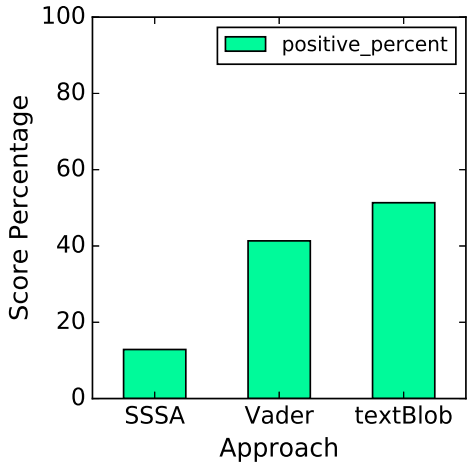
We are comparing our proposed SSSA algorithm with VADER and TextBlob sentiment analysis algorithms. VADER and TextBlob both are specialised for social media where they fit our analysis purpose as our dataset's nature is social network.

- Vader: VADER presents a rule-based model for sentiment analysis and its effectiveness is studied along with state-of-the-art techniques and it is shown that it outperforms them on social media related datasets. VADER assorts the lexical features with grammatical and syntactical rules to measure sentiment and its intensity (Gilbert 2014). This lexical based technique has MIT license and is available on NLTK package for end users as well.
- TextBlob: TextBlob is a python library for natural language processing and contains sentiment analysis feature. It stands on top of NLTK (Bird & Loper 2004) and Pattern (Smedt & Daelemans 2012). TextBlob like VADER hold an MIT license.

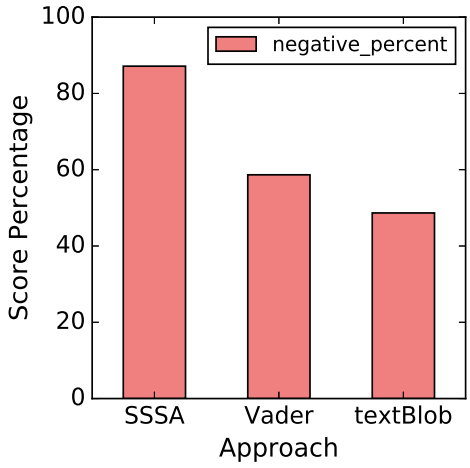
Performance study

In this section, we analyse SSSA's performance and compare it with VADER and TextBlob. We measure sentiment of the dataset and compare the algorithms for each hashtag and also for the entire dataset and validate with the ground truth. First, we present the sentiment analysis result for a subset of positive and negative hashtags and evaluate the performance of the algorithms on each hashtag. Then, we evaluate the algorithms on the whole dataset.

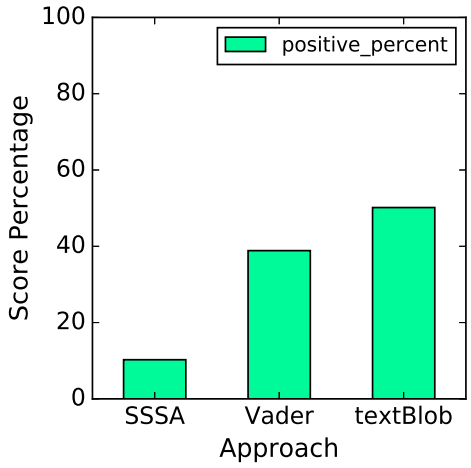
Figure 2.5 shows a snapshot of the sentiment analysis measured for top 40 hashtags by each of the algorithms. However, to analyse the details of the results, we look into each



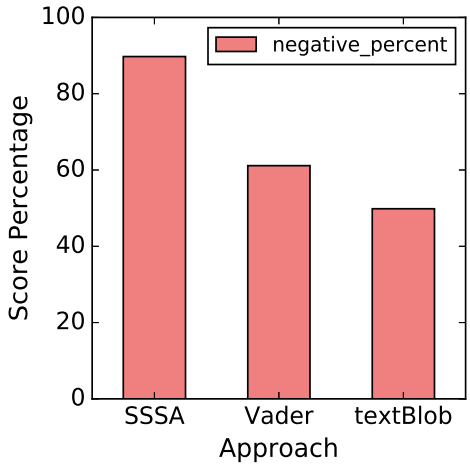
(a) hashtag ”nevertrump”



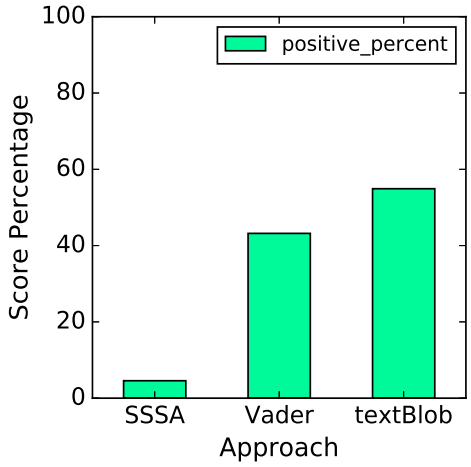
(b) hashtag ”nevertrump”



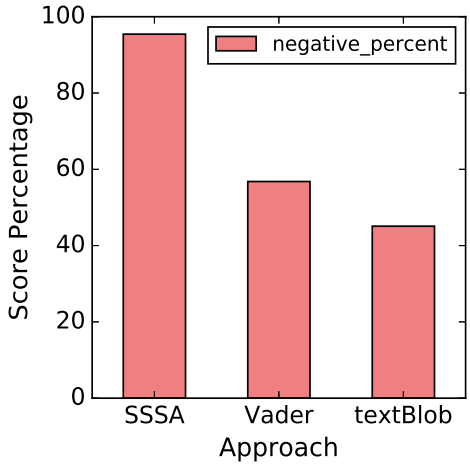
(c) hashtag ”neverhillary”



(d) hashtag ”neverhillary”

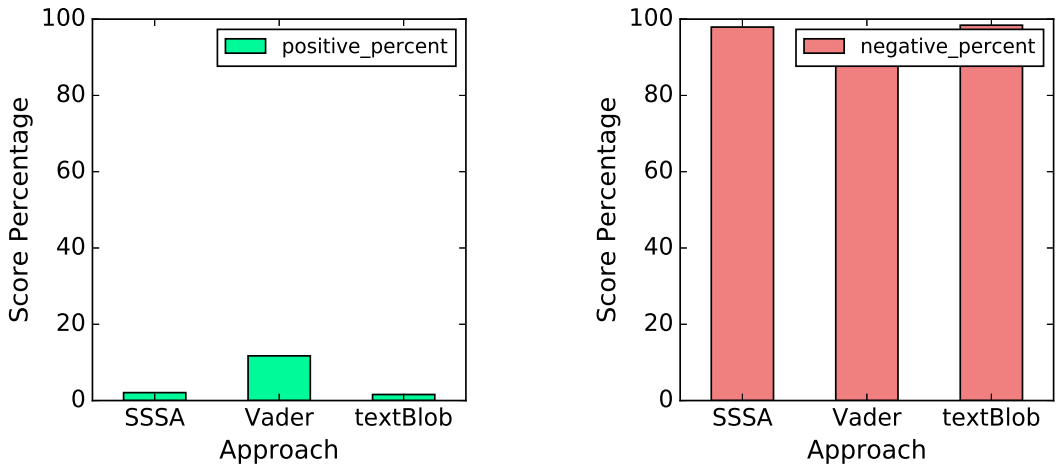


(e) hashtag ”notmypresident”



(f) hashtag ”notmypresident”

Figure 2.7: Negative hashtags and the sentiment results by SSSA, VADER, and TextBlob



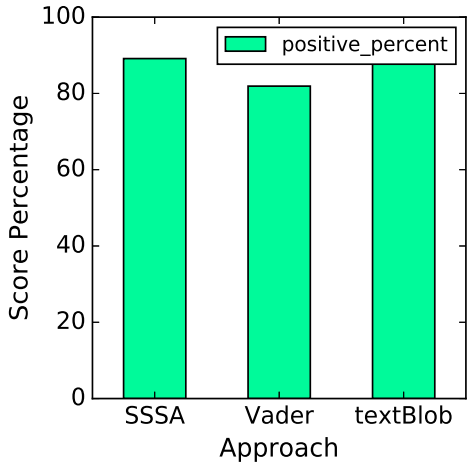
(a) hashtag "nastywoman"

(b) hashtag "nastywoman"

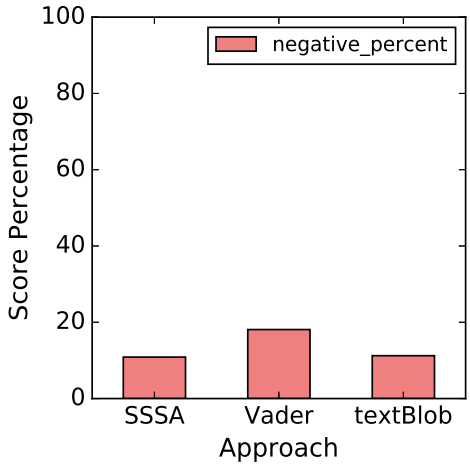
Figure 2.8: Continued - negative hashtags and the sentiment results by SSSA, VADER, and TextBlob

of the hashtags individually. Figure 2.7 shows 4 top hashtags with negative sentiment. For instance, hashtag "nevertrump", Figure 2.7b, carrying negative sentiment toward candidate Donald Trump. SSSA has the best sentiment estimation for tweets with "nevertrump" hashtag which is over 80% negativity while VADER estimate just under 60%. TextBlob had the worst estimation for "nevertrump" with estimating just under 50% of those tweets as negative. The same hashtag is also made for the other candidate, Hillary Clinton, as "neverhillary". Similarly, "neverhillary" is a negative hashtag for Hillary Clinton and tweets with this hashtag are labeled as negative. SSSA outperform VADER by over 20% more accuracy in estimating those tweets as negative and the outperformance is over 30% in compare with TextBlob. The next hashtag is "notmypresident" and SSSA estimated 90% negativity for this category of tweets where VADER estimated under 60% and TextBlob's estimation was under 50% negative. In Figure 2.8, for hashtag "nastywoman", the three algorithms estimated over 90% of tweets as negative.

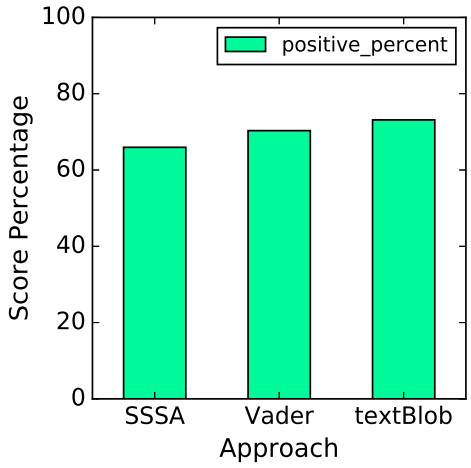
We also examined the algorithms performance on positive hashtags. Figure 2.9 shows the results on three hashtags "votetrump", "votehillary", and "makeamericagreatagain". For hashtag "votetrump" SSSA and TextBlob are equally outperform Vader and estimated



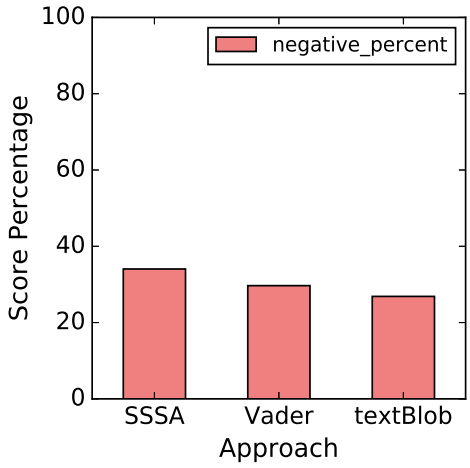
(a) hashtag ”votetrump”



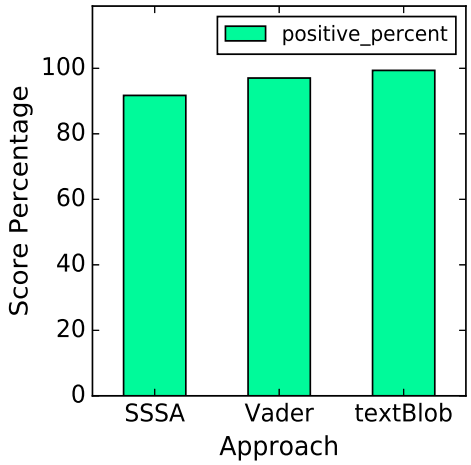
(b) hashtag ”votetrump”



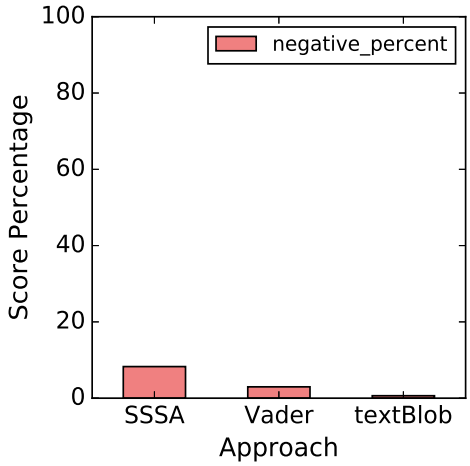
(c) hashtag ”votehillary”



(d) hashtag ”votehillary”



(e) hashtag ”makeamericagreatagain”



(f) hashtag ”makeamericagreatagain”

Figure 2.9: Positive hashtags and the sentiment analysis results by SSSA, VADER, and TextBlob

over 90% of the tweets as positive. The estimation for ”votehillary” and ”makeamerica-greatagain” is positive by the three algorithms although TextBlob is outperforming the competitors.

Finally, we measure the sentiment for the entire hashtag dataset and analyse the accuracy of estimations. Figure 2.6 shows the overall results on the hashtag dataset. According to the results, SSSA's sentiment estimation is 64.2% accurate and outperforms VADER by 4.4%. SSSA outperforms TextBlob as well by a big margin of 24%.

2.5 Conclusion

This chapter proposed a scalable semantic-based sentiment analysis methodology. Also a promising automatic tweet tagging system developed to label positive, negative, and neutral tweets. We first calculate the semantic score of each word. These scores are used to identify whether a tweet is subjective or objective. The subjective tweets will be classified using our polarity classifier. We performed two sets of experiments, one on mobile phones dataset, and the next on 2016 US presidential election dataset.

In the first experiment, the results show that our proposed SSSA is scalable on big text corpuses and its computational time is reduced by one third in compare to conventional emoticon-based and hashtag-based approaches. Also, the polarity classification accuracy improved 2.2% and 10% in compare to EBSA and HBSA respectively. The accuracy improvement was 3.42% and 7.26% in average over the SSSA1 and SSSA2.

The second experiment tested SSSA on a very large dataset of 2016 US election dataset and compared with two sentiment analysis algorithms for social networks. SSSA outperforms VADER and TextBlob and had sentiment estimation accuracy of 64.2%. The main contribution of this chapter is proposing a scalable approach for sentiment analysis for big tweet corpuses, which is more precise, and development of an accurate automated tweet

tagging system. For the future work, we will test SSSA on larger datasets from different domains.

Chapter 3

TSIM: Topic-based Social Influence Measurement for Social Networks

3.1 Introduction

[1] [2] After sentiment analysis, the next important component of the thesis is influence measurement for social network users. Social influence is studied in social sciences in many forms, however, it is a new research direction in online social networks and haven't been studied along with sentiment analysis. Although, social influence has been an area of interest for researchers in sociology and more recently in computer science, still there is no agreement on its definition. A very early definition for influential people is "individuals who were likely to influence other persons in their immediate environment" (Katz 1957). Social influence has either been studied to identify influential users (opinion leaders or

[1] **Aso Hamzehei** et al., TSIM: Topic-based Social Influence Measurement for Social Networks. *In Proceeding of the Fourteenth Australasian Data Mining Conference (AusDM)*, Canberra, Australia. 6-8 December 2016. Vol. 170.

[2] **Aso Hamzehei**, et al., Topic-based Social Influence Measurement for Social Networks. *Australasian Journal of Information Systems (AJIS)*, November 2017, Vol. 21.

authorities), topical or topic-based influential users (Riquelme & González-Cantergiani 2016).

Social science studies, e.g.(Katz & Lazarsfeld 1955), have acknowledged the fact that the social influence of individuals is not identical. Katz (Katz 1957) introduced three main factors that are related to an individual's social influence such as: Who one is, what one knows, and whom one knows. The individual's social influence can be much more easily observed on social media while it is confirmed that the social influence factors are similar in social networks to those in the real society (Libai et al. 2010, Eccleston & Griseri 2008). For example, Eirinaki et al (Eirinaki et al. 2012) introduced two factors (popularity and activity) as factors related to social influence on Online Social Networks (OSN).

One of the main measures studied for influence is information diffusion which measures how important a user is in spreading information in the network. This is equivalent to identify central and hub nodes in the network (Jin & Wang 2013, Hajian & White 2011). Opinion leaders and discussion starters also have been studied as a measure of social influence (Jabeur et al. 2012). A user's position in the network (Jin & Wang 2013), content (Hu, Fang & Godavarthy 2013), and activities (Pal & Counts 2011) have been also studied as influence measures. Another aspect in the studies influence has been the scale of affected users by a post on social network or intensity of emotional and cognitive impact (McNeill & Briggs 2014).

According to (Probst et al. 2013), influential users have different influences on different topics and a very influential user is not necessarily influential on all topics. It is indicated in (Kardara et al. 2015) that topic-based influence measures are more effective and functional than the global ones. One of the differences of topic-related influence studies to network structure analysis is that it takes the posts' (e.g., tweets) content into account. When we consider user influence on topics, no longer the whole network needs to be analyzed, which improves the performance of measures.

However, there are drawbacks and shortcomings in the topic-based influence studies. In most of the existing works, they have aimed at making influential user detection more effective in retrieving the top N users only. Less effort is dedicated in discriminating influential from non-influential users. Also, approaches that uses supervised learning (e.g., SVM) suffer from their dependency on labeled data, which is extremely expensive to prepare for the immense data of social networks. Another considerable issue in these studies is their approach evaluation. This is a difficult task as influence is subjective. More importantly, prediction of user influence is remained as a problem to address in the state-of the-art.

Topic-based user influence measurement and identification are important challenges and the focuses of this chapter. This task is significantly important for different applications such as marketing, election campaigns, or recruiting employees for a company. In this work, we measure topic-based user influence on observed topics in which they have shown their interests by posting in social networks. Our approach, called TSIM (Topic-based Social Influence Measurement), incorporates network structure, user generated contents, users history of activities, and network users engagement in user's activity. Our approach represents users with their topic interests and their social influence on each observed topic.

In more detail, our contributions are:

- We propose a novel topic-based influence measurement approach to integrate the user-topic relationships, topic content information, and social connections between users into the same principled model.
- Instead of considering user-to-user influence and global user influence, the proposed model considers individuals influence and interests in a topic, which gives the capability of predicting one's influence on a new topic.

- Finally we have prepared a unique dataset from real-world social networks for testing and evaluating the proposed approach that contains all the social media related metadata.

The remainder of this chapter is organized as follows: We first discuss existing approaches for topic-based influence analysis in Section 5.2. We then present the background in Section 4.3. Next, we define the research problem, and then propose our approach and algorithms in Section 4.4. We describe our dataset and discuss the results in Section 5.5. Finally, we conclude the chapter in Section 4.6.

3.2 Related Work

One of the main approaches to study user influence in social networks has been through network structure as well as user's position and connectivity in the network. The traditional centrality measures such as closeness and betweenness are measured for users, to discover how well connected a user is to the rest of users in the network and whether a user is acting as a hub (Romero et al. 2011). The major adopted algorithms for network structure based influence measurement include PageRank (Haveliwala 2002) and HITS (Kleinberg 1999). Numerous works have applied PageRank algorithm variations on social network graph to rank user influence according to the network structure. An example of PageRank algorithm variations is the work by Kwak et al (Kwak et al. 2010), in which they ranked users by applying PageRank on follower/following graph in Twitter (along with number of followers and number of retweets). The network structure is relatively static compared to the activities of users in social networks. Some studies have included the social network related meta data (in case of Twitter, the meta data are retweets, mentions, and likes) (Hajian & White 2011).

Topic-based Influence. Following the influence studies (overall user influence) on social

Table 3.1: Key notations

Symbol	Description
t	a topic
$F_{i,j}$	influence of user u_i in t_j
$F_f(i,j)$	follower strength influence measure for user i in t_j
$F_a(i,j)$	activity influence measure for user u_i in t_j
$F_e(i,j)$	engagement influence measure for user u_i in t_j
$F_c(i,j)$	centrality influence of measure for user u_i in t_j
X_i	rank of user u_i calculated through PageRank
I^{op}	operation of identifying authorities
L	asymmetric adjacency matrix representing directed edges
$P_{i1,i2}$	probability of user u_{i1} engage with user u_{i2}
$D_{out}(i)$	number of user that points to user u_i in graph G
N_d	number of words in document d

networks, less studies have shed light on topic-based influence. More recently, topic-based influence studies have combined content of user posts with link-based metrics. Haveliwala (Haveliwala 2002) proposed a topic-sensitive extension of PageRank to rank query results in regards to the query topics. The idea of topic-sensitive PageRank was later used and adjusted for social networks such as Twitter for ranking topic-based user influence. Topical authorities were also studied in (Pal & Counts 2011) by Pal et al. They proposed a Gaussian-based ranking to rank users efficiently. They used probabilistic clustering to filter feature space outliers and showed that mentions and topical signals are more important features in ranking authorities. In (Kong & Feng 2011), Kong et al intended to identify and rank users that are posting quality tweets. They defined a topic-based high quality tweet with the author's topic-specific influence, topic related author's behavior. They applied their proposed metric on graph of following and retweets. Xiao et al (Xiao et al. 2014) aimed at detecting topic related influential users by looking at hashtag user communities where hashtags are pre-identified from news keywords. They proposed RetweetRank and MentionRank as content-based and authority-based influential users. Similarly, Hu, Fang & Godavarthy (2013) worked on detecting topical authorities with the assumption that retweeting propagates topical authority. Montangero and Furini

(Montangero & Furini 2015) also measured Twitter topic-based user influence where they identify topics by hashtags. Although hashtags can reveal the tweet's topic correctly, over 80% of tweets do not have hashtags. These results are neglecting the majority of tweets and can mislead a topic-based user influence, as 4 out of 5 of user's tweets are not considered for measuring her influence. In (Cataldi & Aufaure 2015), they estimated Twitter user influence for topics of conversations based on PageRank. For that purpose they built a topic information exchange graph to take the information diffusion and degree of information shared into account for user influence estimation. They manually considered seven topic categories and later assign each tweet to those categories through an n-gram model. However, their approach is unable to identify topics in the lower level of the main categories. For example, if someone is detected as influential in the sports category we do not know which sport the influence belongs to. In (Weng et al. 2010), they offered TwitterRank, a PageRank extension, that measures user influence by calculating topical similarities of users and their network connections. For topic identification, they used the unsupervised text categorization technique, LDA, by aggregating all tweets of a user into a document. Although this approach is presented as topic-sensitive, this approach cannot discriminate the user influence for the topics. In (Sung et al. 2013) they proposed another extension of PageRank, and unlike (Weng et al. 2010), it does not need predefined topics for topic-based user influence. In (Cano et al. 2014) a PageRank-based user influence rank algorithm introduced that the user links have weights based on their topics of interest similarities. In (Liu et al. 2014), their topic-based influence framework considers retweet frequency and link strength. The link strength is estimated by Poisson regression-based latent variable model on user's frequency of retweeting each other. In (Welch et al. 2011), they found out that topical relevance is better detectable through the retweet link rather than following links. They used two variations of PageRank algorithm to on retweet and following graphs for that purpose. In a recent work by Katsimpras et al. (2015), they proposed a supervised random walk algorithm for topic sensitive user ranking. As it is obvious from the algorithm name, it needs labeled data which is not very practical in

many cases specially with the volume of social networks.

It is worth mentioning that similar works exist that are only after the identification of global influencers instead of influencers for specific topics. An example of such works is (Barbieri et al. 2012) where they extended the Linear Threshold Model and Independent Cascade Model to be topic-aware, the topics are still obtained based on the network structure, while totally ignoring the valuable content information.

3.3 Background

Next, we give preliminaries for probabilistic topic modeling and pagerank.

3.3.1 Probabilistic Topic Modeling

Given a set of documents denoted by $D = [d_1, \ldots, d_q]$, Topic Modeling generates a set of t topics denoted by $\mathcal{T} = [t_1, \cdots, t_j]$. Each topic is related to a weighted representation over m words denoted by $t_j = [w_1 \cdots w_m]$, where w_j is the weight representing the contribution of word w_m to topic t_j. Probabilistic topic modeling , such as Latent Dirichlet Allocation (LDA), represents a low dimensional space of corpus by detecting a set of latent topics. The basic idea of Probabilistic Topic Modeling is having a Z hidden variable for each word's co-occurrence in the collection of documents. Z can range among j topics where each topic is a distribution over a fixed vocabulary. Given a corpus, a document may contain multiple topics and the words are assumed to be generated by those topics. A probabilistic topic model can be generated over a process as follows:

1. Obtain a distribution over topics to generate a document (in LDA this distribution is drawn from a Dirichlet distribution with a corpus-specific hyperparameter α)

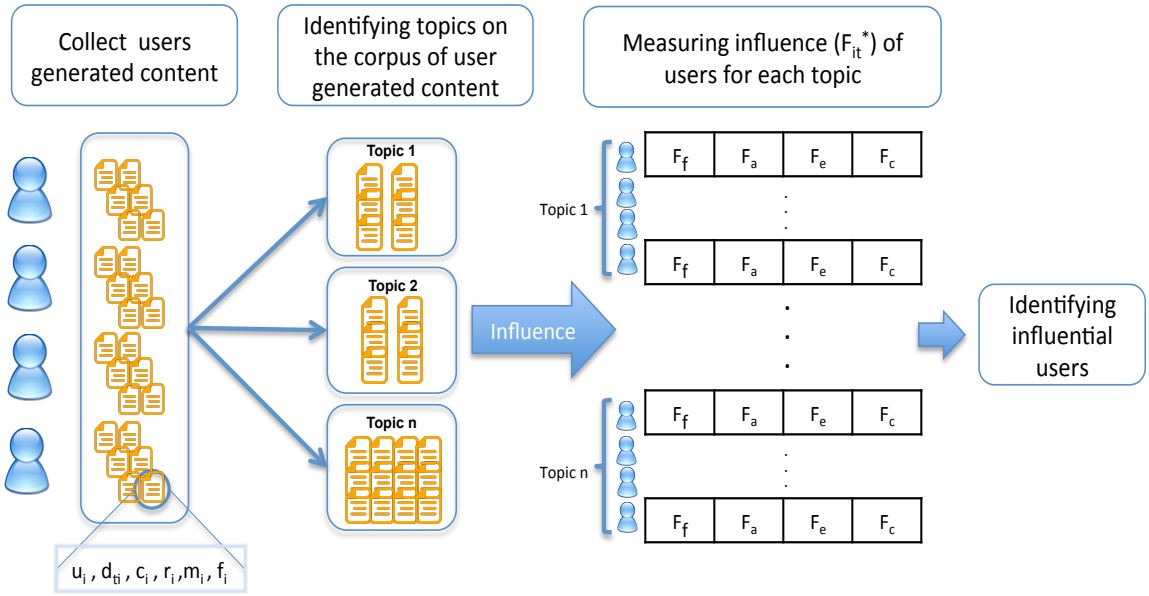


Figure 3.1: TSIM WorkFlow.

2. Then for each word to be generated;

   (a) Assign topics by drawing upon the document-specific distribution over topics

   (b) Finally, generate a word from distribution of topics over words in dictionary, which means words of each document come from a mixture of topics.

We aim to use probabilistic topic modeling to represent items as a set of topics and also detect social network users interest by applying topic modeling on their timelines.

3.3.2 PageRank

PageRank is a webpages ranking algorithm that calculate rank X_i for vertex v_i based on the rank of other vertices in the graph that point to vertex v_i. Assume $G(\mathbb{V}, E)$ denotes a directed graph, where the set $\mathbb{V}$ of vertices consists of i users and users relationships are the edges set E. Considering u_i as a user equal to vertex v_i in the graph G, the directed

edge $(i1, i2)$ exists if user u_{i1} is connected to user u_{i2}. The directed vertices of the graph G contained in the asymmetric adjacency matrix $L = (L_{i1,i2})$, where $L_{i1,i2} = 1$ if $u_{i1} \rightarrow u_{i2}$ and $L_{i1,i2} = 0$ otherwise. Out-degree $D_{out}(i)$ is the number of users that points to user u_i.

$$X_i = \sum_{(j,i) \in E} D_{out}(j)^{-1} X_i \tag{3.1}$$

The above equation is a recursive function that gives any vertex points to vertex v_i, a fraction of the rank inversley.

In PageRank, each out-going link from v_i is weighted by $1/o_i$, thus every node has the same total out-going weights. Each node has a total of one vote. PageRank uses an idea that a "good" node should connect to or be pointed to by other "good" nodes. However, instead of mutual reinforcement, it adopts a web surfing model based on a Markov process in determining the scores:

$$x = I^{op}(x) \tag{3.2}$$

where the I^{op} is an authority that is pointed to by many hubs and the I^{op} operation is defined to be

$$I^{op}(.) = L^T D_{out}^{-1} \equiv P^T. \tag{3.3}$$

This amounts to rescale the adjacency matrix L such that each row is sum-to-one. Thus, $P = (P_{i1,i2})$ is a stochastic matrix, since $\sum_v P_{i1,i2} = 1, P_{i1,i2} \geq 0.P_{i1,i2}$ represents the probability of a web surfer making a transition from webpage v_{i1} to v_{i2} . Starting from any webpage v_i, a surfer goes to any one of the hyperlinked webpages with equal probability $1/o_i$.

At any moment, millions of people are using the social networks. PageRank assumes the users follow the random surfing model in viewing and engaging with the rest of network.

They will reach the equilibrium (stationary) distribution under general conditions. If a node has a high probability in the equilibrium distribution, that means more nodes will point to that node. Therefore, the equilibrium distribution of users in social network is a measure of a node's importance, which is the authority score in PageRank. The equilibrium distribution x is determined by

$$P^T x = \lambda x \tag{3.4}$$

and x satisfies $\sum_k x(k) = 1$. One can obtain the solution iteratively. Note that $\lambda = 1$ if the Markov process has an equilibrium distribution x. PageRank models two types of random jumps on the Internet.

(i) Link-tracking jump: a user often follows other users in the network by simply clicking on them; this is modeled by $L^T D_{out}^{-1}$.

(ii) Link-interrupt jump: a user sometimes observe to engage with a user that they are not already connected to each other. PageRank models such link-interrupt jump with a simple uniform distribution $(1-\alpha)/n$. The full stochastic matrix of transition probability is

$$P^T = I^{op}(.) = \alpha L^T D_{out}^{-1} + (1-\alpha)(1/n)ee^T \tag{3.5}$$

where $\alpha = 0.8 \sim 0.9$. Here $e = (1, ..., 1)^T$; thus ee^T is a matrix of all 1's (Arasu et al. 2002).

3.4 Topic-based Social Influence Measurement

In this section, we introduce a topic-based social influence measure in social networks. We quantify individual's topic-based influence through users' generated content, her po-

sition in the network, her related activities, and other user's engagement in her generated contents. We first define the problem and then explain our approach.

3.4.1 Problem Definition

Assume $G(\mathbb{v}, E)$ denotes a social network graph, where users are the vertex set $\mathbb{V}$ and users relationships are the edges set of E. Assume that users publish a set of texts $D = [d_1, d_2, \ldots, d_q]$, and talk about different topics $\mathcal{T} = [t_1, t_2, \ldots, t_j]$. Each user text (post) d_q holds one or more topics and receives engagement from other users by replying, liking, or re-publishing it. The engagement of other users in a post can reveal the influence of that particular post among its audience.

We denote $A = \{a_{ii'}\}$ as the $n \times n$ matrix which shows the social ties among users in the social network G. For the pair of users i and i', $a_{ii'} \in [0, 1]$ shows the weight of the relationship between users u_i and $u_{i'}$, which we treat as the influence of user i on user i' (the higher the value of $a_{ii'}$, the higher the corresponding influence). The matrix A is not symmetric, as the influence of user i on user i' is not necessarily equal to influence of user i' on user i. We also assume that user post is visible to all users in G.

Quantifying the topic-based influence of each user based on social ties and other users' engagement in social networks, we can identify the influence of user i on topic j, represented as F_{ij}. Then we have matrix $F = [F_{ij}]_{i \times j}$ that represents influence of all the users in all identified topics.

3.4.2 Our Approach

To measure social influence on an observed topic in a social network, we propose TSIM, Topic-based Social Influence Measure, which measures topic-based individuals influence

in social networks. In a nutshell, our model contains two main phases:

- Identifying topics on social networks according to users generated contents, and
- Measuring individuals influence for the detected topics.

Our approach unlikePageRank based influence measures such as Cataldi & Aufaure (2015), Cano et al. (2014), and Katsimpras et al. (2015), identify topics, take users activities in each topic account, and quantify engagement from the network in each topic's user activity, as well as user's position in the network graph.

Figure 3.1, shows our approach's work flow. First, we collect user-generated content from the social media. For each user-generated content, we collect related information such as list of users that have re-published the content (R_i)and list of users that have engaged in that content (M_i) as well as metadata connected to the content. We identify the topics by applying probabilistic topic modeling, LDA, on all the user-generated text. Each topic contains a set of posts with all their related information and metadata, such as; content, replies, and republishing. For each tuple of $(user_i, topic_j)$, we measure the influence of user u_i on topic t_j as F^*_{ij} which comprises of four measures F_f, F_a, F_e, F_c. The details of influence measurement shown in Algorithm 1 and Section 4.4.5. The measure F^*_{ij} identifies user influence for the identified topics and users can be ranked according to their F^*_{ij} score for each topic.

3.4.3 Influence Measurement

We define social influence in a social network as importance of a user in the social network graph, user's activities, and involvement of others in the user's posts. Social influence can be analyzed through different modalities network structure and user's position in the network, scale of a user's post diffusion in the network, a user's activities and engagement

Algorithm 1 Influence Measurement

Input: List of topics, collection of user posts for each topic, interaction graphs, number of friends of each user.

Output: Matrix of user influence on each topic.

1: **for** topic in *topics* **do**
2: **for** user in *users* **do**
3: $F_f(i) \leftarrow \#friends$
4: $F_a(i,j) \leftarrow \sum_{d_i \in D_t} \delta(d_i)$
5: $F_e(i,j) \leftarrow \sum_{d_i \in D_t} (\delta(R_i) + \delta(M_i))$
6: $F_c(i,j) = PR(u_i, G(D_t))$
7: $F^*_{ij} \leftarrow$
8: aggregation of $F_f(i,j), F_a(i,j), F_e(i,j), F_c(i,j)$
9: Return matrix of user influence on topics

in the social network, and message content that a user broadcast in the network(Embar et al. 2015).

From the network structure, we identify influence related attributes, such as user friends and centrality of user in the social network. From the content of broadcasted text, we can identify one or more topics, thus, the influence of that user on different aspects. For instance, in Twitter, a post can contain user mentions, receive replies, and get retweeted by other users. All this information can reveal social influence of a user.

Let denote D_t as the set of collected texts related to topic t_j from the set of topics $\mathcal{T}$. Each text d_i contains a set of attributes as $(u_i, c_i, R_i, M_i, f_i)$ where u_i is the author of the text, c_i is the text, R_i is the list of users republished the text, M_i is the list of mentions for that text, and f_i is the number of followers of the text author.

We define the following dimensions for measuring social influence of a user on a topic as

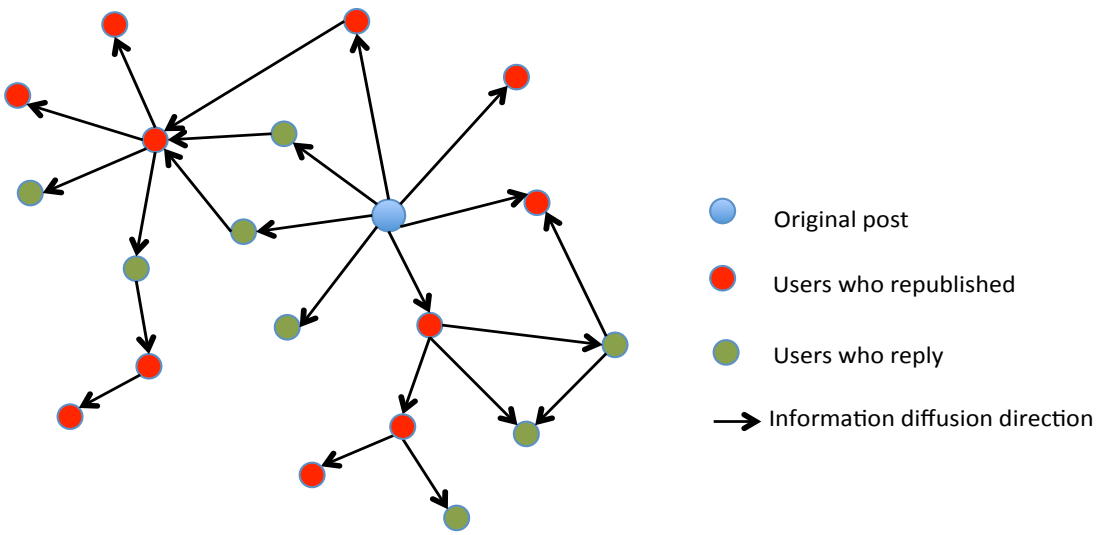


Figure 3.2: Retweet and Mention Graph.

following:

Follower scale: This measure depicts the number of friends a user has in the network. This value is constant across all topics for a user and is independent of topics. It shows the strength of social ties of a user. Although the number of social connections can be an indicative of influence, it does not carry information on any specific topic. The following influence measures are more topic-specific.

Topic Activity: This measure captures topic-related activities of a user. $F_a(i, j)$ denotes influence of user u_i in terms of activities related to topic t_j and we define it as:

$$F_a(i, j) = \sum_{d_i \in D_t} \delta(d_i) \tag{3.6}$$

where $\delta(d_i)$ is 1 if d_i belongs to texts set for topic t and is 0 otherwise. It intuitively measures the volume of topic t_j-related activities of user u_i.

Topic-based Attractiveness: This measure indicate how other users are attracted to

user$i$'s post. It takes other users' feedback on user u_i's activities into account. We define it as

$$F_e(i,j) = \sum_{d_i \in D_j} (\delta(R_i) + \delta(M_i)) \tag{3.7}$$

where $\delta(R_i)$ is the number of times d_i is republished by other users and $\delta(M_i)$ is the number of mentions or replies of d_i.

Network centrality: Centrality of a user is another indicator of her influence in a social network. PageRank was introduced first for ranking webpages for search engines, and can be used here to calculate topic-specific centrality of users in the social graph. Figure 3.2 shows an interaction graph of users on a post generated by user 1. To that end, we perform PageRank on the induced graph of interactions on a specific topic t_j. The interaction graph is a better representative of the topical relevance of two users rather than friendship graph (Welch et al. 2011). We denote it as:

$$F_c(i,j) = PR(u_i, G(D_t)) \tag{3.8}$$

where $G(D_t)$ is a graph corresponding to users over documents set D_t for topic t. $PR(u_i, G(D_t))$ indicates the PageRank score of user u_i in the graph $G(D_t)$. In this work, we reconstruct the interaction graph, (e.g., retweet and mention graphs from Twitter), to measure topic specific centrality of users by PageRank.

Aggregating Influence Scores: The four influence measures described above F_f, F_a, F_e, F_c will be aggregated to form a single influence score F^* for user u_i in topic t_j. For the first attempt, we averaged the measures which gives every measure the same share in the overall influence score. For the future works, we investigate other methods for aggregating the measures.

3.5 Results and Experiments

In this section, we discuss the the details of conducted experiments. It includes the data and the influence measurement performed by our proposed method.

3.5.1 Dataset

To validate our proposed method, we collected a unique dataset from Twitter using the Twitter Search API. We targeted the Machine Learning domain and identified core 500 users that have mentioned machine learning as a keyword in their profile description. To choose the users, we selected a set of machine learning users as seeds and crawled among their friends and friends of friends for other machine leaning-related users. For the prepared list of users, we gathered their timeline tweets which for most of the users covers their tweets for the last 5 years. For each tweet, we also, collected the related metadata such as the list of users who have replied to each tweet (mention list) and the list of users who have retweeted each tweet (retweet list). The final dataset contains 101363 tweets with their related metadata, mention lists, and retweet lists. The network that is built on retweet list contains 301870 nodes.

3.5.2 Evaluation

Our experiments contains a main task of user influence measurement on the identified topics from the tweet corpus.

We evaluate the measured user influence through expert opinion and user citations on the topics that the user has published in scientific conferences and journals. We collected publications through Google scholar for validation. The community of study is intentionally chosen as researchers then we are able to cross-validate our results through the users

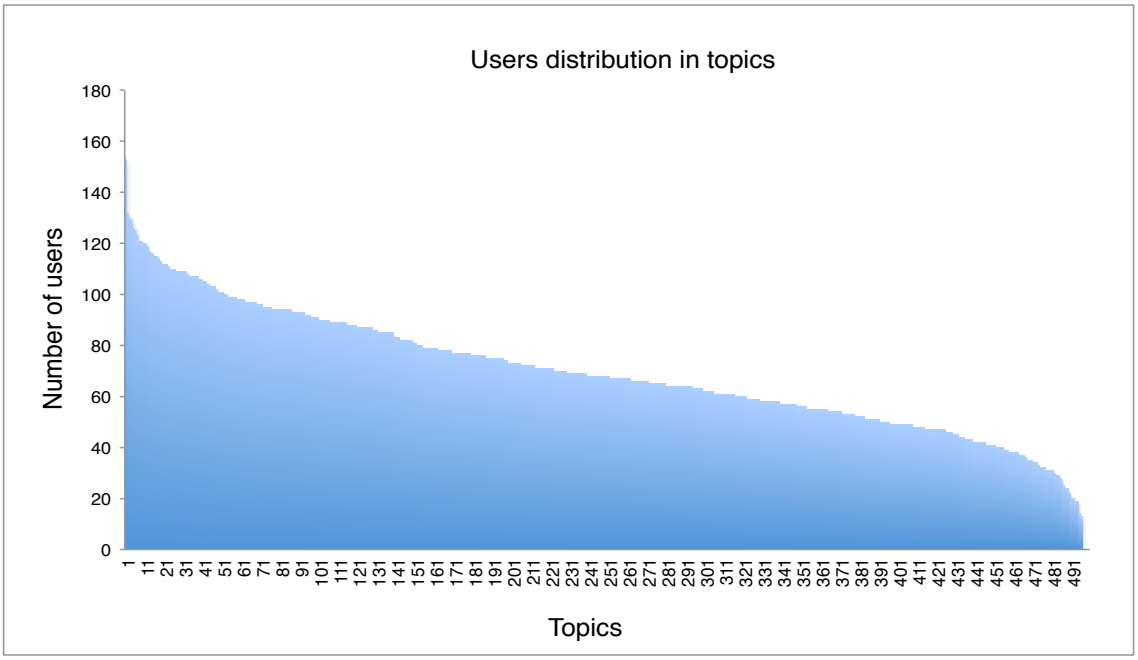


Figure 3.3: number of users belong to topics.

influence in research community measured by topics of their publications and citations.

3.5.3 Topic-based Influence Measurement

Next, we proceed with identifying topics from the collection of all tweets and then measuring influence. The number of topics generated by LDA can affect the quality of features that will be used in TSIM. We determined a number of topics through cross validation that could receive higher recall in user influence prediction. In our proposed approach, we perform probabilistic topic modeling for identifying the topics in the tweets dataset.

The user tweets gathered from their timelines, belong to the identified topics with a probability. We set the probability threshold to 0.1 to consider whether a tweet belongs to a topic. Each tweet is mapped to at least one topic. Now that for each topic we have a collection of related tweets with their mention and retweet lists, we can measure user influence for them. In Section 4.4.5, we defined influence based on 4 measures; follower

Table 3.2: A sample from the influence matrix *before* aggregating the 4 influence measures of user u_i in topic t_j.

User	*Topic1*	*Topic2*
vnfrombucharest	[0, 0, 0, 0]	[0, 0, 0, 0]
CharlieDataMine	[0.12, 0.01, 0.34, 0.16]	[0, 0, 0, 0]
sepehr125	[0, 0, 0, 0]	[0, 0, 0, 0]
sDataManagement	[0.12, 0.02, 0.35, 0.67]	[0.05, 0.04, 0.15, 0.67]
yisongyue	[0, 0, 0, 0]	[0, 0, 0, 0]

Table 3.3: continued. - A sample from the influence matrix *before* aggregating the 4 influence measures of user u_i in topic t_j.

User	*Topic3*	*Topic4*
vnfrombucharest	[0.1, 0.02, 0.51, 0.008]	[0, 0, 0, 0]
CharlieDataMine	[0.1, 0.014, 0.511, 0.163]	[0.28, 0.198, 0.38, 0.16]
sepehr125	[0, 0, 0, 0]	[0, 0, 0, 0]
sDataManagement	[0.6, 0.042, 0.512, 0.674]	[0, 0, 0, 0]
yisongyue	[0, 0, 0, 0]	[0.07, 0.05, 0.27, 0.08]

Table 3.4: A sample from the influence matrix *after* aggregating the 4 influence measures of user u_i in topic t_j.

User	*Topic1*	*Topic2*	*Topic3*	*Topic4*
vnfrombucharest	0	0	0.159	0
CharlieDataMine	0.157	0	0.197	0.254
sepehr125	0	0	0	0
sDataManagement	0.29	0.227	0.457	0
yisongyue	0	0	0	0.117

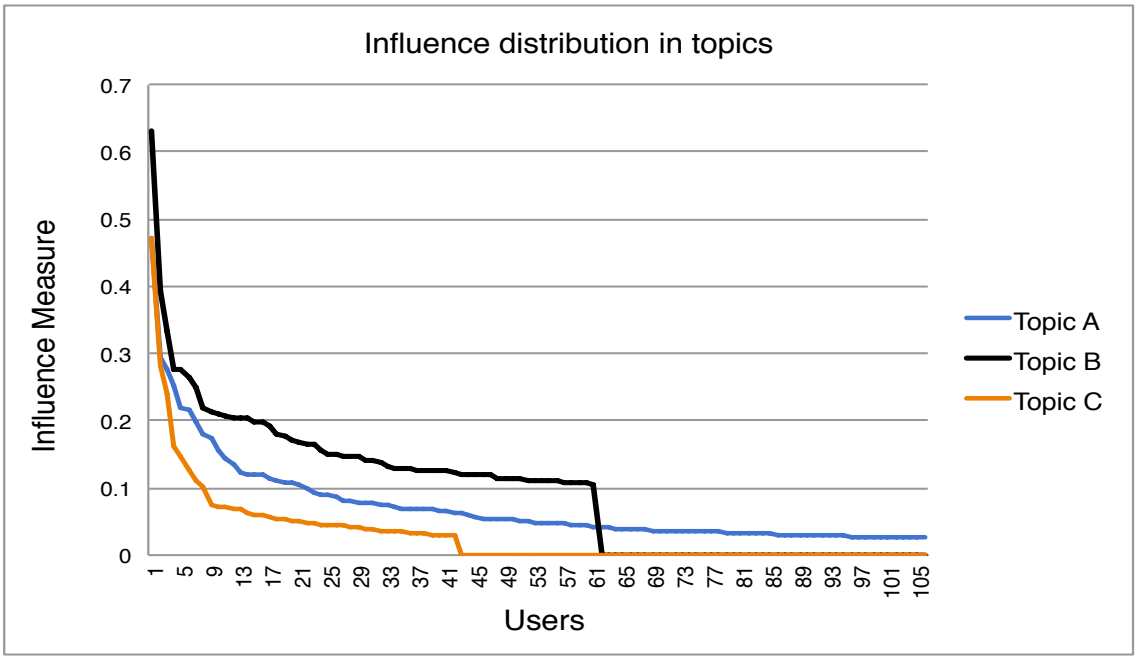


Figure 3.4: User's influence measure for three sample topics.

strength, activity, engagement, and network centrality. Follower strength will be taken from the number of users follow the user u_i on Twitter. Activity represents the number of tweets user u_i has in topic t_j. Engagement is the sum of number of mentions and retweets for all of user u_i's tweets in topic t_j. For measuring network centrality, we build the retweet graph for each topic separately from the corresponding retweet list and measure centrality of that user node through PageRank algorithm. Tables 3.2 and 3.3 show a small sample of the 4 calculated measures of topic-user influence. The zero scores mean that user u_i did not have any tweet for that corresponding topic. The non-zero scores are normalized to lie in the range of [0,1] and higher score means higher influence for that topic. The measured influence scores are aggregated and a sample of aggregated scores is shown in Table 3.4.

Users are not active on all topics. Some topics can be more popular with more active users and some topics can be less popular with less number of active users in social networks. We observed that the top 5% of most popular topic have 23% of active users in average.

Table 3.5: A sample of topics and their 5 top influencers measured by our proposed topic-based influence measurement system.

Deep Learning		*Text Mining*		*Programming Languages*	
Screen Name	*F*	*Screen Name*	*F*	*Screen Name*	*F*
kdnuggets	0.63	randal_olson	0.62	analyticbridge	0.70
analyticbridge	0.49	analyticbridge	0.55	randal_olson	0.49
deeplearning4j	0.33	jmgomez	0.53	DataScienceCtrl	0.41
KirkDBorne	0.31	IBMbigdata	0.51	BernardMarr	0.35
DataScienceCtrl	0.31	kdnuggets	0.49	eddelbuettel	0.34

Table 3.6: Table 3.5 continued - a sample of topics and their 5 top influencers measured by our proposed topic-based influence measurement system.

NLP-BigData		*Neural Networks*		*Social Networks*	
Screen Name	*F*	*Screen Name*	*F*	*Screen Name*	*F*
jmgomez	0.51	kdnuggets	0.93	analyticbridge	0.58
randal_olson	0.48	KirkDBorne	0.43	kdnuggets	0.56
analyticbridge	0.45	smolix	0.34	mjcavaretta	0.47
stanfordnlp	0.43	mapr	0.32	CharlieDataMine	0.44
bigdata	0.36	mjcavaretta	0.30	jure	0.35

Table 3.7: Table 3.6 continued - a sample of topics and their 5 top influencers measured by our proposed topic-based influence measurement system.

Recommender Systems		*BigData-Hadoop*		*Database*	
Screen Name	*F*	*Screen Name*	*F*	*Screen Name*	*F*
xamat	0.70	analyticbridge	0.69	analyticbridge	0.58
analyticbridge	0.54	mapr	0.60	randal_olson	0.55
kdnuggets	0.40	BernardMarr	0.58	IBMbigdata	0.36
jmgomez	0.36	odbmsorg	0.56	OracleAnalytics	0.32
KirkDBorne	0.32	infochimps	0.53	MarkLogic	0.30

Table 3.8: Table 3.7 continued - a sample of topics and their 5 top influencers measured by our proposed topic-based influence measurement system.

Artificial Intelligence		*R and Stats*		*Visualization*	
Screen Name	*F*	*Screen Name*	*F*	*Screen Name*	*F*
analyticbridge	0.57	kdnuggets	0.62	analyticbridge	0.90
ML_toparticles	0.55	analyticbridge	0.54	DataScienceCtrl	0.45
DataScienceCtrl	0.37	randal_olson	0.47	hmason	0.42
IBMbigdata	0.24	DataScienceCtrl	0.36	KirkDBorne	0.35
kdnuggets	0.21	paulblaser	0.36	paulblaser	0.31

The percentage of active users for least 5% popular topics was just 4%. Figure 3.3 shows the distribution of users in the topics sorted based on the number of active users in the topics. A user is active in a topic if has engaged in a topic by posting about it.

Moreover, the influence of users in each topic is not equally distributed. On average, topics have 13.8% of active users. It means 86.2% of users have zero influence on topics. In average 6% of users have influence of greater than 0.1 and just 1.4% of users have influence of greater than 0.2. This indicates that our proposed influence measure is capable of distinguishing very influential users from other for topics. The statistics provided are presented in Table 3.9.

Figure 3.4 shows influence measure for three selected topics. In this figure, topic A is selected from the top 5% popular topics. Topic B is selected from topics with number of active users close to the average and Topic C is from the least 5% popular topics. The figure shows how influence measure exponentially decreases from high influential users to non-influential users.

Tables 3.5, 3.6, 3.7, and 3.8 show the top 5 influencers for selected topics. The sample of topics presented in the tables contain machine learning topics such as Neural Networks, Deep Learning, Big Data, Social Networks, Text Mining, NLP, Database, Visualization, and more specific topics such as Hadoop. For the task of validation of the influence results, there is no standard method in the literature to validate the algorithm output. One of

Table 3.9: : Statistical information on users, topics, and influence measures

Average percentage of users active in topics	13.8%
Average of influence measures of active users in topics	0.125
Average of influence measures of all users in each topic	0.015
Percentage of users have influence of greater than 0.1 in topics	6%
Percentage of users have influence of greater than 0.2 in topics	1.4%

the reasons we have chosen the machine learning and data science community on Twitter as our community of study was the wide availability of experts in the domain that allows us to verify the identified influential users through our algorithm. We manually verify the top topic-based influential users through expert opinions, their Twitter, and Google scholar accounts. For example, for the topic NLP, Stanford NLP group appeared in the top 5 influential accounts on Twitter. For ”Recommender Systems” topic, Xavier Amatriain, who is known for his works on recommender systems, received a high influence score. Also, for the topic ”Neural Networks”, Alex Smola was in the top 5 influencers who have extensively published on neural network topic. In the topic ”Social Networks”, Jure Leskovec, who is well-known in the social networks community, was among the top influencers.

3.5.4 Implications and Applications

This section describes the real-world implication and applications of our model. Identifying topic-based influential users is similar to the problem of finding experts and authorities. Spotting the elite group of users for topics can improve available systems such as search engines. The query result for both contents and users can be returned and ranked using the score provided by our system.

One of the main applications of this work is in Marketing. Marketing campaigns can be implemented through the influential users in the related topic to have more productive and

cost effective campaign. Influential users act as hubs in the network and have a central position in the network in terms of information diffusion, also they attract and engage more users into their conversations.

Our model is able to detect the new and surprising topics. This capability gives the strength to our model that works in real-world and detect new topics and related influential users. As a result, there wouldn't be a need for manually defining the topics and consequently the recent and new topic would not be missed. TSIM, also can be applied to detect topics at what period get viral and who are influential in those topic in different period of time.

3.6 Conclusions

In this study, we have presented a Topic-based Social Influence Measurement, TSIM, to measure topic-based user influence in social networks. We have identified topics from user posts on social networks, and measured each user's influence on each topic. TSIM is then used to calculate user influence for the observed topics. Our main contributions include:

- the proposal of an effective method to measure topic-based influence for social network users
- opening a new discussion for user influence prediction in social networks that has not been explored in the literature.

Finally, we have tested TSIM using a unique dataset that we collected from Twitter, which we are making it available online.

In future work, we are interested to measure topic-based user influence over time, and study how influence changes over time. Prediction of user influence on unobserved topics is also currently under our investigation. We will also investigate other methods to combine influence measures.

Chapter 4

Collaborative Topic Regression for Predicting Topic-Based Social Influence

4.1 Introduction

[1] So that now we can measure social influence for a given topic, we are interested in predicting social influence in a newly appeared topic. In terms of influence measurement and prediction, there are topic-based and global user influence prediction. One of the key differences of *topic-based influence* studies to the global user influence studies is that it takes the user-generated content into account in addition to network structure. On the other hand, global user influence is based on the network structure and dismiss the topics in the measurement. Also, there are influence maximization and propagation, which aim to find a group of users that have the highest power of information propagation. However, influence maximization is *not* the focus of this chapter. To the best of our knowledge,

[1] **Aso Hamzehei**, et al., Collaborative Topic Regression for Predicting Topic-Based Social Influence. *Machine Learning (Special issue of ACML18 Journal Track)*. 2019. (Hamzehei et al. 2019)

there is no available study to predict topic-based user influence for a *new* topic, for which there is no observed data.

Topic modeling is a main phase in topic-based influence prediction. LDA as a standard topic model sees documents as a mixture of topics. and each topic is represented by set of words made by a distribution of words over the bag of words. These topics are drawn from the patterns of words co-occurrence within the document. Number of issues will raise if LDA is applied in short text corpus due to the low frequency of words and sparsity of co-occurrence of words. It is difficult for LDA to infer the document/word correlations in short text corpus and sense of words become more ambiguous (Cheng et al. 2014). As a result, specialized topic models are required to infer topics from short text corpus given those limitations.

We repeatedly use *observed topics* and *unobserved topics* terms in this chapter. Whether a topic is observed or unobserved is defined as follows. The observed topics are the discovered topics in the historical data that our model is trained on and unobserved topics are those topics that our model is predicting their user influence. For example, if we train our model with user influence on US 2012 election topic, that topic is observed topic, while if we want to predict user influence on an upcoming topic like US 2016 election, it is unobserved topic.

Topic-based user influence prediction is an important challenge and the main focus of this chapter. This task is significantly important for different applications such as marketing and election campaigns. In this work, we predict topic-based user influence on unobserved topics based on observed influence of users on the topics extracted from history of their posts on social network. We propose a Social Influence Collaborative Topic Regression (SICTR) approach to learn user, topic, and social factor latent spaces for estimating user social influence on an unobserved topic in social network. To infer the topics of social networks, SICTR, unlike CTR, is using a generative biterm topic model where words co-occurrence are explicitly modelled throughout the corpus.

Our approach represents users with their topic interests and their social influence on each observed topic. In more detail, our contributions are:

- We propose a novel topic-based influence prediction approach to integrate the user-topic relationships, topic content information, and social connections between users into the same principled model. Our model is inspired by the collaborative topic regression (CTR) model (Wang & Blei 2011), which has been successfully applied to article recommendation, and combines both user-topic and topic content information for influence prediction. SICTR improves the shortcoming of CTR in social network and influence prediction.
- Instead of considering user-to-user influence and global user influence, the proposed model considers individual users' influence and interests in a topic, which gives the capability of predicting one's influence in a new topic.
- The usefulness of considering content of topics and co-occurrence in the user-topic matrix are confirmed by our experiments on topic-based user influence prediction: topic-aware methods show better performance over other approaches that just consider a generic item and neglect its characteristics (content-based, e.g., LDA; and item-based, e.g., collaborative filtering methods).
- Finally, We have prepared and used a unique dataset from real-world social network to evaluate our proposed approach.

The remainder of this chapter is organized as follows. We first discuss existing approaches for topic-based influence analysis in Section 5.2. Next, we define the research problem that we focus on, and then propose our approach and algorithms in Section 4.4. We describe our dataset and discuss the results in Section 5.5. Finally, we conclude the chapter in Section 4.6.

4.2 Related Work

One of the main approaches to study user influence in social networks has been through network structure as well as user's position and connectivity in the network. The traditional centrality measures such as closeness and betweenness are measured for users, to discover how well connected a user is to the rest of users in the network and whether a user is acting as a hub (Romero et al. 2011). The major adopted algorithms for network structure based influence measurement include PageRank (Haveliwala 2002) and HITS (Kleinberg 1999). Numerous works have applied PageRank algorithm variations on social network graph to rank user influence according to the network structure. An example of PageRank algorithm variations is the work by Kwak et al. (2010), in which they ranked users by applying PageRank on follower/following graph in Twitter (along with number of followers and number of retweets). The network structure is relatively static compared to the activities of users in social networks. Some studies have included the social network related meta data (in case of Twitter, the meta data are retweets, mentions, and likes) (Hajian & White 2011).

Topic-based Influence. Following the influence studies (overall user influence) on social networks, less studies have shed light on topic-based influence. More recently, topic-based influence studies have combined content of user posts with link-based metrics. Haveliwala (2002) proposed a topic-sensitive extension of PageRank to rank query results in regards to the query topics. The idea of topic-sensitive PageRank was later used and adjusted for social networks such as Twitter for ranking topic-based user influence. Topical authorities were also studied in Pal & Counts (2011). They proposed a Gaussian-based ranking to rank users efficiently. They used probabilistic clustering to filter feature space outliers and showed that mentions and topical signals are more important features in ranking authorities. Kong & Feng (2011) intended to identify and rank users that are posting quality tweets. They defined a topic-based high quality tweet with the author's topic-specific influence, topic related author's behavior. They applied their proposed met-

ric on graph of following and retweets. Xiao et al. (2014) aimed at detecting topic related influential users by looking at hashtag user communities where hashtags are pre-identified from news keywords. They proposed RetweetRank and MentionRank as content-based and authority-based influential users. Similarly, Hu, Fang & Godavarthy (2013) worked on detecting topical authorities with the assumption that retweeting propagates topical authority. In Sung et al. (2013) they proposed another extension of PageRank, and unlike Weng et al. (2010), it does not need predefined topics for topic-based user influence. Also in Cano et al. (2014), they introduced a PageRank-based user influence rank algorithm that the user links have weights based on their topics of interest similarities. In Liu et al. (2014), their topic-based influence framework considers retweet frequency and link strength. The link strength is estimated by poisson regression-based latent variable model on user's frequency of retweeting each other. In Welch et al. (2011), they found out that topical relevance is better detectable through the retweet link rather than following links. They used two variations of PageRank algorithm to on retweet and following graphs for that purpose. Montangero & Furini (2015) also measured Twitter topic-based user influence where they identify topics by hashtags. Although hashtags can reveal the tweet's topic correctly, over 80% of tweets do not have hashtags. These results are neglecting the majority of tweets and can mislead a topic-based user influence, as 4 out of 5 of her tweets are not considered for measuring her influence. In Cataldi & Aufaure (2015) and Bingol et al. (2016), they estimated user influence for topics based on PageRank. For that purpose they build a topic information exchange graph to take the information diffusion and degree of information shared into account for user influence estimation. They manually considered seven topic categories and later assign each tweet to those categories through an n-gram model. However, their approach is unable to identify topics in the lower level of the main categories. For example, if someone is detected as influential in the sports category we do not know which sport the influence belongs to. In Weng et al. (2010), they offered TwitterRank, a PageRank extension, that measures user influence by calculating topical similarities of users and their network connections. For topic identifi-

cation, they used the unsupervised text categorization technique, LDA, by aggregating all tweets of a user into a document. Although this approach is presented as topic-sensitive, this approach cannot discriminate the user influence for the topics. In a recent work by Katsimpras et al. (2015), they proposed a supervised random walk algorithm for topic sensitive user ranking. As it is obvious from the algorithm name, it needs labeled data which is not very practical in many cases specially with the volume of social networks. It is worth mentioning that similar works exist that are only consider the identification of global influencers instead of influencers for specific topics (Barbieri et al. 2012). An example of such works is where they extended the Linear Threshold Model and Independent Cascade Model to be topic-aware, the topics are still obtained based on the network structure, while totally ignoring the valuable content information.

Collaborative Topic Regression. Users and items information is considered valuable to improve recommendation performance. Wang & Blei (2011) proposed Collaborative Topic Regression (CTR) that utilizes user and item information into topic modeling based Collaborative Filtering (CF) models to further improve recommendation performance. CTR is further extended in some studies. For example, Purushotham et al. (2012) and Chen et al. (2014) proposed two models (i.e., CTR-SMF and CTR-SMF2) to incorporate user social network into CTR to further improve item recommendation performance. Wang et al. (2013) proposed a model to incorporate item social relationship into CTR to further improve tag recommendation performance in social tagging systems. More recently, HU et al. (2018) incorporated three source of information, rating, item review, and social connections, to predict rating via latent factors and hidden topics. To address the sparsity problem, Zhao et al. (2018) developed a heterogeneous social-aware movie recommendation system where they included movie poster image as an input as well as textual data, ratings, and social connections. Furthermore, Chen et al. (2017) have introduced a neural network-based recommender system and tried to efficiency issues as neural network-based algorithms are computationally expensive.

In contrast to other works, in this chapter, we take one step further in topic-based influence measurement, by proposing an approach that measures topic-based user influence and predicts user's influence on an unobserved topic.

4.3 Background

Next, we give preliminaries for Probabilistic Topic Modeling and Collaborative Topic Regression.

4.3.1 Biterm Topic Model

Probabilistic topic modeling , such as Latent Dirichlet Allocation (LDA), represents a low dimensional space of corpus by detecting a set of latent topics. The basic idea of Probabilistic Topic Modeling is having a t hidden variable for each word's co-occurrence in the collection of documents. t can range among k topics where each topic is a distribution over a fixed vocabulary. Given a corpus, a document may contain multiple topics and the words are assumed to be generated by those topics.

Given a set of documents denoted by $N_D = \{d_q\}_{q=1}^{D}$, topic modeling generates a set of $\mathcal{T}_K$ topics denoted by $\mathcal{T}_K = \{t_k\}_{k=1}^{K}$. The topics of documents can be represented by a multinomial distribution $\theta = \{\theta_k\}_{k=1}^{K}$ with $\theta_k = P(k)$ and $\sum_{k=1}^{K} \theta_k = 1$ and if consider t as topic variable and $\theta_k = P(t = k)$. Each topic is related to a weighted representation over m words denoted by $t_k = \{w_m\}_{i=m}^{W}$, where w_m is the weight representing the probability that word m belong to topic t_k (e.g. $P(w|t)$). The word distribution over all topics, $P(w|t)$, can be represented by matrix ϕ where row ϕ_k corresponds to topic k and the matrix entries are $\phi_{k,w} = P(w|t = k)$ and $\sum_{w=1}^{W} \phi_{k,w} = 1$. In BTM, documents are seen as a collection of biterms N_B where $B = \{b_m\}_{m=1}^{N_B}$ and $b_m = (w_{m,1}, w_{m,2})$ (Cheng

et al. 2014, Li et al. 2016).

BTM can be generated over a process as follows:

1. Obtain θ, a distribution over topics to generate a document. This distribution is drawn from a Dirichlet distribution with a corpus-specific hyperparameter α.
2. For each topic to be generated
   (a) Draw ϕ_k from a Dirichlet distribution with hyperparameter β.
3. Then for each biterm to be generated;
   (a) Assign topics by drawing upon the distribution over topics
   (b) Finally, generate $w_{m,1}, w_{m,2}$ from distribution of topics over biterms in dictionary, which means biterms of each document come from a mixture of topics.

We aim to use probabilistic topic modeling to extract topics from the social networks and also as a part of SICTR process to create topic latent spaces.

4.3.2 Matrix Factorization (MF) and Collaborative Topic Regression (CTR)

CF analyzes the relationships between users and their associations with items by relying on historical user behavior (e.g., movies rating), without requirement of explicit user profiles. A basic approach of CF is neighborhood-based methods which analyze the relationship between items or users. For the item-based approach, the rating of a user on an item j is estimated based on her ratings on similar items, while user-based approach estimates item j's rating by looking at the rating behavior of other users with similar interests. Another CF approach is known as latent factor model, (e.g., matrix factorization

(Koren et al. 2009)), which estimates a rating by utilizing both user and item patterns. It factorizes user-item matrix into a user-specific matrix and a item-specific matrix. The objective function of a matrix factorization model can be formulated as follows,

$$\pounds = \sum_{i,j} (U_i^T V_j - r_{i,j})^2 + \lambda_U \|U\|_F + \lambda_V \|V\|_F \tag{4.1}$$

in which the first term is the difference between observed and prediction and the rest are regularization terms.

CTR (Wang & Blei 2011) is proposed on top of the matrix factorization and utilizes probabilistic topic modeling. It assumes that items (documents such as news and movie reviews) are generated by a topic model which is represented as topic latent vector v_j. In CTR, users are represented by topic interests. Similar to matrix factorization, CTR computes latent parameter of users u_i and items v_j. Latent variable ϵ_j captures the differences between topics for a user based on the users ratings on items. Equation 4.2 draw the ϵ_j as:

$$\epsilon_j \sim N(0, \lambda_v^{-1} I_k) \tag{4.2}$$

where N is probability density function of the Gaussian distribution with mean zero and variance equal to λ_v^{-1} and I is the indicator function that is equal to 1 if user i rated item j and equal to 0 otherwise.

CTR assumes that item latent vector v_j is close to topic proportion θ_j so that $v_j = (\theta_j + \epsilon_j)$ and draw it as:

$$v_j \sim N(\theta_j, \lambda_v^{-1} I_k) \tag{4.3}$$

The generative process of our SICTR model is inspired by CTR (Wang & Blei 2011).

4.4 Topic-based Social Influence Prediction

In this section, we introduce a topic-based social influence prediction model in social networks to predict individual's social influence on a new unobserved topic. We quantify individual's topic-based influence and model their topic-based influence through our proposed model. We first define the problem and then explain our approach.

4.4.1 Problem Definition

Assume $G(\mathbb{V}, E)$ denotes a social network graph, where users are the vertex set $\mathbb{V}$ and users relationships are the edges set of E. Assume that users publish a set of texts $D = [d_1, d_2, \ldots, d_q]$, and talk about different topics $\mathcal{T} = [t_1, t_2, \ldots, t_j]$. Each user text (post) d_q holds one or more topics and receives engagement from other users by replying, liking, or re-publishing it. The engagement of other users in a post can reveal the influence of that particular post among its audience.

Quantifying the topic-based influence of each user based on social ties and other users' engagement in social network, we can identify the influence of user u_i on topic t_j, represented as F_{ij}. Then we have matrix $F = [F_{ij}]_{i \times j}$ that represents influence of all the users in all identified topics.

Given a list of users, topics, and social influence of each user on those topics, we are interested in predicting an unknown value in $F_{ij} = [F_{1j}, F_{2j}, \ldots, F_{ij}]^{\mathcal{T}}$; the social influence of user u_i on a new topic t_j where $\mathcal{T}$ is the set of topics. Specifically, we aim to estimate user influence on an unobserved topic, based on observed influence of users on the topics from their history of activities and generated contents.

We expect that an influential user in topic t can be influential on a similar topic t'. Assuming that there are patterns among users with similar topic-based influenc weights, the

prediction can be performed with CF algorithms. However, the content-based methods use only the content information for recommendation. For example, if we want to predict influence for topic t_j , we can use the influence from the nearest neighbor in $\mathcal{T}$, the set of topics, based on the topics content similarity. We can also treat each topic as a label and use multi-label methods to train classifiers based on content information. Co-occurrence based methods use only the user-topic matrix F for prediction. For instance, if two topics t and t' co-occur for many users, and t is associated with u_i, we expect similar influence of t' to u_i . Both content-based methods and co-occurrence based approaches neglect useful information. As a result, they cannot achieve satisfactory performance in social influence prediction.

4.4.2 Our Approach

To measure and predict social influence on unobserved topics in a social network, we propose SICTR (stands for Social Influence-based Collaborative Topic Regression), which predicts topic-based influence. In a nutshell, our model performs a two-part representation of the users: (i) latent feature representation of the users according to the social network, and their connections to other users, and (ii) latent feature representation of the users based on the topics they are active in. Our method adopts Collaborative Topic Regression, CTR, a well-known method that combines CF with topic modeling, to learn a model that uses the latent topic space to explain both the observed ratings and the observed words. We propose SICTR to predict topic-based user influence in social networks. SICTR computes the latent parameter of users U and topics $\mathcal{T}$.

Social network-based representation. We want to derive a k-dimensional feature U from the social network g to represent users. Let $U \in R^k$ be latent user metric with column vector U_i for user-specific latent feature vector. We have user and factor feature vectors after placing zero-mean spherical Gaussian prior on them as follows:

Table 4.1: Key notations

Symbol	Description
$\mathcal{T}$	the set of topics extracted from Social Networks
$U, \lvert U \rvert_n$	the set and the number of users, respectively
V	the set of topics generated by SICTR
u	user latent vector
v	topic latent vector
t	a topic
k	number of latent dimensions
F_{it}	influence of user i in t
$F_f(i,t)$	follower strength influence measure for user i in t
$F_a(i,t)$	activity influence measure for user i in t
$F_e(i,t)$	engagement influence measure for user i in t
$F_c(i,t)$	centrality influence of measure for user i in t
α	offset term
β	topic bias parameter
λ_u	regularization parameter for u
λ_v	regularization parameter for v
θ_t	k-dimensional topic distribution for t
ϕ_k	word distribution for topic k
ω_{tn}	n^{th} word of document in topic t
z_{tn}	the topic for the nth word in topic t
N_d	number of words in document d
ϵ_t	topic t's latent offset
c_{it}	precision parameter for F_{it}
I_k	k-dimensional identity matrix

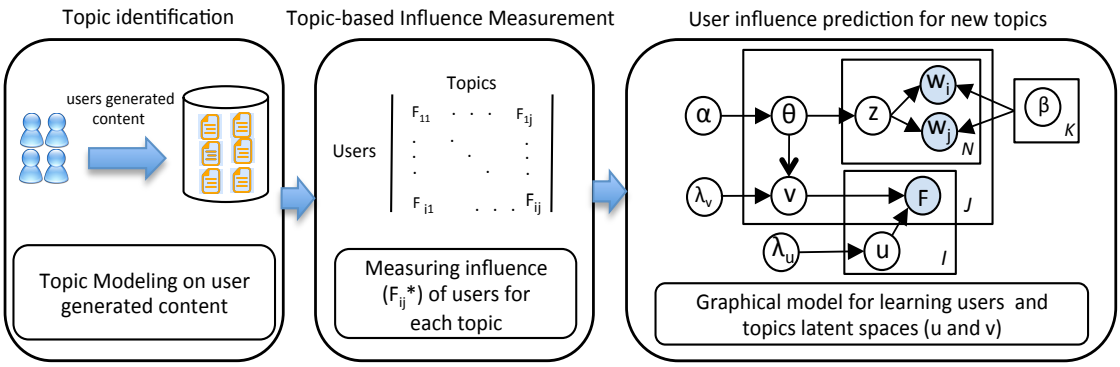


Figure 4.1: The SICTR framework.

$$p(U \mid \sigma_U^2) = \prod_{i=1}^{m} N(U_i \mid 0, \sigma_U^2 I) \tag{4.4}$$

where $N(x \mid \mu, \sigma^2)$ is the probability density function of the Gaussian distribution with mean μ and variance σ^2, and I_k is a k-dimensional identity matrix.

Topic-based representation. In our model, SICTR, items are the extracted topics from the corpus of user generated content in a social network, and users are represented by their topics of interests. SICTR predicts user influence on a topic according to similarity of items and other users' influence on similar topics. We identify the topics by applying probabilistic topic modeling on all the user-generated text. Each topic contains a set of social networks posts with all their related information and metadata, such as; content, replies, and republishing. For each tuple of $(user_i, topic_j)$, we measure the influence of user u_i on topic t_j in Section 4.4.5. That way, SICTR generates a topic latent space and a user latent space.

An important part of SICTR is generating topic latent vector $v_j = (\theta_j + \epsilon_j)$, where ϵ_j captures users interest in topic t_j and it assumes topic latent vector v_j is close to topic proportion θ_j. The expectation of F_{ij} is a linear function of θ_j,

$$E[F_{ij}|u_i, \theta_j, \epsilon_j] = u_i^T(\theta_j + \epsilon_j).$$

Moreover, from SICTR we know that item latent vector v_j is close to topic proportion θ_j and it generates item latent vector as $v_j = (\theta_j + \epsilon_j)$ where $\epsilon_j \sim N(0, \lambda_v^{-1} I_k)$ is equivalent to $v_j \sim N(\theta_j, \lambda_v^{-1} I_k)$ then

$$P(V \mid \sigma_v^2) \sim N(\theta_j, \lambda_v^{-1} I_k) \tag{4.5}$$

where V is the latent space for the set of topics $\mathcal{T}$ and $\lambda_v = \sigma_F^2 / \sigma_V^2$

Taking BTM into account, through Bayesian inference:

$$\begin{aligned} & p(U, V \mid F, \sigma_F^2, \sigma_U^2, \sigma_V^2) \\ & \propto p(F \mid U, V, \sigma_F^2) \\ & \times p(U \mid \sigma_U^2) p(V \mid \sigma_V^2) \end{aligned} \tag{4.6}$$

Figure 4.1 shows the SICTR framework. It contains three main steps: collecting data and identifying topics, measuring user influence for each topic, and predicting user influence for a new topic. It shows the graphical model of SICTR for learning users and topics latent spaces. Also, Algorithm 2 describes the generative process of SICTR.

4.4.3 SICTR Comparison with CTR

We compare SICTR with CTR to better understand how our model utilises topic modeling in collaborative topic regression and how it tackles short text corpus problem. Given a matrix of rating, CTR computes latent representation of items and users. CTR is using LDA to extract topics for explaining the users interest in items. It includes the latent topic space of the items into a model where observed ratings and learnt topics are combined in

the recommendation systems. The topic modeling that is used in CTR, for each word of a document, it draws a topic k from topic distribution θ_d and then draw a word w from topic k. This means that the topic k of word w is dependent to the other words of the same document. This is not ideal for short text documents and social networks contents and affect poorly on estimation of k, θ_d, and topic-word distribution. This phenomenon, directly influence the CTR results when the items of the user/item matrix contain a bunch of short text documents to draw their topic latent spaces. However, to alleviate this shortcoming, SICTR uses a Biterm Topic Modeling (BTM) approach to learn the topic latent spaces of the items of the matrix. BTM unlike LDA, generate topic from biterms in the corpus rather than words in documents. In this way the data sparsity issue is alleviated as θ is learnt from a global topic distribution.

4.4.4 Learning the Parameters of SICTR

We use an EM-style algorithm to learn the parameters (Purushotham et al. 2012). Maximization of the posterior is equivalent to maximizing the complete log-likelihood of U, V, $\theta_{1:t}$, F given U, V, and β.

$$\begin{aligned} L = \frac{-\lambda_u}{2}\sum_i u_i^T u_i - \frac{-\lambda_v}{2}\sum_t (v_t - \theta_t)^T (v_t - \theta_t) \\ + \sum_t \sum_n log(\sum_k \theta_{tk}\beta_{k,w_{tn}}) - \sum_{it} \frac{c_{it}}{2}(F_{it} - u_i^T v_t)^2 \end{aligned} \tag{4.7}$$

where $\lambda_u = \sigma_F^2/\sigma_U^2$, $\lambda_v = \sigma_F^2/\sigma_V^2$ and Dirichlet prior (α) is set to 1. We optimize this function using gradient ascent by iteratively optimizing the CF on social network variables u_i, v_t and topic proportions θ_t. For u_i, v_t, maximization follows similar to matrix factorization. Given a current estimate of θ_t, taking the gradient of L with respect to u_i, v_t and setting it to zero helps to find u_i, v_t in terms of U, V, C,F, λ_v, λ_u. Solving the corresponding equations will lead to the following update equations:

$$u_i \leftarrow (VC_iV^T + \lambda_u I_k)^{-1}(VC_iF_i) \tag{4.8}$$

$$v_t \leftarrow (UC_tU^T + \lambda_v I_k)^{-1}(UC_tF_t + \lambda_v \theta_t) \tag{4.9}$$

where C_i is diagonal matrix with c_{it}; $t = 1 \cdots J$ as its diagonal elements and $F_i = (F_{it})_{t=1}$ for user i. For each topic t, C_t and F_t are similarly defined. Note that c_{it} is precision parameter for influence matrix F_{it}. Equation 4.9 shows how topic proportions θ_t affect the topic latent vector v_t, where λ_v balances this effect. Given U and V, we can learn the topic proportions θ_t . We define $q(z_{tn} = k) = \phi_{tnk}$ and then we separate the topics that contain θt and apply Jensen's inequality:

$$\begin{aligned} L(\theta_t) &\geq -\frac{\lambda_v}{2}(v_t - \theta)^T(v_t - \theta_t) \\ &+ \sum_n \sum_k \phi_{tnk}(log\theta_{tk}\beta_{k,w_{tn}} - log\phi_{tnk}) \\ &= L(\theta_t, \phi_t) \end{aligned} \tag{4.10}$$

The optimal ϕ_{ink} satisfies $\phi_{tnk} \propto \theta_{tk}\beta_{k,wtn}$. Note that we cannot optimize θ_t analytically, so we use projection gradient approaches to optimize $\theta_{1:t}$ and other parameters U, V, and $\phi_{1:t}$. After we estimate U, V, and ϕ, we can optimize for β,

$$\beta_{kw} \propto \sum_t \sum_n \phi_{tnk} 1\left[w_{tn} = w\right] \tag{4.11}$$

4.4.5 Influence Measurement

We define social influence in a social network as importance of a user in the social network graph, user's activities, and involvement of others in the user's posts. Social influence

can be analyzed through different modalities network structure and user's position in the network, scale of a user post's diffusion in the network, a user's activities and engagement in the social network, and message content that a user broadcast in the network (Embar et al. 2015).

From the network structure, we identify influence related attributes, such as user's friends and centrality of user in the social network. From the content of broadcasted text, we can identify topics, thus, the influence of user on broadcasted aspects. For instance, in Twitter, a tweet can contain user mentions, receive replies, and get retweeted by other users. All this information can reveal social influence of a user.

As a reminder of the Influence measures, let denote $D = \{D_t\}_{t=1}^{\mathcal{T}}$ as the set of collected texts, where D_t is texts related to topic t where there are $\mathcal{T}$ topics. Each text d_i contains a set of attributes as $(u_i, c_i, R_i, M_i, f_i)$ where u_i is the author of the text, c_i is the text, R_i is the list of users republished the text, M_i is the list of mentions for that text, and f_i is the number of followers of the text author.

We define four influence measures namely; follower scale F_f (the number of friends a user has in the network); Topic Activity F_a (topic-related activities of a user); Topic-based Attractiveness F_e (how other users are attracted to useru_i's post); and Network centrality F_c (centrality of a user in the graph of active users in topic t_j). The four influence measures described above F_f, F_a, F_e, F_c will be aggregated to form a single influence score F^* for user u_i in topic t_j.

For more details on the Influence measure please refer to Hamzehei et al. (2016) and Hamzehei et al. (2017).

Algorithm 2 SICTR generative process

1: Measure users influence for each identified topic from the corpus
2: For each user u_i from the user's collection of posts of topic t_j, draw user latent vector of her topics of interest $u_i \sim N(0, \lambda_u^{-1} I_k)$
3: **for** topic (each topic extracted from the corpus and represented as a bag of words) in j **do**
4: Draw topic proportions $\theta_j \sim Dirichlet(\alpha)$
5: Draw topic latent offset $\epsilon_j \sim N(0, \lambda_v^{-1} I_k)$, set the topic latent vector as $v_j = \epsilon_j + \theta_j(d)$
6: **for** word in w_{jn} **do**
7: Draw topic assignment $t_{jn} \sim Multinomial(\theta)$
8: Draw word $w_{jn} \sim Multinomial(\beta_{t_{jn}})$
9: For each user-topic pair (u_i, t_j), draw, F_{ij}, the rate of social influence of user u_i for topic t_j

4.4.6 User Influence Prediction on a New Topic

Having the parameters learned, our SICTR model can be used for in-matrix and out-matrix prediction. In-matrix prediction refers to predicting a user influence on a topic where influence rates of some other users are available. Out-matrix prediction refers to predicting user influence on a topic that no influence data is available (totally new topic).

For a given observed set of documents, D, prediction of influence of a user on a topic can be predicted as the expected value of:

$$E\left[F_{ij}|D\right] \approx E\left[u_i|D\right]^T \left(E\left[\theta_j|D\right] + E\left[\epsilon_j\right]\right) \tag{4.12}$$

In-matrix can be predicted by using point estimate of u_i, θ_j and ϵ_j to approximate their expectations (recall $v_j = \theta_j + \epsilon_j$),

$$F_{ij}^{*} \approx (u_i^{*})^T(\theta_j^{*} + \epsilon_j^{*}) = (u_i^{*})^T(v_j^{*}) \tag{4.13}$$

For out-matrix prediction, as a topic is new and no historical influence is available from users then $E[\epsilon_j] = 0$. As a result, we can predict a user's influence on a new unobserved topic as:

$$F_{ij}^{*} \approx (u_i^{*})^T\theta_j^{*} \tag{4.14}$$

Putting everything together, we obtain our proposed SICTR method (Algorithm 2).

4.5 Results and Experiments

In this section, we discuss the the details of conducted experiments. It includes the data, the influence measurement, and topic-based user influence prediction by SICTR.

4.5.1 Dataset

In order to conduct our experiments, we collected a dataset from Twitter and Google Scholar. The details of each dataset is is as following:

• **Twitter dataset:** To validate our proposed method, we collected a unique dataset from Twitter using the Twitter Search API. We targeted Machine Learning community and identified 500 users that have mentioned machine learning or data science as a keyword in their profile description. To choose the users, we selected a set of well-known machine learning users as seeds and crawled among their friends and friends of friends for other machine learning-related users. For the prepared list of users, we gathered their timeline

tweets which for majority of the users it covers their last five years tweets. For each tweet, we also, collected the related meta-data such as the list of users who have replied to each tweet (mention list) and the list of users who have retweeted each tweet (retweet list). The final dataset contains 101363 tweets with their related metadata, mention lists, and retweet lists. The network that is built on retweet list contains 301870 users. The dataset was collected on 4 May 2016.

• **Google Scholar dataset:** If we consider number of citations of a research paper as a measure of influence of that paper and author in that research topic, then it is safe to claim that a user with a high number of citations in a topic is very influential in that topic. So, if the same author is an active Twitter user and tweets on the same topic, we expect that her influence will be visible in Twitter as well.

For validating our influence measurement results, we prepared this Google Scholar dataset on the same users that are in the Twitter dataset described above. Specifically, we spotted 254 of our Twitter users in Google Scholar and extracted their top 20 papers with the highest number of citations. Each extracted paper contains author names, title, abstract, publication date, number of citations, Google Scholar link, and PDF link. The dataset was collected on 14 November 2016.

4.5.2 Evaluation

Our experiments contain two main evaluation tasks. We first measure topic-based user influence on the corpus. Second, we predict user's influence for unobserved topics.

User influence measurement on the identified topics from the tweet corpus: User influence is an abstract concept and evaluating user influence is very subjective, and thus challenging. That means, for the task of validation of the influence results, there is no ground truth. Moreover, there is no standard method in the literature to validate the al-

gorithm output. (Montangero & Furini 2015) compared their method to the number of followers of the users of interest. (Cataldi & Aufaure 2015) used opinions from experts to evaluate their proposed influence measure. Lastly, (Katsimpras et al. 2015) used PageRank scores for as their evaluation measure.

We evaluate the measured user influence results through expert opinion and a secondary dataset prepared for cross-validation. This dataset is prepared from Google Scholar. For the same Twitter users included in our study, we collected information about their research publications, as described in the previous subsection. We identify a user's influence for each topic based on her citations. For example, for the topic of 'Social Network Analysis', we measure user influence separately from Twitter and scientific publications (Google Scholar). Then, we cross validate the results from both datasets.

We have chosen the machine learning and data science research community on Twitter as our community of study for two main reasons. First, there is wide availability of experts in the domain, which allows us to verify the identified influential users through our algorithm. Second, as these users have research publications, we can measure their influence on the topics they have published by considering their number of citations per topic. This is based on the assumption that the measured user influence is correlated between the two datasets.

Prediction of user influence on unobserved topics through our proposed SICTR: We report recall, precision, accuracy and F-score. Due to the uncertainty of the meaning of zero influence, recall will be the main performance measure while we still report precision, accuracy, and F-score. Zero influence mean either the user has not been influential in a topic or her activities are not represented in our dataset. For experimental analysis, we split the dataset into 80% train and 20% test datasets. If we present each user by topics that are estimated as influential from SICTR, recall corresponds to the number of topics that the user u_i is predicted as influential over the total number of topics that the user u_i

recognized as influential:

$$recall = \frac{\textit{\# of topics the user is predicted as influential}}{\textit{\# of topics the user is influential in}} \tag{4.15}$$

4.5.3 Topic-based Influence Measurement

Next, we proceed with identifying topics from the collection of all tweets and then measuring user influence for each topic. Number of topics is one of the inputs for topic modeling. As our aim in this chapter is to predict topic-based user influence, we choose a number of topic for topic modeling that gives the best performance to SICTR.

In our proposed approach, we perform probabilistic topic modeling in two different rounds. The first round is for identifying the topics in the tweets dataset and the second round is to create topics latent space to take topics similarity into account for influence prediction.

The user tweets gathered from their timelines, belong to the identified topics with a probability. We set the probability threshold to 0.1 to consider whether a tweet belongs to a topic. Each tweet is mapped to at least one topic. Now that for each topic we have a collection of related tweets with their mention and retweet lists, we can measure user influence for them. In Section 4.4.5, we defined influence based on 4 measures; follower strength, activity, engagement, and network centrality. Follower strength will be taken from the number of users follow the user u_i on Twitter. Activity represents the number of tweets user u_i has in topic t_j. Engagement is the sum of number of mentions and retweets for all of user u_i's tweets in topic t_j. For measuring network centrality, we build the retweet graph for each topic separately from the corresponding retweet list and measure centrality of that user node through PageRank algorithm.

4.5.4 SICTR Prediction Results and Comparisons

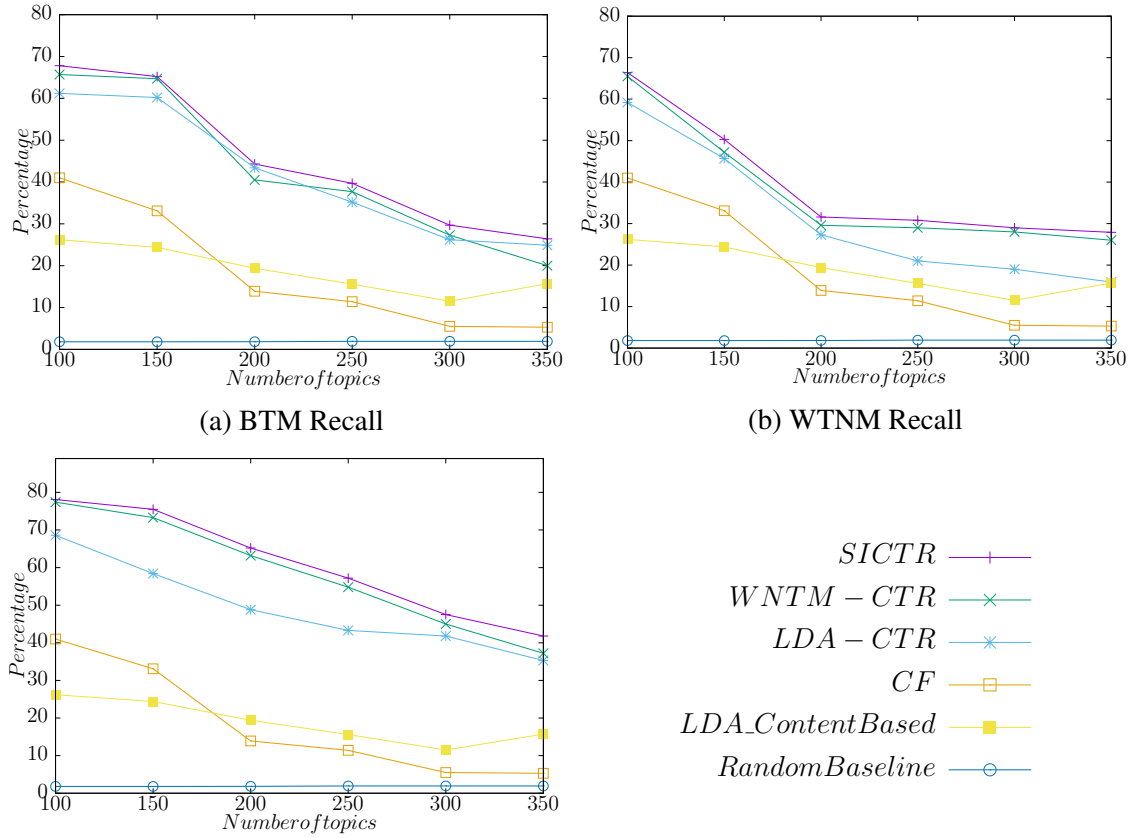
Here, we present the SICTR, in-matrix prediction results and compare them with WNTM-based approach, LDA-based approach, CF, a content-based model, and a random baseline. In WNTM-based approach (Zuo et al. 2016), CTR is using WNTM for generating topic latent vectors. In LDA-based approach, CTR is using LDA for topic vector generation. For the CF model, per-user and per-topic latent vectors are fixed with the influence values in the matrix. While, content-based model behaves as a content-only model. For content-based model, the per-user latent vector is generated by LDA and is fixed to influence entries in the matrix and per-topic latent vector is only based on the words of the topics. The random baseline is a random model, where a user randomly is predicted as influential for a topic. The random model predicts a user as influential using the probability of appearance of influential users in the user-topic influence matrix. As we mentioned before, in-matrix prediction considers influence prediction for the topics that already exist in the data and influence of at least one user is available for them.

We split the data into train and test datasets. For measuring performance, we use 5-fold-cross-validation. We make sure every topic appears in all the folds so that each topic appears both in the train and test data.

SICTR experimental settings.

SICTR contains parameters that need to be tuned to evaluate the proposed approach with the benchmarks. In the models, the a and b are tuning parameters for the parameters c_{ij} and d_{ij} The parameter λ_v is the precision parameter to balance the diverging of topic latent vector from the topic proportion θ_j. Similar to CTR, we increase λ_v to penalty v_j diverging from θ_j. If we set $v_j = \theta_j$ for per-topic latent vector, SICTR behaves as a content-based method and the topic vector θ_j will be just based on the words in the topic j. To find the best value of the parameters, we use grid search. For comparisons of the three models, the parameters are set to $k = \frac{number\ of\ topics}{4}$, $\lambda_v = 0.01$, $\lambda_u = 0.01$, a= 1, b

= 0.01. Moreover, we analyze the SICTR performance for different values of λ_v, number of topics, and influence threshold.



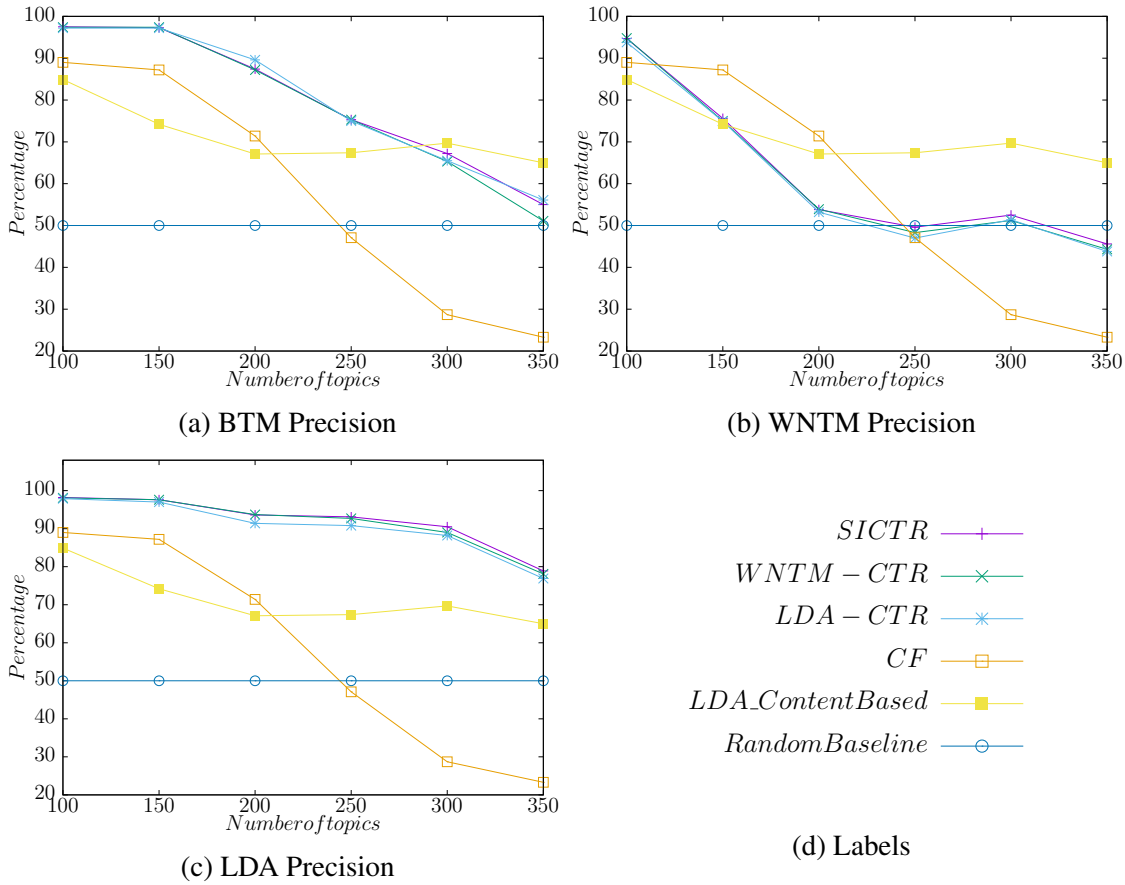
(a) BTM Recall

(b) WTNM Recall

(c) LDA Recall

Figure 4.2: Recall comparisons between SICTR, WNTM-base model, LDA-base model, CF, Content-based model, and Random baseline. The x-axis shows experiments for different number of topics. The influence threshold is set to 0.3

Effect of sparsity on influence prediction. The matrix of topic-based user influence is a sparse matrix. The matrix sparsity can be increased with higher influence threshold. By increasing the influence threshold, fewer users will be identified as influential and as a result, the influence matrix becomes more sparse. Also, by increasing the number of topics, the topic-based influence matrix becomes more sparse. Figure 4.2 shows the effect of number of topics on recall. When the matrix becomes more sparse, recall decreases. Accordingly, it is shown in Figures 4.3 and 4.4 that precision and accuracy for SICTR also decrease when the matrix becomes more sparse.



(a) BTM Precision

(b) WNTM Precision

(c) LDA Precision

(d) Labels

Figure 4.3: Precision comparisons between SICTR, CF, LDA, and Random baseline. The x-axis shows experiments for different number of topics. The influence threshold is set to 0.3

Effect of influence threshold on SICTR performance. The topic-based user influence is in the range of zero to one. Zero indicates no influence over that topic and one means the user has the highest influence for all four influence measures in that particular topic. Higher influence threshold results in less number of users being identified as influential, and, consequently, makes the influence matrix more sparse. In the experiments, when influence threshold is increased, our model predicts influential users more precisely. However, due to sparsity, the recall measure decreases.

Effect of Lambda. For this experiment, we fix the number of topics to 300, the influence threshold to 0.3, the number of factors to 50, and the remaining parameters as explained

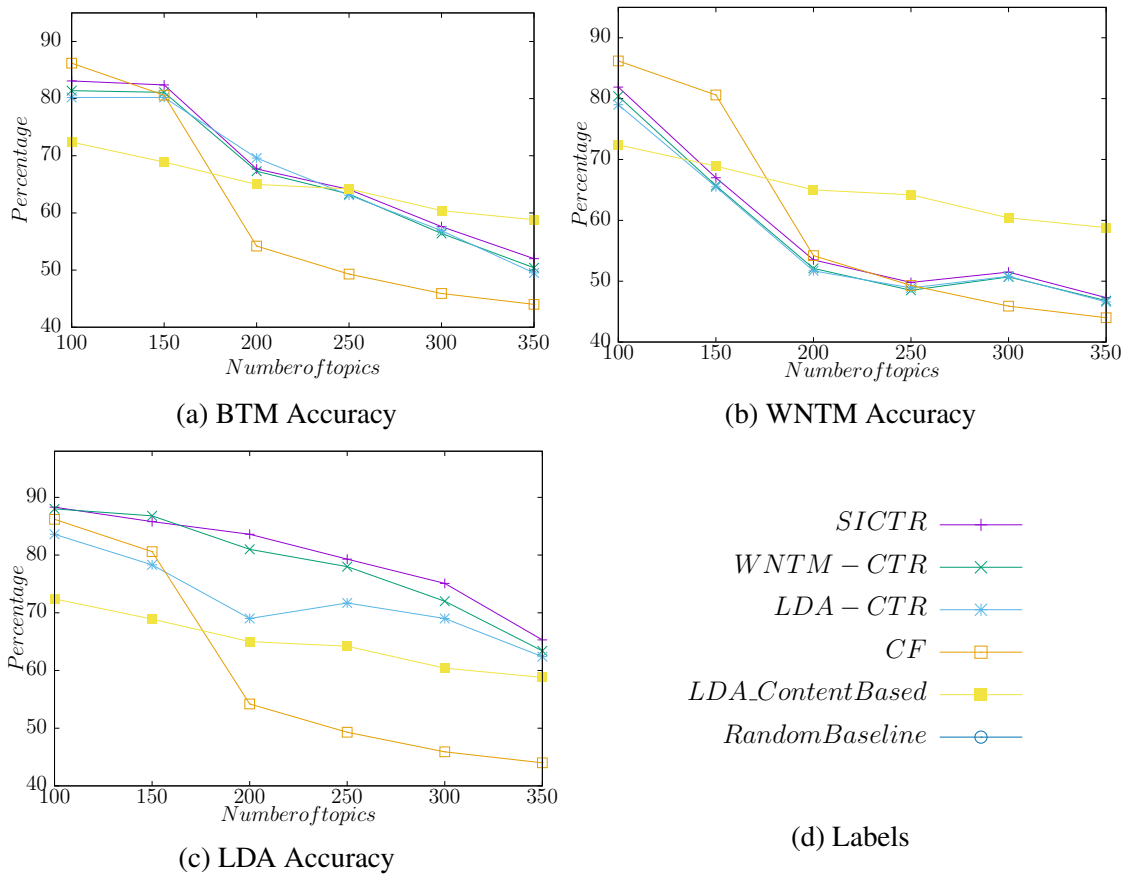
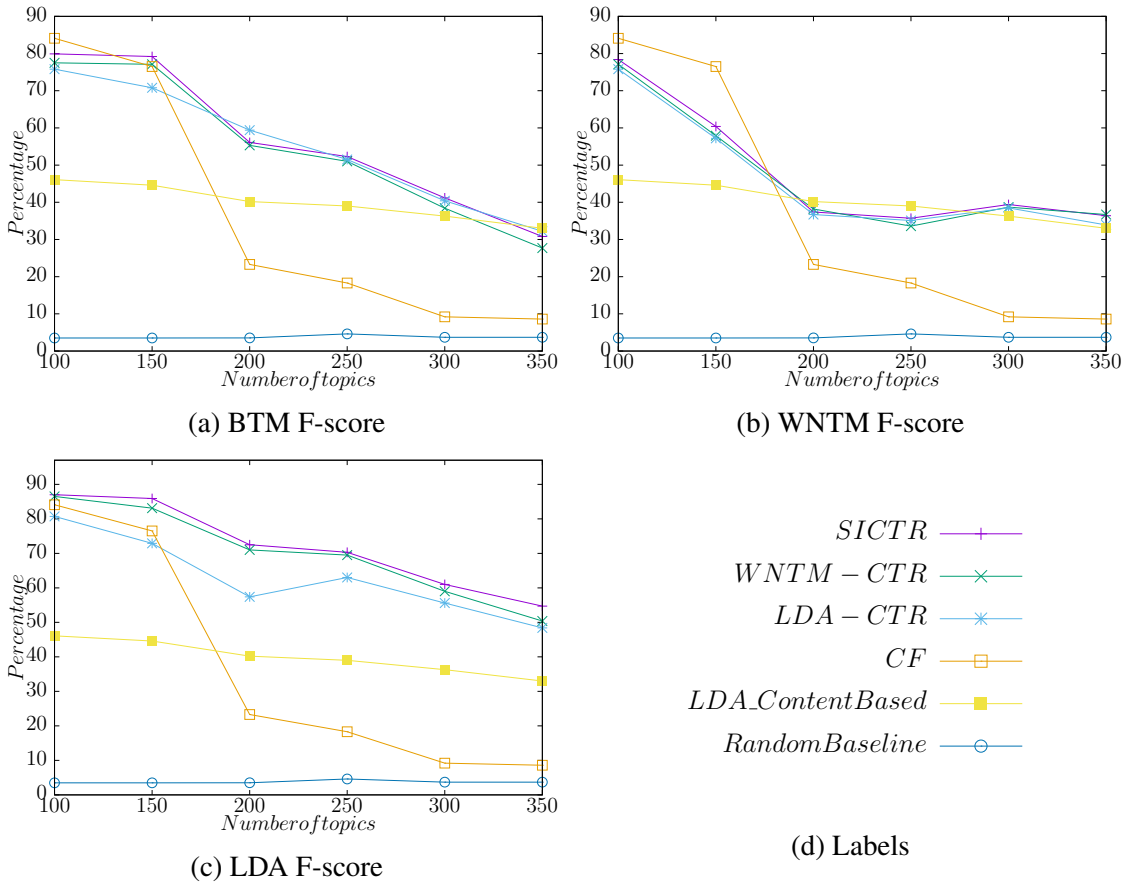


Figure 4.4: Accuracy comparisons between SICTR, CF, LDA, and Random baseline. The x-axis shows experiments for different number of topics. The influence threshold is set to 0.3

in experiment settings. We vary $\lambda_v \in [0.01, 0.1, 1, 10, 100, 1000]$ to evaluate the SICTR performance for different λ_v values. As λ_v is the precision parameter to balance the divergence of the topic latent vector from topic proportion θ_j and is specific for SICTR, its change does not affect CF and content-based model performance. The results show that SICTR outperforms the CF by 29% in the best case scenario with $\lambda_v = 100$ and beats the CF in all experiments. This suggests that taking topic features into account improves the influence prediction performance. Also, it indicates that a user's influence on a new unobserved topic can be estimated by similarity of the new topic with the previously observed topic-based user influences.



(a) BTM F-score

(b) WNTM F-score

(c) LDA F-score

(d) Labels

Figure 4.5: F-score comparisons between SICTR, CF, LDA, and Random baseline. The x-axis shows experiments for different number of topics. The influence threshold is set to 0.3

Effect of number of topics on SICTR, CF, and LDA. We evaluate SICTR overall performance against WNTM-baed model, LDA-based model, CF, Content-based model, and random baseline. Figures 4.2, 4.3, 4.4, and 4.5 show the performance of the SICTR and benchmarks in terms of recall, precision, accuracy and F-score. Each of the figures contain three subfigures. The difference between the three subfigures is in the topic modeling approach that is used for identifying the corpus' topics. For example, in Figure 4.2, the subfigure 4.2a shows the results for recall measure where the corpus topics (items) in SICTR, WNTM-CTR, and LDA-CTR are identified through BTM.

In Figure 4.2, it is clear that SICTR is outperforming the benchmarks on recall. The

highest recall rate is achieved on 100 topics. The recall rate decreases as the number of topics grows due to increase in sparsity of the influence matrix. The second best performer in Recall measure is WNTM-CTR. Both SICTR (BTM-based) and WNTM are using topic modeling approaches that are developed for social networks. So, it is not out of expectations that they perform better than others. CF and Content-based model are the worst performers after the random baseline.

It is worth mentioning that the prediction results is highly dependent on the quality of topics generated in the first round of topic modeling. in Figure 4.2, each subfigure is the experiment result on a different topic modeling approach used in the first round to extract topics from the corpus. In subfigure 4.2b, the poor topics generated by WNTM impacted the prediction results. As SICTR and CTR related models in the benchmark are highly dependent on the first round of topic modeling, the content-based and CF outperformed the CTR based models in the experiment performed on WNTM topics.

Figure 4.3 shows precision metric results of SICTR and the benchmarks on the three sets of topics generated from BTM, WNTM, and LDA. The highest achieved precision is observed for results on 100 topics. The precision figures explain the quality of topic modeling as well. It can be seen that topics generated by BTM and LDA are leading to better precision by the models. The highlight of this subsection is that SICTR is slightly outperforming the benchmarks in terms of precision.

The results of accuracy metric can be found in Figure 4.4. The accuracy of predicting influential users decreases when the number of topics increases, similar to recall and precision. Regardless of the topic modeling approach used in the pre-modeling stage, the SICTR has been more accurate against the benchmarks. WNTM-based and LDA-based models are second and third best performers. For instance, in Figure 4.4c when number of topics is set to 150, accuracy measure is 88% for SICTR while it is 86% for WNTM-based model and 72% for LDA-based model respectively.

Furthermore, we report the F-score measure for this experiment. Figure 4.5 presents F-score and it shows the outperformance of SICTR in this evaluation as well. However, Figure 4.5b shows otherwise when number of topics is less than 170. Similarly, the same trend was observed in Figure 4.2b where CF was performing better on low number of topics when the topic modelling from the first round has been poor. According to Figures 4.2b and 4.5b, CF's precision, accuracy, and F-score is higher than competitors when pre-modelling topics are in poor quality. Under the same circumstances, CF's outcome deteriorates when the number of topics increases.

4.5.5 Implications and Applications

This section describes the real-world implications and applications of our model. Identifying topic-based influential users is similar to the problem of finding experts and authorities. Spotting the elite group of users for topics can inform and improve available systems, such as search engines. The query result for both contents and users can be returned and ranked using the score provided by our system.

One of the main applications of this work is in Marketing. Marketing campaigns benefit from targeting the influential users in the related topic, as this often leads to a more productive and cost effective campaign. Influential users act as hubs in the network and have a central position in the network in terms of information diffusion, and also attract and engage more users into their conversations. The influence prediction power of SICTR brings the ability to form a stronger campaign by predicting users influence for the new product.

Unlike the work in the literature, our model is able to detect *new* and *surprising* topics. This makes our work capable of working in the real-world and detecting new topics and related influential users. As a result, it eliminates the need for manually defining the topics, and at the same time it does *not* miss recent and new topics (which could have

been missed via a manual process). SICTR can also be applied to a time-evolving setting, where it can detect when topics get viral and which users are influential in those topics at different periods of time.

4.6 Conclusions

In this study, we presented a collaborative topic regression model, SICTR, to predict topic-based user influence in social networks. We identified topics from user posts on social networks, measured each user's influence on each topic, based on the influence definition we proposed, and used SICTR to predict user influence for unobserved topics. The main contributions our work include:

- proposing a topic-based user influence prediction approach in social network that incorporates both network structure and user-generated content for topic-based influence measurement and prediction,
- opening a new discussion for user influence prediction in social networks that has not been explored in the literature.

We tested our topic-based influence measurement system and prediction model using a unique dataset that we collected from Twitter and Google Scholar. Our experiments show that SICTR is effective for predicting influential users for a new topic. For instance, SICTR can be valuable for marketing campaigns by selecting the most influential people for for a new product's marketing campaigns. In future work, we are interested in studying how topic-based user influence changes over time and improves the topic-based influence prediction. We will also investigate alternative methods for combining various influence measures.

Chapter 5

A Case-study: Weight-based Sentiment Analysis

5.1 Introduction

[1] In this chapter, we utilise the contribution of previous chapters in sentiment analysis and social influence measurement to introduce a novel and innovative weight-based sentiment analysis approach. Through an election case-study, 2016 US Presidential Election, we experiment how aspect-based influence of users can improve sentiment-analysis-base election prediction.

Social media contents, e.g. tweets, regardless of their content and intention, often carry information about their author's sentiment or opinion (Bollen, Mao & Pepe 2011). It has created great opportunities and considerable promises to capture, measure, and extract public sentiment, mood state, and opinion by analysing them in a more rapid, accurate

[1] **Aso Hamzehei**, Raymond Wong, Fang Chen. Weight-based Sentiment Analysis: A New Approach for Prediction. (Under Submission)

and cost-effective way. In fact, data obtained from social networks can be mined and analysed extensively in a wide range of applications and disciplines such as marketing research companies, public opinion organisations, and other text mining entities (Mostafa 2013). Also, a growing body of literature is available that have shown correlation between public mood and sentiment measured from social media with financial market and stock market changes(Mao et al. 2011, Bollen & Mao 2011, Ruiz et al. 2012), election results (Bermingham & Smeaton 2011, Miller 1995), commercial sales (Mishne et al. 2006), influenza epidemics (Culotta 2010). It is claimed that computational indicators of public sentiment may have predictive power in regard to socio-economic phenomena (Asur & Huberman 2010). It is assumed that social media may have predictive power mainly for incidents that are public opinion driven.

Collective and aggregated sentiment analysis can be considered as a practical use of sentiment analysis. Through aggregated sentiment analysis, businesses can track their customers response and feedback to their services and products. The aggregated sentiment analysis also practically can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets. Such predictions traditionally could be conducted through polling systems which are highly expensive with limited data points. Due to ubiquitous social networks and the advancement of text mining and sentiment analysis, the polling approach can be replaced with more cost-effective, timely, and comprehensive approaches (O'Connor et al. 2010, Asur & Huberman 2010).

In this chapter, we propose an innovative approach, Weigh-based Sentiment Analysis (WSA), to predict events that are public opinion driven e.g. elections. We weight users' opinion and sentiment according to their influence value in a respective topic. WSA proposes a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. Such predictions, traditionally, are conducted through polling systems which are highly expensive with limited data points. In WSA, the focus is on the impact of influential users

in prediction through sentiment analysis. In particular, the goal is to observe whether influence-based aggregated sentiment analysis is outperforming generic aggregated sentiment analysis in prediction. Our approach has two major components: topic-based influence measurement and sentiment analysis. Both together form WSA, the aggregated weighted topic-based sentiment analysis. For evaluating WSA, a dataset collected from Twitter for the US 2016 presidential election. The dataset contains 280 million tweets.

The remainder of this chapter is organised as follows: We first discuss existing approaches for sentiment analysis and its related topics in Section 5.2. We then present the background in Section 5.3. Next, we define the research problem, and then propose our approach and algorithms in Section 5.4. We describe our dataset and discuss the results in Section 5.5. Finally, we conclude the chapter in Section 5.6.

5.2 Related Work

Sentiment analysis is a technique to measure sentiment of a given textual content. It has been mainly about the literal text sentiment and author of the text has been ignored in the process. Studying the text's author can help the process in understanding the author's sentiment. To do so, some research have been conducted to study the social relationships of a text's author to improve his/her measured sentiment.

Tan et al. (2011) proposed that connected users could possibly have similar opinion. They experimented their hypothesis on Twitter and extracted follower and following networks of users. They learned a semi-supervised model and concluded that connections and network of a user can improve the classified sentiment. To measure aggregated sentiment of an aspect, Neves-Silva et al. (2016) studied influence propagation of opinion to combine with sentiment analysis in there classifier. They claimed that they have included ambiguity, reach, influence and relevance and have reached 80% accuracy. However, their

results were preliminary and not elaborative. On a similar note, Hu, Tang, Tang & Liu (2013) utilised social connections to reduce noise in the sentiment measurement of social networks short texts. They modeled tweets sentiments relations via graph Laplacian. It means their efforts conducted under the assumption that author's network can explain tweets sentiment in the classification process. Their results demonstrated significant improvement in classification results.

In aspect-level collective sentiment analysis, Nguyen et al. (2012) utilised network information to predict sentiment direction without going into tweet level sentiment measurement. They ignored micro-analysis of users' sentiment and their network, rather modeled social media dynamics through features extracted from the network. They showed that they can predict the sentiment direction with accuracy of 80%. An innovative work by Zhu et al. (2014) has proposed a sentiment clustering framework where they involve users, tweets, and features as three dimensions of the clustering approach. Their tri-clustering framework is claimed to provide better quality of user-level and tweet-level sentiment analysis.

5.2.1 Sentiment Analysis with Deep Learning

Deep learning as one of the recent emerging techniques of machine learning has been applied in many domains including sentiment analysis. Long Short-Term Memory (LSTM) and its variations have been a promising candidate to measure sentiment in short texts from social media. Wang et al. (2015) introduced an LSTM recurrent network model where they predict tweets sentiment. Their model is based on simulating words' interactions in a compositional process. The essence of their model is in a gate structure for word embeddings that is more flexible than simple recurrent neural networks. Through their experiments on short noisy social media data, they presented that their model outperforms multiple feature-based approaches. One the same note, Ren et al. (2016) proposed

a convolutional neural network model to classify tweets sentiment. This model utilised both tweet content and context features in the same model.

To take the syntactic property of natural language into the modeling, a tree-LSTM is introduced. It is built on a tree-structure network topology. An example of tree-LSTM is implemented by Tai et al. (2015). They have shown that tree-LSTM outperforms others when it comes to modeling words and sentences semantic connections. In a recent work, Feng et al. (2018) introduced a context-aware LSTM model that consider the other of tweets and words in a tweet. Attention mechanism was used to give different weight to tweets and words.

5.2.2 Sentiment Analysis with Naive Bayes

Naive Bayes classifiers are used in estimating sentiment of documents and microblogs contents. Naive Bayes calculates probability by measuring frequency of words. Prasad (2010) experimented Naive Bayes to classify sentiment of social media content and concluded that Naive Bayes as positive in terms of performance. Naive Bayes classifier introduced and tested for aspect-based sentiment analysis by Mubarok et al. (2017). To handle real-time stream data in sentiment analysis, Jung et al. (2016) incorporated Laplace Smoothing with Binarized Naive Bayes classifier to improve accuracy and utilised SparkR to speedup the process in their approach for real-time usage.

5.2.3 Elections Prediction via Sentiment Analysis

When it comes to applications of sentiment analysis in prediction, elections have been an interesting event to study. A few studies have tested their sentiment analysis models on elections. Anuta et al. (2017) experimented whether sentiment analysis and social networks can be a replacement for traditional election polls. Bovet et al. (2016) intended

to predict 2016 US presidential election in a temporal perspective and concluded that sentiment analysis on twitter can be a predictor for public opinion in a national level. Similar effort is performed in predicting recent French election and they were successful in predicting Macron's victory (Wang & Gan 2017). Paul et al. (2017) proposed a topic sensitive spatio-temporal sentiment analysis framework. This system is able to analyse aspect-based sentiment for a given location and time. Their case study was 2016 US presidential election. They suggested that their system can be used in other domains as well.

5.2.4 Stock Market Prediction

Alongside the news effect on stock price movement, public and investor sentiment is identified as a more important factor on stocks price volatility. It is validated that market is driven by investor's decisions and investors make decisions based on their sentiment and bullishness (Qian & Rasheed 2007, Rao & Srivastava 2014, Bollen, Mao & Zeng 2011). We also have learned from psychology and customer behavior research that emotion is a factor in decision making process of human being (Dolan 2002). The substantial role of sentiment in financial decisions is also verified by behavioral finance research (Nofsinger 2005). Therefore, it can be assumed that more comprehensive and accurate measurement of public and investors sentiment and mood could result in a better stock market prediction.

Recent researches in the area of sentiment analysis have attempted to develop techniques for public sentiment indicators measurement directly through social networks and blogs. (Bollen, Mao & Zeng 2011) showed that Dow Jones Industrial average can be predicted by collective public sentiment on Twitter.

One of very few studies on engaging multi source of information for sentiment analysis for stock market prediction is (Mao et al. 2011). They used social network (Twitter),

news, search engine data, and sentiment surveys as their datasets for measuring different sentiment indicators such as bulishness, investor intelligence, investor sentiment, tweet volume of financial search terms, negative news sentiment and google search volume of financial search terms.

There are other studies like (Oh & Sheng 2011) that studied investors blogs such as stock-twits.com and concluded that stock microblogs content seem to have strong correlation with stock movements. Furthermore, (Liu et al. 2015) studied the social network influence on stock return co-movement. They showed that companies with official twitter account have higher co-movement than companies without official page.

5.3 Background

5.3.1 Deep Learning Sentiment Analysis

One of the well-exercised applications of Deep Learning (DL) is in Natural Language Processing (NLP) and more recently in Sentiment Analysis. Among the DL techniques, Recurrent Neural Networks (RNN) are more preferred for sentiment analysis due to its capability of taking inputs with variable length (Irsoy & Cardie 2014). RNNs are able to handle sequence of data and for this reason they can be utilised for text classification. However, RNNs are not the best with long sequence of data. Long Short-Term Memory (LSTM) is an alternative kind of recurrent neural networks without the shortcomings of RNNs. Here we describe LSTM for sentiment analysis(Tang et al. 2015, Cambria et al. 2018).

LSTM computes representation of textual document from representation of its building blocks (e.g. words are building blocks of sentences). To measure a sentence' sentiment with LSTM, we need the sentence representation as features. Each word representation

will be a low dimensional vector of real values called word embedding. All words together form a word embedding matrix. Assume X as an input sequence, $X = \{x_1, x_2, ..., x_n\}$. LSTM process each input at each time tep t and holds a hidden vector h_t, a memory vector m_t, and an output. The memory vector updates the input state and the output is generated based on the past state of the input x_i. At each time step t, we have the following computations:

$$\begin{aligned} g_i &= \sigma(W_i[h_{t-1}; W_t] + I_i) \\ g_f &= \sigma(W_f[h_{t-1}; W_t] + I_f) \\ g_o &= \sigma(W_o[h_{t-1}; W_t] + I_o) \\ g_c &= \sigma(W_c[h_{t-1}; W_t] + I_c) \\ m_t &= g_f \odot m_{t-1} + g_i \odot g_c \\ h_t &= tanh(g_o \odot m_t) \end{aligned} \tag{5.1}$$

where $\odot$ is element-wise product, σ is logistic sigmoid function, $W_i, I_i, W_f, I_f, W_o, I_o, W_c$, I_c are recurrent weight and projection parameters, and g_i, g_f, g_o, g_c are computed gates by LSTM. The gates help LSTM to learn useful features. The g_i gate keep features in memory vector m_t, g_f remove the past vector, and g_o is the output gate and controls what to read from the new vector in memory and inject it into the hidden layer h_t.

These composed textual document representations are the features generated by LSTM that needed for sentiment classification (dos Santos & Gatti 2014, Cambria et al. 2018).

5.3.2 Naive Bayes Sentiment Analysis

Naive Bayes classifier is a probabilistic approach based on Bayes' theorem to classify high dimensional inputs such as textual documents. It calculates the word/class posterior

probability of the classes in documents. Posterior probability is estimated by Bayes' rules where a document will be labeled with a class that has greater posterior probability. Bayes' rule is based on the condition that words are independent of the other words in posterior probability (Song et al. 2017, Liu et al. 2013). According to the Bayes' rule we can define the classifier as:

$$c'(d) = \underset{c \in C}{argmax} P(c) \prod_{w \in W} P(w|c) \tag{5.2}$$

where c' is the estimated class for document d. C is the set of all labels and W is the set of words. $P(c)$ is the prior probability of c and $P(w|c)$ is the Bayes' condition.

A version of Naive Bayes is Multinomial Naive Bayes which is designed for modelling language and assumes conditional independence between words. It is light and scalable fro bigger datasets a and classes. It is more often applied in textual document classification and sentiment analysis. The Multinomial Naive Bayes classifier is defined as follows:

$$c'(d) = \underset{c \in C}{argmax} \left[logP(c) \sum_{w \in W} f_w logP(w|c) \right] \tag{5.3}$$

where W is the words vector and f is the words frequency vector. The log is the main contributor to the scalability of Multinomial Naive Bayes. $P(c)$ is the prior probability and is estimated as:

$$\begin{aligned} P(c) &= \frac{D_c}{D} \\ P(w|c) &= \frac{\sum_{w \in W} f_w}{\sum_{d \in D} \sum_{w \in W} f_{dw}} \end{aligned} \tag{5.4}$$

where D_c is the number of documents in class c and D is the total number of documents. Also, f_w is the frequency of word w int he class c and f_{dw} is the frequency of all words in

class c.

5.4 Weight-based Sentiment Analysis (WSA)

In this section, we introduce an innovative Weight-based Sentiment Analysis approach for prediction. The proposed approach is to predict events that are mainly public opinion driven. In such events, we include individual's influence as a coefficient to their sentiment on the related topic and perform prediction through the aggregated and weighted sentiment analysis. We first define the problem and then explain our approach.

5.4.1 Problem Definition

Assume $G(\mathbb{V}, E)$ denotes a social network graph, where users are the vertex set $\mathbb{V}$ and users relationships are the edges set E. Assume that users publish a set of texts $D = [d_1, d_2, \ldots, d_q]$, and talk about different topics $\mathcal{T} = [t_1, t_2, \ldots, t_j]$. Each user text (post) d_q holds one or more topics as well as author's sentiment for the topic or topics of the document. Each user text (post) d_q receives engagement from other users by replying, liking, or re-publishing it. The engagement of other users in a post can reveal the influence of that particular post among its audience.

Quantifying the topic-based influence of each user based on social ties and other users' engagement in social network, we can identify the influence of user u_i on topic t_j, represented as F_{ij}. Then, we have matrix $F = [F_{ij}]_{i \times j}$ that represents influence of all users in all identified topics. Given a list of users, topics, social influence of each user on those topics F_{ij} , and sentiment of users in each of their social network posts S_{id}, we are interested in measuring weight-based sentiment of users in topics $WSA_{ij} = F_{ij}S_{id}$. The weight of a user in a particular topic comes from the influence of that user in that topic.

If we consider D_t as the collection of documents for topic t, the common available approach in the literature is to measure the overall sentiment of the network in that topic, where it measures sentiment for every single document in that collection. It considers the percentage of positive and negative sentiment toward that topic in the given network. But, in WSA, the influence of user u_i in a topic is the coefficient of the measured sentiment, which means the higher the influence of a user is in a topic, the more her/his sentiment share will be in the overall sentiment value for topic t_i. This is based on the assumption that a central node in the network is more impactful to the overall sentiment value than a node with few connections and edges. For the case of election, if two candidates both have equal fans, it is more likely that the candidate with more influential fan wins the election.

5.4.2 Our Approach

In literature, there have been several studies on sentiment analysis with focus mainly on text/document polarity classification. Very few research is conducted in aggregated sentiment analysis such as Hu, Tang, Tang & Liu (2013) and Neves-Silva et al. (2016). However, to best of our knowledge, in sentiment analysis no study has considered author's topic-based influence to refine aggregated sentiment analysis outcome.

To measure overall sentiment of a topic, we propose WSA, a Weight-based Sentiment Analysis approach for prediction. For measuring the overall sentiment of a topic, WSA discriminates user opinion. It uses user's influence in a particular topic as a weight and coefficient for user's sentiment in calculating the overall sentiment of the topic. In a nutshell, our approach contains three main parts: (i) measuring topic-based influence of users in the network, (ii) measuring sentiment of every document generated by users in the network, and (iii) calculate overall weight-based sentiment of a topic for the entire network. We propose WSA for prediction purposes where public opinion is the main

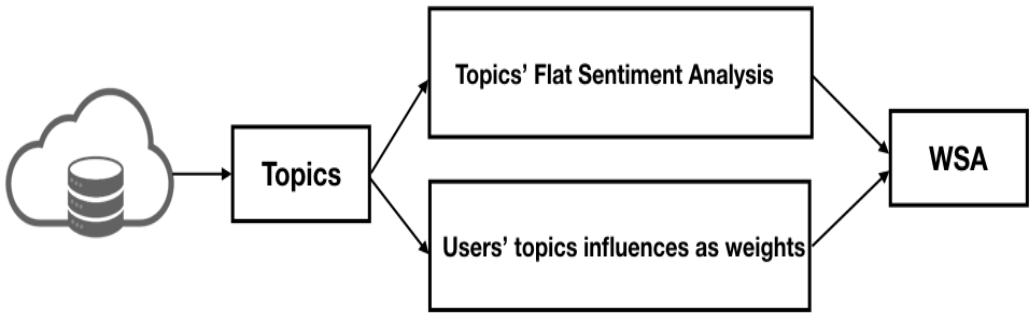


Figure 5.1: WSA Framework

drive for an incident.

Topic-based Weights

Figure 5.1 shows the process of measuring weighted sentiment analysis. The first step is to identify topics in the dataset. Knowing the topics, we are interested in measuring every user's influence in an identified topic. We accept the definition of influence introduced by Topic-based Social Influence Measure (TSIM) in Chapter 3. According to TSIM, social influence for each tuple of $(user_i, topic_j)$ can be measured through F^*_{ij} which comprises of four measures F_f (follower count), F_a (activity), F_e (engagement), F_c (centrality) for user u_i in topic t_j. F^*_{ij} stands as the weight of user u_i in topic t_j. It means, in the overall sentiment of a topic, a user with higher influence in that topic will have a higher portion and higher impact on the calculated value for the aggregated topic-based sentiment. The detail of the TSIM can be found in Chapter 3.

Sentiment Analysis

A component of WSA is the conventional sentiment analysis which here we refer to it as Flat Sentiment Analysis. There is no weighting involved in flat sentiment analysis and

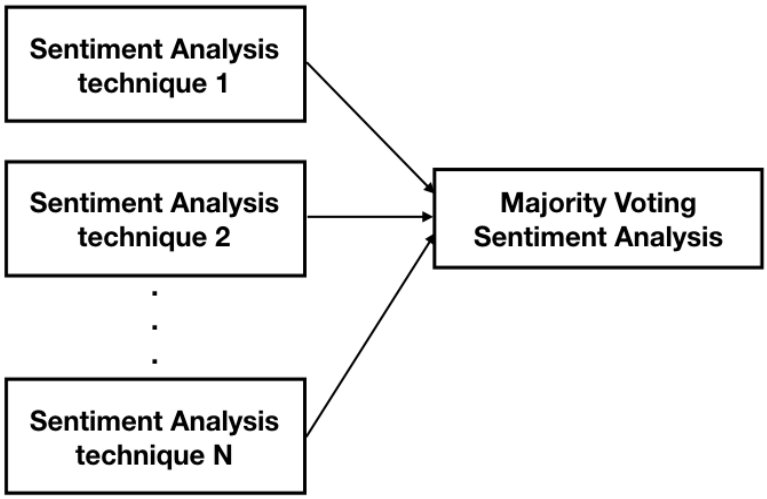


Figure 5.2: Majority voting approach for WSA

users are not discriminated in overall sentiment analysis result. Flat sentiment analysis is simply about measuring sentiment of a given document regardless of who is the author of the document, his/her network, or his/her influence in the topic of the document. To measure sentiment of documents, we use SSSA (proposed in Chapter 2) along with Naive Bayes sentiment analysis, Deep Learning sentiment analysis, Vader, and TextBlob. We propose a majority voting sentiment analysis approach that listen to all the used five algorithms and assign a positive/negative label to a document if majority of the algorithms agree on the same label. Figure 5.2 shows the majority voting approach.

WSA

A user could have generated multiple contents in a single topic. However, in aggregated sentiment analysis, every user has one sentiment value regarding a topic. To do so, for users who have generated multiple documents regarding a particular topic, we average her measured sentiment in all her documents regarding topic t and will assign one sentiment value to the user for each topic. In this way, a user with multiple documents will not be over presented in the aggregated sentiment value calculated for a topic. Equation 5.5

shows each user's sentiment regarding a topic:

$$S_{u_i} = \frac{\sum_{d_{u_i}=1}^{m} S_{u_i d_m}}{m} \tag{5.5}$$

where $S_{u_i d_m}$ is user's sentiment in her set of documents in a topic.

For a given dataset with multiple users, the equation 5.6 calculates the sentiment of a topic in the dataset as:

$$S_{t_j} = \frac{\sum_{u_i=1}^{i} S_{u_i}}{i} \tag{5.6}$$

Now, to have a weighted sentiment, we add influence to the equation as coefficient of users' sentiments:

$$WSA = \frac{\sum_{u=1}^{i} F_{u_i} (\frac{\sum_{d_{u_i}=1}^{m} S_{u_i d_m}}{m})}{\sum_{u=1}^{i} F_{u_i}} \tag{5.7}$$

where F_{u_i} is the user's influence.

Algorithm 3 explains the the WSA process.

5.5 Experiments

In this section, we explain and discuss the details of conducted experiments of the proposed weight-based sentiment analysis approach presented in the previous section. It includes the dataset, influence measurement, flat sentiment analysis, and weight-based

Algorithm 3 Weight-based Sentiment analysis

Input: Topic's social network dataset.

Output: Overall topic-based weighted sentiment value.

1: Identify topics though topic modelling
2: **for** user in *users* **do**
3: measure F_i^*
4: **for** document in *dataset* **do**
5: measure document's sentiment S_d
6: $WSA_{id} = F_i S_{id}$
7: $WSA = \frac{\sum WSA_i}{\sum F_i}$
8: Return the overall sentiment of topics

sentiment analysis on the Election data. This section provides details on how our proposed approach performs for prediction and its performance is compared with the available state-of-the-art sentiment analysis techniques. We compare WSA with flat sentiment analysis approaches. This comparison is conduced on the whole dataset as well as shorter periods of weekly analysis. Moreover, we study WSA performance in identified topics along with flat sentiment analysis approach. In the flat sentiment analysis approach, we use both supervised and unsupervised approaches and their performance are presented in details.

The experiments are performed on a machine with 24 cores of Intel Xeon X5690 (48 threads) and 192 GB of Ram. The implementation is taken place in Python, using the key libraries of Pandas, Scikit-Learn, Numpy, and specifically the Python high performance computing library, DASK (Dask Development Team 2016), for multiprocessing. DASK is a library that uses Numpy and Pandas data structures and provides functionalities for cluster computing, multiprocessing, and parallel processing in Python for large-scale computations.

Table 5.1: Hashtags used for annotating tweets sentiment

Labels	*Hashtags*
Trump positive	maga, draintheswamp, votetrump, makeamerica-greatagain, maga3x, americafirst, votetrumppence16, womenfortrump, trumpstrong, latinosfortrump, blacksfortrump, votetrump2016, women4ttump, al-waystrump, presidenttrump, makeamericasafeagain, teamtrump, mikepence, lovetrumpshate, trumpsarmy, trumpwon
Trump Negative	nevertrump, trumpandhillarywannafuckeachother, dumptrump, trumptapes, trumptaxes, stoptrump, trumpisanabuser, trumplies
Hillary Positive	imwithher, strongertogether, gohillary, imwither, hillyes, votehillary
Hillary Negative	neverhillary, crookedhillary, wikileaks, trumpand-hillarywannafuckeachother, lockherup, hillaryfor-prison, makehillarybrokeagain, corrupthillary, clin-toncorruption, nastywoman, hillaryforprision, sick-hillary, clintoncrimefamily, bringbernieback, killary, lyinghillary, hillaryemails, clintonemails, clinton-cash, hillaryshealth, billclintonisarapist, clintonscan-dals

5.5.1 Dataset

The case study in this chapter for testing WSA is the US 2016 Presidential Election data on Twitter. We collected the election tweets from 19 September 2016 (one week before the first election debate program) until election day, 10 November 2016. It includes 139.5 million tweets and 13.7 million users. The tweets are from four categories of tweets, retweets, quotes, and replies. The dataset contains important metadata such as favorites count, retweet count, replies, hashtags, and mention's list. Also, at the end of the election, we collected the retweet list of all tweets that we streamed during the election. To make a ground truth dataset, we selected s subset of the dataset from the tweets with obvious positive and negative hashtags (Hashtags generally are considered as author's annotation for a tweet). Figure 5.1 shows the list of positive and negative hashtags used for labeling

the ground truth dataset for sentiment analysis algorithms evaluation. We also annotated the sentiment of 150 thousands of randomly selected tweets by Rosette, a commercial service for social network sentiment analysis and we used this small labelled dataset for evaluations as well. The details of the dataset is explained in Section 7.0.3.

5.5.2 Sentiment Analysis on Election Topics

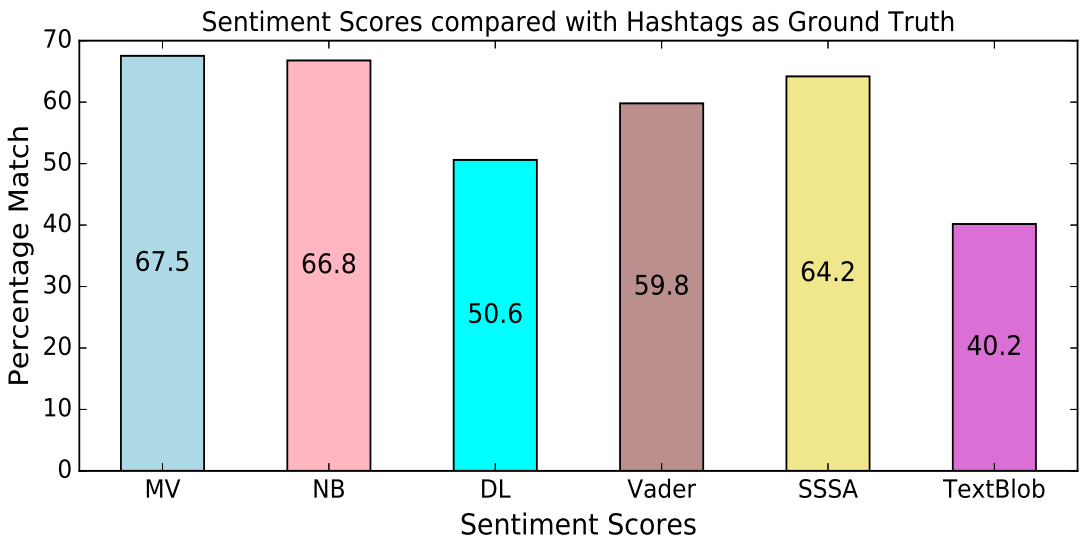


Figure 5.3: Sentiment analysis techniques and the majority voting performance evaluated with ground truth.

Sentiment analysis is the core of WSA. The sentiment analysis approach that we proposed in this chapter for WSA is a majority voting approach where we listen to multiple sentiment analysis techniques to label a tweet. Figure 5.2 shows the majority voting sentiment analysis approach. The majority voting approach comprises of SSSA (proposed in Chapter 2), Naive Bayes sentiment analysis (explained in Section 5.3.2), Deep Learning sentiment analysis (explained in Section 5.3.1), Vader (explained in Section 2.4.2), and TextBlob (explained in Section 2.4.2).

Among the above mentioned algorithms, NB and DL are supervised and need a labeled

dataset to create NB and DL models. For this reason, we have randomly selected 150K tweets and annotated them for their sentiment value via Rossete cloud (*Rossete Sentiment Analysis* 2018), a commercial sentiment analysis platform, through their API. We used the annotated dataset to train NB and DL models to measure sentiment of the election dataset. However, SSSA, Vader, and TextBlot do not need labeled data to measure a document's sentiment.

Figure 5.3 shows the sentiment analysis result of the algorithms mentioned above on the ground truth dataset which is 25% of the main election dataset. The ground truth dataset is labeled directly by using positive or negative hashtags that defined by tweets' authors. Those hashtags can be found in Table 5.1.

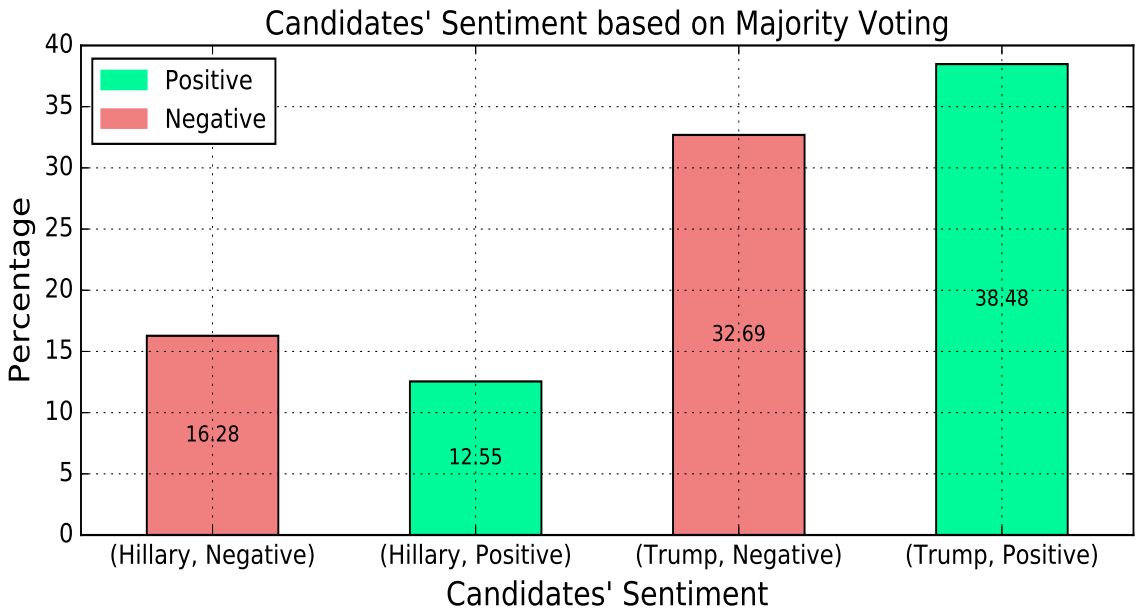


Figure 5.4: Majority voting results for both presidential candidates. Results are presented in terms of positive and negative values. The values are the portion of each candidate in the whole dataset.

According to the Figure 5.3, the majority voting approach achieved the highest accuracy of 68.47% in sentiment determination. The majority voting, label a tweet as positive if three out of the five algorithms agree on the positivity of that tweet. The second and third

best performers are NB and SSSA with 66.41% and 64.23% respectively. The worst performer is TextBlob with 40%. Due to better performance of the majority voting, the rest of the result reported in this chapter will be only based on the majority voting sentiment analysis results.

Now that we have settled with majority voting, we will look into the candidates sentiment in the whole dataset. Figure 5.4 shows the sentiment measurement for the both candidates in the entire dataset. According to Figure Figure 5.4, 28.83% of tweets are about Hillary Clinton and 71.13% about Donald Trump. Among the tweets, 16.28% were negative and 12.55% were positive for Hillary Clinton. So, there were more negative tweets than positives for Hillary Clinton. On the other hand, 32.69% of tweets were negative and 38.48% were positive for Trump. Based on this observation, there is more positive tweets than negative ones on Trump, while there are more negatives than positives for Hillary Clinton.

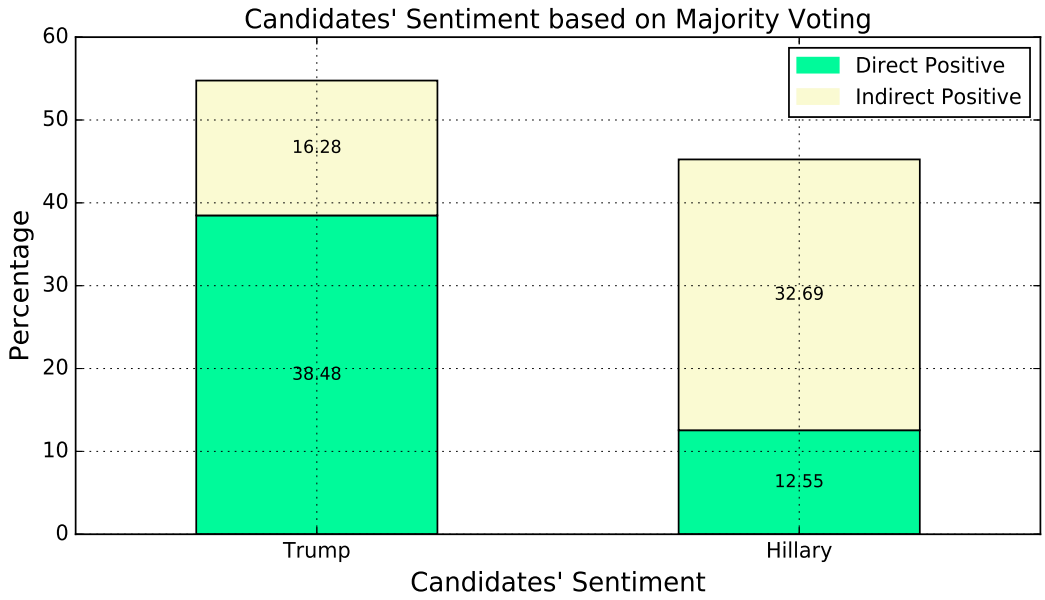


Figure 5.5: Majority voting results for both presidential candidates. Results are based on direct and indirect sentiment values.

Assume an opponent of candidate A could potentially be a fan of candidate B. With this

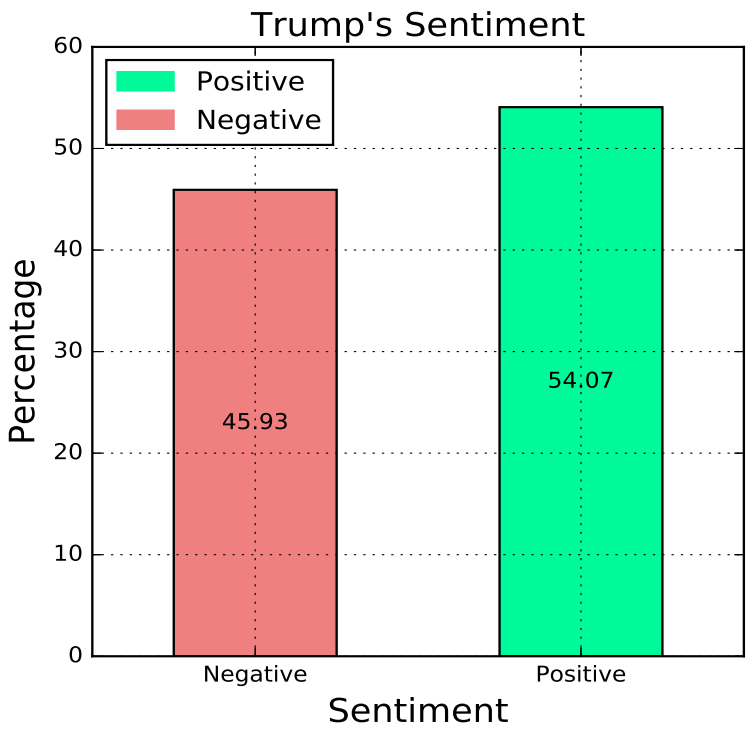


Figure 5.6: Majority voting results on tweets related to Donald Trump. Results are based on only direct sentiment values

assumption, we can consider the negative tweets of candidate A as positive tweets for the opponent and vice versa. We call this negative sentiment of candidate A as indirect positive sentiment for candidate B. In Figure 5.5, we present the candidates sentiment given the above assumption. The figure shows both direct and indirect sentiments for both candidates. According to the figure, the Trump's positive sentiment is above 50% at 54.76% and Clinton's positive tweets are under 50% at 45.24%.

To understand the positivity and negativity of tweets collection of each candidate, we look into individual candidates related tweets and check how sentiment is distributed between positive and negative. Figure 5.6 shows the distribution of sentiment among tweets related to Trump. It depicts that 54.07% of the tweets about Trump were positive about him and 45.93% were negative. However, according to Figure 5.7, there has been more negative tweets about Clinton than positive ones. The negative tweets were 56.47% and 43.53%

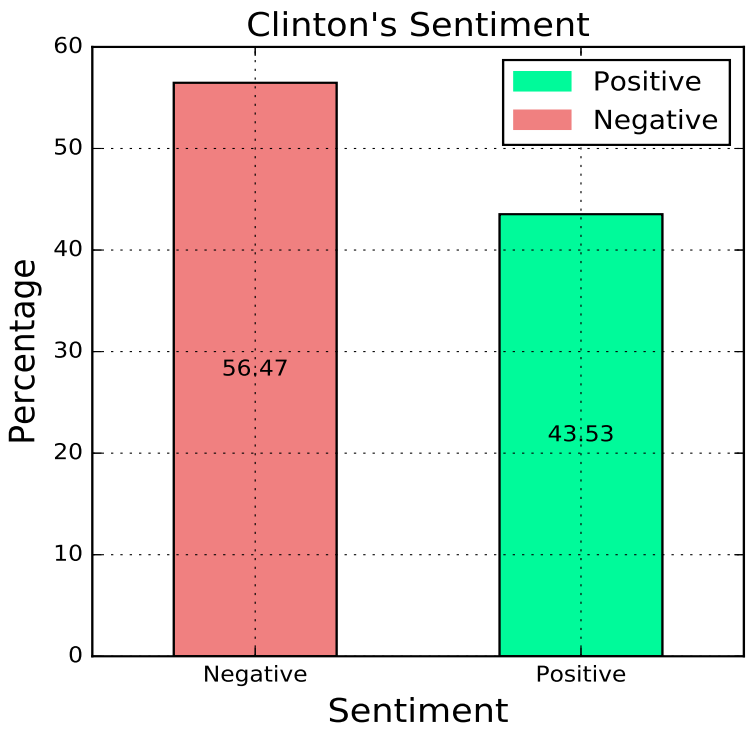


Figure 5.7: The percentage of positive and negative sentiment in the dataset of tweets about Hillary Clinton. Results are generated from WSA with majority voting they are based on only direct sentiment values

positive tweets about her.

5.5.3 Weight-based Sentiment Analysis

Table 5.2: Prediction accuracy of Trump and Clinton in both flat sentiment and WSA approaches

Candidate	Flat Sentiment	WSA
Hillary Clinton	40.3%	42.2%
Donald Trump	50.4%	50.2%
Neutral	9.3%	7.6%

In this subsection, we present the election prediction results for the proposed weight-based sentiment analysis approach and compare it with conventional flat sentiment analysis.

The sentiments are measured for the whole dataset for both candidates. In these results the neutral tweets are dismissed where they count around 9.3% of the whole dataset. So, the percentages presented in the results are from the polar dataset.

Table 5.2 is summarising the sentiment results for both candidates. Based on the flat sentiment analysis, the percentage of positive tweets for Trump is 50.4% of and for Clinton is 40.3% and 9.3% of the tweets where estimated as neutral for the candidates. In the flat sentiment analysis, tweets weights are equal. However, when we included the weights that calculated through measuring the users' influence, the gap between candidates became closer. With WSA, Trumps positive percentage falls by 0.2% and Clintpn's positivity raise by 1.9%. The WSA result is closer to the ground truth of the election outcome (*Official 2016 Us Presidential General Election Results* 2016). According to the official results, Clinton received 48.18% of general election and Trump received 46.09% of the votes. However, in terms of electoral college, Trump won 30 states which is 57% of electors. Considering either of the general election or electoral college, WSA results are closer to the ground truth than conventional flat sentiment analysis.

It worth mentioning that Table 5.2 is based on same sentiment result pool. In Flat sentiment there is no weighting in the measured sentiment whereas in WSA the influence-based weights could adjust impact of a measured sentiment in the prediction result. To observe how each influence measure is affecting the prediction results, refer to Table 5.3.

5.5.4 Influence Measurement on Election Dataset

The core building block of WSA is weights of sentiments. The weights are the users' influence in the topic of the document. The calculated weights are set as the sentiment's coefficients and could change the overall measured sentiment of a topic. The measures of influence are defined according to TSIM (described in 3)as Follower count, Activity, Engagement, and PageRank. We experimented WSA, where we used each of the TSIM

Table 5.3: WSA results for both candidates where the weights of WSA is calculated based on a single influence measure. They are compared with flat sentiment values.

Influence metrics	*Candidates*			
	Hillary Clinton		*Donald Trump*	
	Flat Sentiment	*WSA*	*Flat Sentiment*	*WSA*
Follower Count	40.3%	37.5%	50.4%	51.4%
Activity	40.3%	41.6%	50.4%	53.54%
Engagement	40.3%	41.8%	50.4%	53.67%
PageRank	40.3%	42.96%	50.4%	47.67%
TSIM (all metrics)	40.3%	42.2%	50.4%	50.2%

sub-measures as an independent influence measure for the weight in WSA. The experimental results are presented in Table 5.3.

In the Table 5.3, the four measures are experimented independently, each as a weight for WSA and compared with WSA with TSIM and flat sentiment analysis results. For follower count, Clinton's sentiment with WSA is 37.5% which is less than flat sentiment of 40.3% and WSA with TSIM of 42.2%. Tump's WSA value with follower count is 51.4% which is higher that both WSA with TSIM and flat sentiment. This means that trump's supporters had more follower count in twitter than Clinton's supporters.

However, in case of other measures both candidates have higher sentiment percentage. For activity, Clinton has 1.3% more positive sentiment and Trump has 3.14% more positive sentiment, both higher than flat sentiment. This means that Trump's supporters have been more active in twitter than Clinton's supporters. Both candidate's sentiment from WSA with activity is higher than WSA with TSIM. The reason is that authors of neutral tweets have been less active in the election dataset therefore; neutral tweets percentage is dropped from 9.3% to 4.86% and is added to the candidate's positive sentiments. Also, it can be inferred that Trump's supporters have been more active in twitter than Clinton's supporters.

Similar to activity measure, experiments on engagement shows that both candidates have

higher positive sentiment percentage than flat sentiment. For engagement, Clinton has 1.5% more positive sentiment and Trump has 3.27% more positive sentiment than flat sentiment. Like activity, the reason is that authors of neutral tweets have received less engagement from users in the election dataset. As a result, neutral tweets' percentage is dropped from 9.3% to 4.6% and is added to the candidate's positive sentiments. Also, it can be inferred that Trump's supporters have been more active in twitter than Clinton's supporters.

PageRank is the last measure in the weights experiments. The higher percentage of PageRank for a candidate means has has been more central in the retweet graph of the topic which shows a candidate is retweeted by more users in that topic. Unlike engagement and activity, PageRank has been in favour of Clinton in WSA and her positive sentiment is 2.66% more than her flat sentiment. Trump's sentiment is dropped by 2.73%. PageRank shows that Clinton's supporters have been more central in the retweet graph of the election network.

One of the possible manipulations in any election could be done by fake accounts where they generate large volume of content in favour of a candidate. Our influence measures eliminate the effect of those users in WSA as our approach gives more weight to users who attracted more audience. It is not possible for a fake user to receive a large number of follower count, large weight of engagement, as well as be retweeted in a large scale.

5.5.5 Weekly Weight-based Sentiment Analysis

In this section, we perform weekly analysis on the dataset with WSA and flat sentiment analysis. We divide the dataset into weekly subsets and measure flat sentiment and WSA starting from 19 Sep 2016 for each week. We compare the results with the election polls. The results for Clinton's sentiment by flat sentiment analysis show that from the first week of the dataset toward the election day, Clinton has a constant positive sentiment from range

Table 5.4: WSA results for both candidates on weekly datasets starting from 19 Sep 2016 to 10 Nov 2016. The values are presented for both WSA and flat sentiment analysis.

	Hillary Clinton		*Donald Trump*	
	Flat Sentiment	*WSA*	*Flat Sentiment*	*WSA*
Week 1	44.97%	44%	55.03%	56%
Week 2	46.06%	46.28%	53.94%	53.72%
Week 3	46.44%	44.83%	53.56%	55.17%
Week 4	44.97%	43.39%	55.03%	56.61%
Week 5	44.29%	42.64%	55.71%	57.36%
Week 6	43.38%	41.84%	56.62%	58.16%
Week 7	46.14%	43.50%	53.86%	56.50%
Week 8	42.70%	39.21%	57.30%	60.79%

of 42.7% to 46.4%. However, WSA is showing that Clinton in Twitter has a downward positive sentiment trend. According to WSA's sentiments, Clinton hit a bottom 39.21% positive sentiment in her last week of the election period. For Clinton, WSA results shows approximately a 3% less positive sentiment compare to flat sentiment analysis, although the difference was just bellow 1% in the first week. The flat sentiment analysis for Trump is in the range of 54% to 57%, where the last week is the most positive week. WSA results show a 1% more positive sentiment compare to flat sentiment in the first week and this difference reach to 3.5% in the last week. According to both flat sentiment and WSA, the last week of the election is the strongest among the 8 weeks of the study for Trump where it is the worst for his rival from the democrats.

The 2016 presidential election polls, performed by various sources, were predominantly biased toward Hillary Clinton. Tables 5.5, 5.6, and 5.7 show 9 national polls results of the election conducted by main sources. The period of the polls presented in the tables are from 19 September 2016 until the election day, same as the duration of our collected twitter dataset. Except of LA Times, the rest of the polls with a margin of +1% to +11% were biased against Trump. One important reason of the poor performance of the polls could be the considerably small study samples. Polls are expensive to perform and very inefficient. Adversely, social networks contains a large portion of opinion of the public

Table 5.5: 2016 presidential election polls for Donald Trump and Hillary Clinton.

	YouGov		*ABC/Wash Post*		*FOX News*	
	Clinton	*Trump*	*Clinton*	*Trump*	*Clinton*	*Trump*
11/1 - 11/7	49%	45%	49%	46%	48%	44%
10/25 - 10/30	48%	45%	48%	47%	46%	45%
10/20 - 10/25	49%	46%	50%	45%	49%	44%
10/10 - 10/15	48%	43%	50%	46%	49%	41%
9/19 - 9/24	45%	44%	49%	47%	49%	44%

Table 5.6: Continued- 2016 presidential election polls for Donald Trump and Hillary Clinton.

	LA Times		*NBC News*		*Reuters/Ipsos*	
	Clinton	*Trump*	*Clinton*	*Trump*	*Clinton*	*Trump*
11/1 - 11/7	44%	**47%**	51%	44%	44%	39%
10/25 - 10/30	43%	**47%**	51%	44%	45%	39%
10/20 - 10/25	44%	**46%**	50%	44%	43%	37%
10/10 - 10/15	44%	**45%**	51%	43%	43%	39%
9/19 - 9/24	43%	**46%**	51%	44%	44%	38%

Table 5.7: Continued- 2016 presidential election polls for Donald Trump and Hillary Clinton.

	Gravis		*Bloomberg*		*CBS News*	
	Clinton	*Trump*	*Clinton*	*Trump*	*Clinton*	*Trump*
11/1 - 11/7	-	-	46%	43%	47%	43%
10/25 - 10/30	50%	50%	-	-	-	-
10/20 - 10/25	50%	50%	50%	41%	51%	40%
10/10 - 10/15	50%	50%	-	-	-	-
9/19 - 9/24	-	-	46%	46%	49%	43%

which is easily accessible.

5.5.6 Implications and Applications

This section describes the real-world implications and applications of the proposed WSA. Assume there are two candidates running to win an election and both have equal number of fans. According to ordinary sentiment analysis, the chance of each candidate winning the election is 50%. However, if we knew the influence of each of those individuals in the fan groups, we may have a different estimation. In this example, if the candidate A has highly influential supporters and candidate B has low influential supporters, most likely candidate A wins as the influential supporters could influence other people to join to the same group. WSA is bringing the quantified influence of users into the sentiment analysis equation and improve prediction results. One of the main applications of WSA is in prediction of public opinion driven events such as elections and stock markets through social networks content and graph of connectivity.

5.6 Conclusion

Aggregated sentiment analysis can be used to measure public mood and opinion for variety of aspects and topics in social networks. However, user information is not utilised to improve the measured sentiments. In this study, we presented a weight-based sentiment analysis approach, where the influence of a content's author is used as a weight for the content's sentiment in the topic's aggregated sentiment. Our proposed approach consists of two main components: Influence measurement and sentiment analysis. To predict a public opinion's driven event such as election, our approach, first measures every user's influence in network for a given topic. Then, WSA measures sentiment of users' contents. To measure overall topic's sentiment, users' influence is used as a coefficient of users'

sentiment in the topic. Therefore, users will not have equal impact on the aggregated sentiment. The more influential a user is, the more weight will have in the aggregated sentiment value.

We evaluated our proposed weight-based sentiment analysis approach using a very large dataset that we collected from Twitter for US 2016 presidential election. Our extensive experiments show that our model is outperforming the state-of-the-art in aggregated sentiment analysis. We have demonstrated that the improvement in prediction result is mainly the contribution of the users' influence as the measured sentiment's coefficient. Elections predictions can be performed with higher accuracy and less cost with our proposed approach. In summary, the effect of users are not equivalent in the network on a topic's aggregated sentiment value and users influence can improve the sentiment classification's accuracy.

Chapter 6

Conclusion

In aggregated sentiment analysis in the literature, every document in a corpus has the same weight and impact on the overall calculated sentiment. However, in the real world people have unequal influence and impact on each other and outcome of an event. When we observe opinions regarding a particular topic, an expert in that topic is expected to have more accurate opinion or sentiment regarding the topic than a non-expert person. For example, a Nobel prize winner for economic is expected to have a more accurate opinion regarding an economic phenomenon than someone without economic science background. Not every influential user in the network can be influential in every topic. For example, let's assume we have two topics of 2019 global economy and 2019 Wimbledon Championships and two users, our economic Nobel prize winner and Roger Federer, the world's most Grnad Slam single title winner in the history of tennis. If we ask both of them about their prediction for 2019 global economy and the winner of 2019 Wimbledon Championships, we expect the prediction accuracy of our Nobel prize winner for economy be more accurate than the tennis professional. We also expect that Roger Federer's prediction for the winner of 2019 Wimbledon Championships be more accurate than our economic Nobel prize winner. Therefore, if we are going to observe users and predict a

topic based on their opinion and sentiment, we should not give every user the same equal weight. We expect to have higher accuracy of prediction if we give more weight to users for their opinions and sentiments on a topic respective to their influence in that topic.

Weight-based Sentiment Analysis (WSA) is a new innovative approach of sentiment analysis measurement. In WSA, sentiment analysis and opinion mining meet with social influence to form a weighted sentiment analysis approach to predict public opinion driven incidents. In this thesis, we propose an innovative aggregated sentiment analysis approach that utilises users social influence in sentiment analysis and discriminate users based on their social influence. Accordingly, users with higher topic-based social influence value will have more impact on the aggregated sentiment value of a topic.

This dissertation consists of two main components: Sentiment Analysis (Chapters 2 and 5) and social influence measurement and prediction (Chapters 3, 4, and 5).

6.1 Summary of Contributions

The contributions of the thesis are as follow:

- First, the thesis introduced a scalable sentiment analysis approach in Chapter 2. We proposed a methodology that has engaged semantic scoring for subjectivity and polarity classification to be applied on a large scale of tweets, which is more scalable and accurate. Our approach reduced the sentiment analysis computational cost to one third in regard to conventional emoticon-based, hashtag-based, and similar semantic-based (SSSA1 and SSSA2) approaches. Also, we developed an automatic system for tagging the tweets with high precision for single target tweets. Our approach is able to filter out news, ads, and noise, automatically. The experiments

presented substantial improvements in scalability and accuracy of sentiment analysis in regard to the benchmarks.

- In Chapter 3, we studied social influence and introduced Topic-based Social Influence Measurement (TSIM). We proposed an approach that incorporates network structure, user-generated content, and user's interactions in the network for topic-based influence measurement. Instead of considering user-to-user influence and global user influence, the proposed model considers individuals influence and interests in a topic, which gives the capability of predicting one's influence on a new topic. The performed experimental analysis showed that the proposed TSIM can effectively measure topic-based user influence.

- In Chapter 4, we proposed a Collaborative Topic-based Social Influence model (SICTR) that incorporates both network structure and user-generated content for topic-based influence prediction. Our novel topic-based influence prediction approach integrated the user-topic relationships, topic content information, and social connections between users into the same principled model. SICTR predicts topic-based user influence on unobserved topics, based on observed topic-based user influence through their generated contents and activities in social networks. SICTR improves the shortcoming of CTR in social network and influence prediction by incorporating short text topic modeling. Instead of considering user-to-user influence and global user influence, the proposed model considers individual users' influence and interests in a topic, which gives the capability of predicting one's influence in a new topic. The usefulness of considering content of topics and co-occurrence in the user-topic matrix are confirmed by our experiments on topic-based user influence prediction: topic-aware methods show better performance over other approaches that just consider a generic item and neglect its characteristics (content-based, e.g., LDA; and item-based, e.g., collaborative filtering methods).

- In Chapter 5, we demonstrated the final contribution of the thesis. We proposed

Weight-based Sentiment Analysis (WSA), an innovative approach where the author's influence is included in sentiment analysis measurement. WSA is a sentiment analysis and opinion mining framework for learning predominant sentiments of social media contents that leads to better performance in prediction tasks. Such predictions, traditionally, are conducted through polling systems which are highly expensive with limited data points. In the last main chapter, we focused on the impact of influential users in prediction through sentiment analysis. In particular, the goal was to observe whether influence-based aggregated sentiment analysis is outperforming generic aggregated sentiment analysis in prediction. Our extensive experiments showed that WSA is outperforming the state-of-the-art in aggregated sentiment analysis as well as predicting elections with higher accuracy. We have demonstrated that the improvement in prediction results is mainly the contribution of the users influence as the measured sentiment's coefficient. Elections prediction can be performed with higher accuracy and less cost with our proposed approach.

In summary, the effect of users are not equivalent in the network on a topic's aggregated sentiment value and incorporating users' influence can improve the sentiment classification's accuracy. In this dissertation, a weight-based sentiment analysis approach is proposed where its components are developed in earlier chapters of the thesis and it is tested as a whole in Chapter 5. We apply more advanced sentiment analysis technique that goes beyond the simple and discrete classification of positive and negative polarity. Collective and aggregated sentiment analysis can be considered as practical use of sentiment analysis. The aggregated sentiment analysis can be used for prediction of aspects that is dependent to public opinion such as elections and stock markets.

6.2 Future Work

For each of the sections in this dissertation, there has been questions unanswered. There could be future works to further investigate the research problems raised in this thesis. We list the most import possible future works in the following:

- In Chapter 2, we proposed a semantic scoring sentiment analysis approach where we combined the advantages of NLP and machine learning for a scalable sentiment analysis approach. One direction to improve our model, is to include word ordering to include negation into measured sentiment value. Also, we used SVM in classification of the labeled data (labeled through our tagging system) where there is room for improvement by considering other methods of sentiment classification. Furthermore, the effect of sarcastic content on SSSA could be investigated in future works.

- In Chapter 3, we proposed TSIM for topic-based social influence measurement. We used LDA for extracting topics. However, according to our findings in Chapter 4, topic modelings for short text will be more suitable for TSIM. Moreover, the influence model could be extended to take spatiotemporal influence into account. It would be valuable and interesting to measure geographical and location-based influence. A person could be influential in a region but not in another. Location-based influence could be very valuable for regional marketing. Also, influence could increase or deteriorate in time for a user which could be included in TSIM. In addition, sentiment aware influence measurement would be another possible extension to TSIM where could identify positive influence or negative influence in atopic.

- In Chapter 4, SICTR predicts social influence for a new topic. One of the limitations of SICTR is identifying the number of topics for topic-based influence model. Having large number of topics cause the user influence matrix to become very sparse. Finding an optimal number of topics could help not only the sparsity problem but

also the quality of topics. One could include a user's friends influence in the model to further investigate whether the close friends' influence helps in a user's influence prediction on a new topic.

- In Chapter 5, we introduced a weighted-based sentiment analysis approach and tested it on an Election dataset. WSA could be tested on other elections from different countries and on different kind of elections although we expect strong positive result from other elections due to points we have discussed in Chapter 5. Moreover, for predicting a particular event, an interesting research direction would be investigating effect of other social networks and blogs other than twitter on prediction results. As an extra point on future works, WSA could include spatiotemporal features in the model which require a geographic-tag rich dataset. Last but not least, it would be insightful to test WSA on other domains such as stock market prediction.

Chapter 7

Appendix 1

The experiments in this thesis are performed on three different datasets. The datasets are collected from different domains from professional networks to event related and fast growing topics. Our datasets are a mix of small and big datasets.

7.0.1 Mobile phones

For the experiment, we streamed tweets related to mobile phones. Our prepared keywords for stream query contain mobile phone related words such as iphone, samsung, nexus, htc, one, android, etc. We used Twitter API to stream the tweets that contain at least one of the key words in the list of keywords. Among the streamed tweets, we kept the English tweets that are not retweets. As a result, we obtained 3717039 English original tweets.

The train/test dataset preparation process is described as follows:

- We have prepared a sampled dataset from the streamed tweets for testing the proposed approach. The dataset contains tweets that have just one target from the list of mobile phones. The sampled dataset from the main streamed corpus is chosen

when only one target from those in our target list appears in a tweet. The comparative tweets are not considered in this work. A comparative tweet has more than one target. For example in this tweet ”I love iphone5 more than HTC one” two targets iphone5 and HTC 1 are comparative. It means positive sentiment over iphone gives negative sentiment to HTC one. It is important to map the determined sentiment to the right target. Comparative tweets are out of the scope of this research and they are filtered out, therefore single target tweets are chosen. Further preparation on the corpus is explained in the pre-processing section in the research methodology. After removing duplicate tweets and affecting the single target condition, total number of 178552 tweets remained.

- The automatic tweet tagging system identified 82768 positive and negative tweets that 80000 (half positive and half negative) of them selected for the experiments. This tagging system is explained in Section 2.3.4. This dataset is used for scalability assessment. Also, it is applied for comparison of our approach with similar semantic-based approaches.

- A dataset consist of 20000 subjective tweets, which derived from the final tweets dataset used for accuracy comparison of our approach with the hashtag-based approach. This dataset contains at least one hashtag.

- A dataset consists of 1062 subjective tweets is prepared and derived from the final tweets dataset. This dataset is used for accuracy comparison of our approach and emoticon-based approach. Tweets in this dataset contain at least an emoticon.

7.0.2 Machine Learning Community

To validate our proposed method, we collected a unique dataset from Twitter using the Twitter Search API. We targeted the Machine Learning domain and identified core 500

Table 7.1: Dataset statistics

Tweet Type	Number of Tweets	Unique Users in Tweets	Unique Users in Hashtags
Retweets	93874368	9248263	0
Tweets	27270502	4108954	1346545
Quotes	8973923	1553011	308211
Replies	9276342	1516375	232692

users that have mentioned machine learning as a keyword in their profile description. To choose the users, we selected a set of machine learning users as seeds and crawled among their friends and friends of friends for other machine leaning-related users. For the prepared list of users, we gathered their timeline tweets which for most of the users covers their tweets for the last 5 years. For each tweet, we also, collected the related meta-data such as the list of users who have replied to each tweet (mention list) and the list of users who have retweeted each tweet (retweet list). The final dataset contains 101,363 tweets with their related metadata, mention lists, and retweet lists. The network that is built on retweet list contains 301870 nodes.

7.0.3 2016 US Presidential Election

New algorithms and approaches need to be tested on big datasets. The United States 2016 presidential Election was an important event with millions of online users to comment on and millions of contents to be generated. Twitter has been the largest platform that public have participated to express their opinions toward the election and candidates. We started streaming the election tweets on twitter from 19 September (one week before the first presidential election debates) until 10 November 2016. The streaming keywords was the candidates names, ”Donald Trump” and ”Hillary Clinton”, key election hashtags, and related phrases to both sides. The collected dataset contains 139.5 million tweets and 13.7 million users.

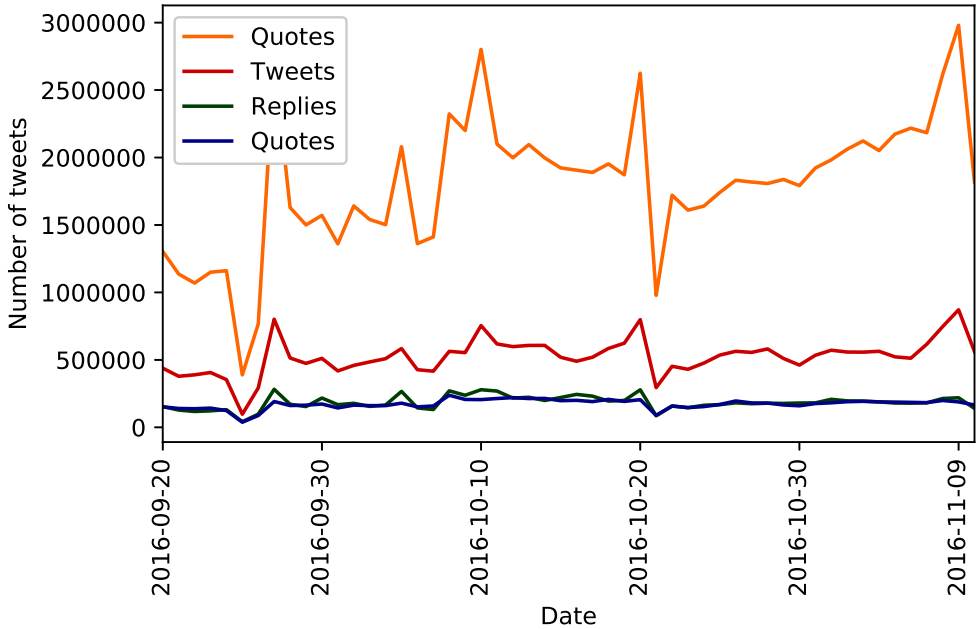


Figure 7.1: Daily count of the number of tweets, quotes, replies, and retweets

Due to the importance of this dataset in the future chapters, we present a more detailed analysis on the dataset to understand the overall structure of dataset. We show how dataset is distributed between tweets types, hashtags, and candidates and also unique users in each category.

Table 7.1 describes how the dataset is distributed between retweets, tweets, quotes, and replies. Majority of tweets are retweets with 92 millions and tweets are 27 millions. The unique users are 9.2 millions for the retweets and 4.1 millions for the tweets. It also shows that 1.3 million users have used hashtags in their tweets. Figure 7.1 shows the number of tweets daily is generated from the first date of streaming until the end of election. The peaks in the figure are happened at the debate nights. Figure 7.2 is the cumulative sum of generated tweets over time which shows the dataset is continuous and without any gap.

Majority of users don't activate their geographical locations. Thus, not many tweets contain the tweet's location. In our dataset, around 4% of tweets have their location param-

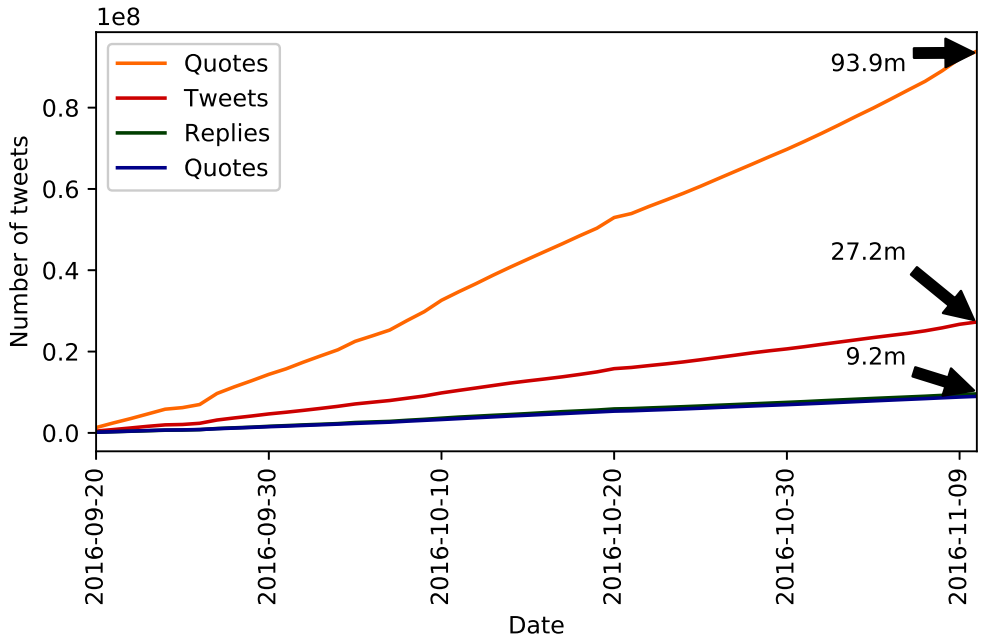


Figure 7.2: Cumulative sum of the number tweets, quotes, replies, and retweets over time

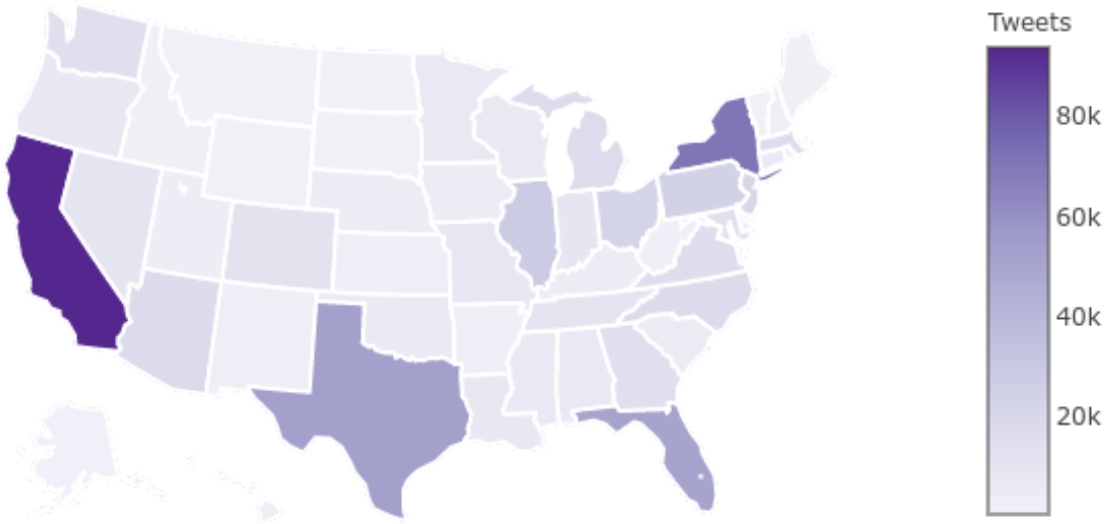


Figure 7.3: Distribution of geo-enabled tweets (from 27 millions tweets)in the states of United States

eter. Figure 7.3 maps the geo enabled tweets into their originated states. From the figure it is obvious that California has the most tweets with geo enabled among all the states followed by New York, Texas, and Florida.

Hashtags are very important metadata in tweets. From Figure 7.4 to Figure 7.7 we describe the distribution of different types of tweets, their hashtags. For top hashtags we look at the number of tweets in each hashtag, the number of their users , their favorite counts, and the number of times they have appeared in retweets.

Figure 7.4 describes that among the total number of tweets in the dataset, 67% are retweets, 20% are tweets, 6% are quotes, and 7% are replies. The second layer of the Figure shows the percentage of hashtags in total dataset. It is learnt that just 25.2% of tweets contain hashtags and this figure is 20% and 10.3% for quotes and replies respectively. Among the tweets with hashtags, 28% of them had hashtag ”Trump” while hashtag ”Clinton” just appeared among 11% of tweets with hashtags.

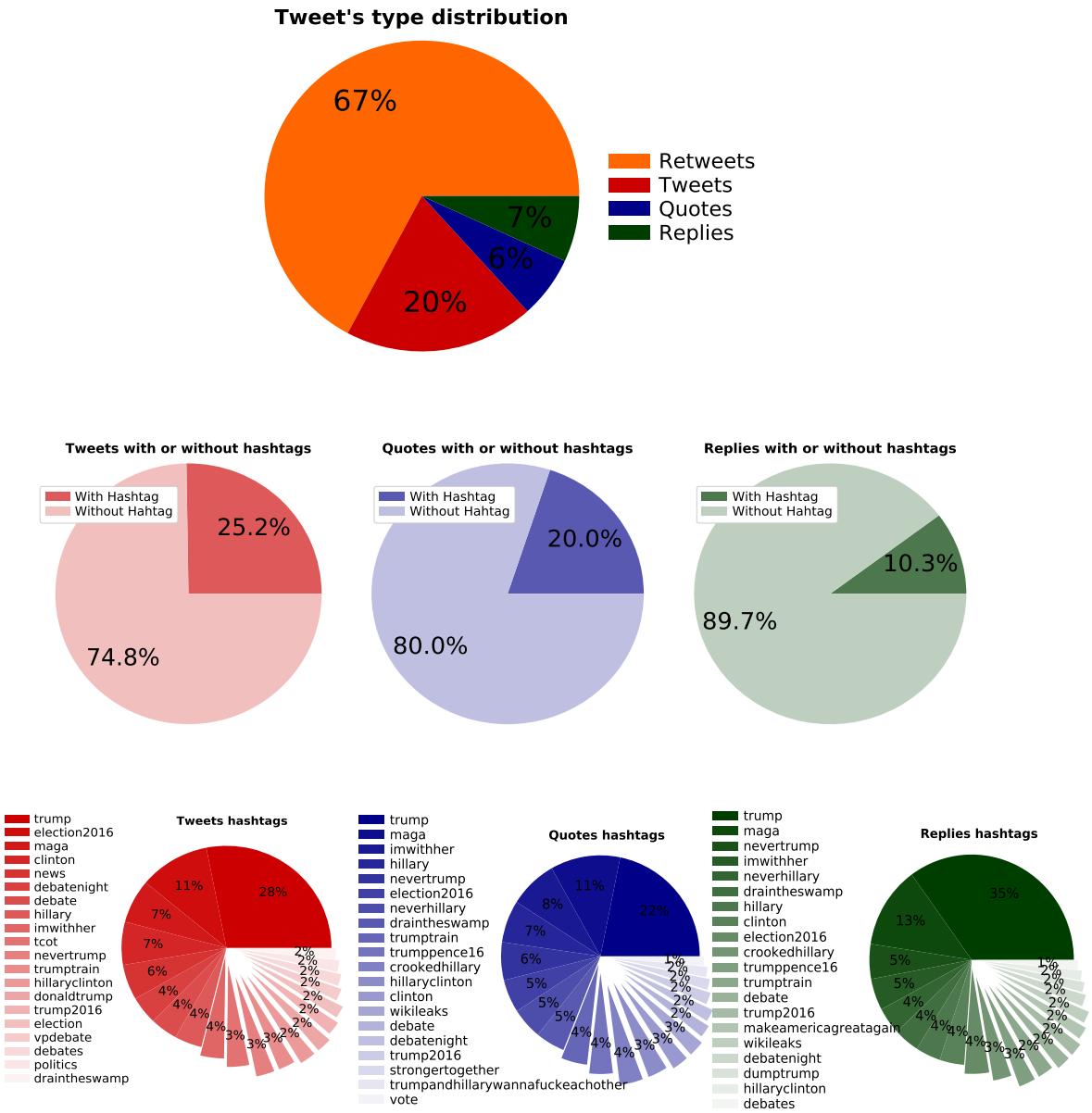


Figure 7.4: Tweets distribution based on their types and hashtags

A user can have one to many tweets in the dataset. It is valuable to know the number of unique users in each category of tweets and hashtags. Figure 7.5 shows that 56% of unique number of use. This percentage is 25% for tweets. Quotes and replies datsets have 9% of unique users. The second layer of the Figure shows the rate of tweets with hashtags compare to those without hashtags. The third layer in Figure 7.5 shows the unique users in hashtags for those tweets that contains hashtags. If we consider tweets with hashtags as a sub-dataset, 21% of unique users have tweeted with hashtag ”Trump” while just 5% of users have included hashtag ”Clinton” in their tweets. The figures are similar for the quotes and replies.

An important metadeta in tweets datasets is favorites which explains how other users have positively reacted to a particular tweet. Figure 7.6 shows the percentage of favorites counts for the categories. This figure doesn’t include retweets as retweets cannot be favorited. The figure shows that 79% of favorites are for tweets and just less than 22% for quotes and replies combined. Among the hashtags, ”Trump” has the higher favorite count in all categories. The noticeable observation here is that ”Trump” hashtag has 28% of tweets (Figure 7.4) while the favorite counts for that hashtag is 15% of the dataset.

Tweets can be retweeted and the higher number of retweets for a tweet shows it is importance among the users. Figure 7.7 shows the retweet sum for the categories of tweets and their hashtags. The higher layer of the Figure shows that tweets are more retweeted that quotes and replies. ”Trump” hashtag contains 19% of retweets in the tweets category while this number is much lower for ”Clinton” around 6%.

7.0.4 Summary

In this chapter, we explained the three datasets that are used in the next chapters. It is described in details that what the datasets are about, their domain, how they are collected and a brief statistics on them. The datasets are from three different domains namely:

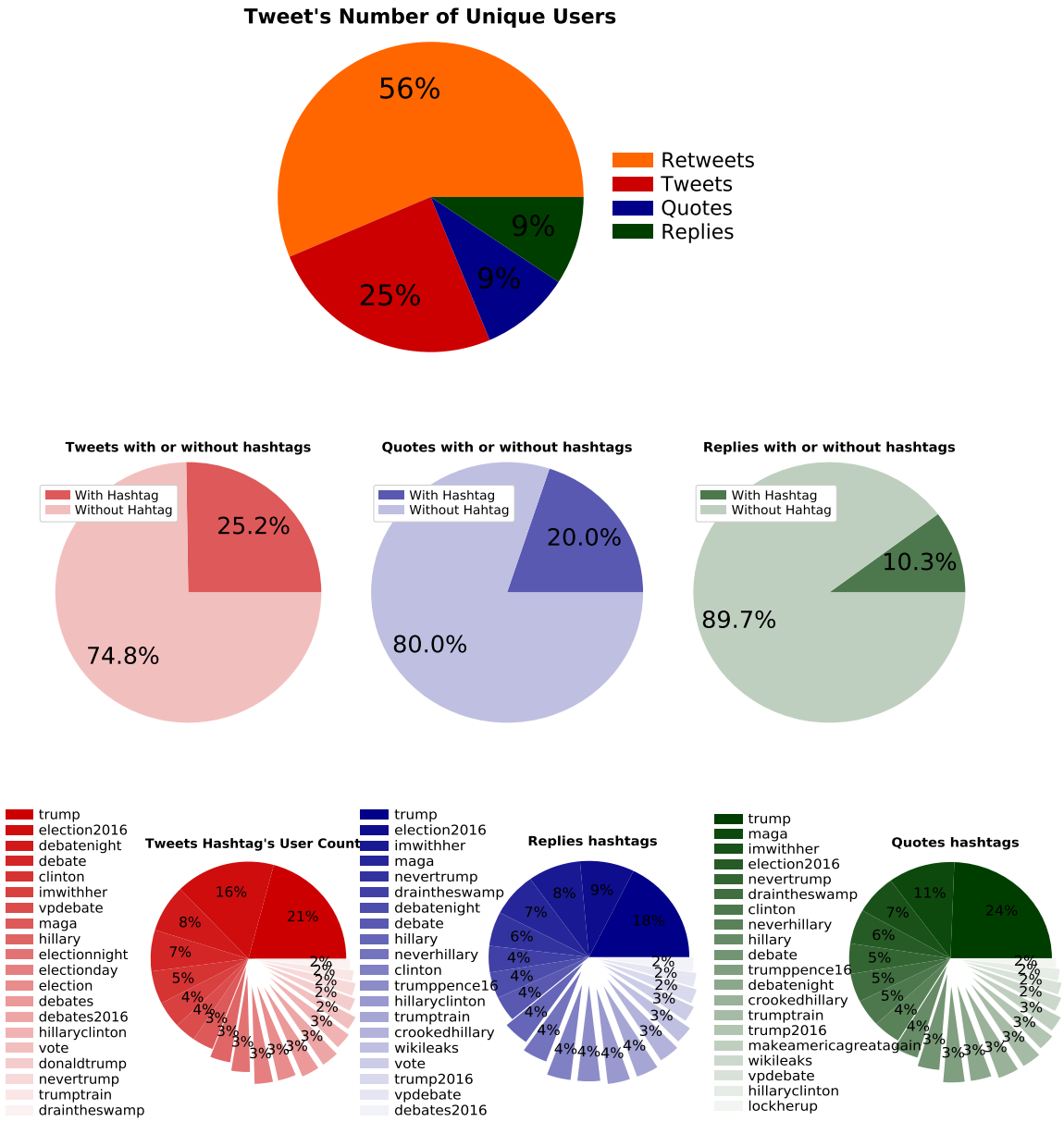


Figure 7.5: Users distribution in different types of tweets and hashtags

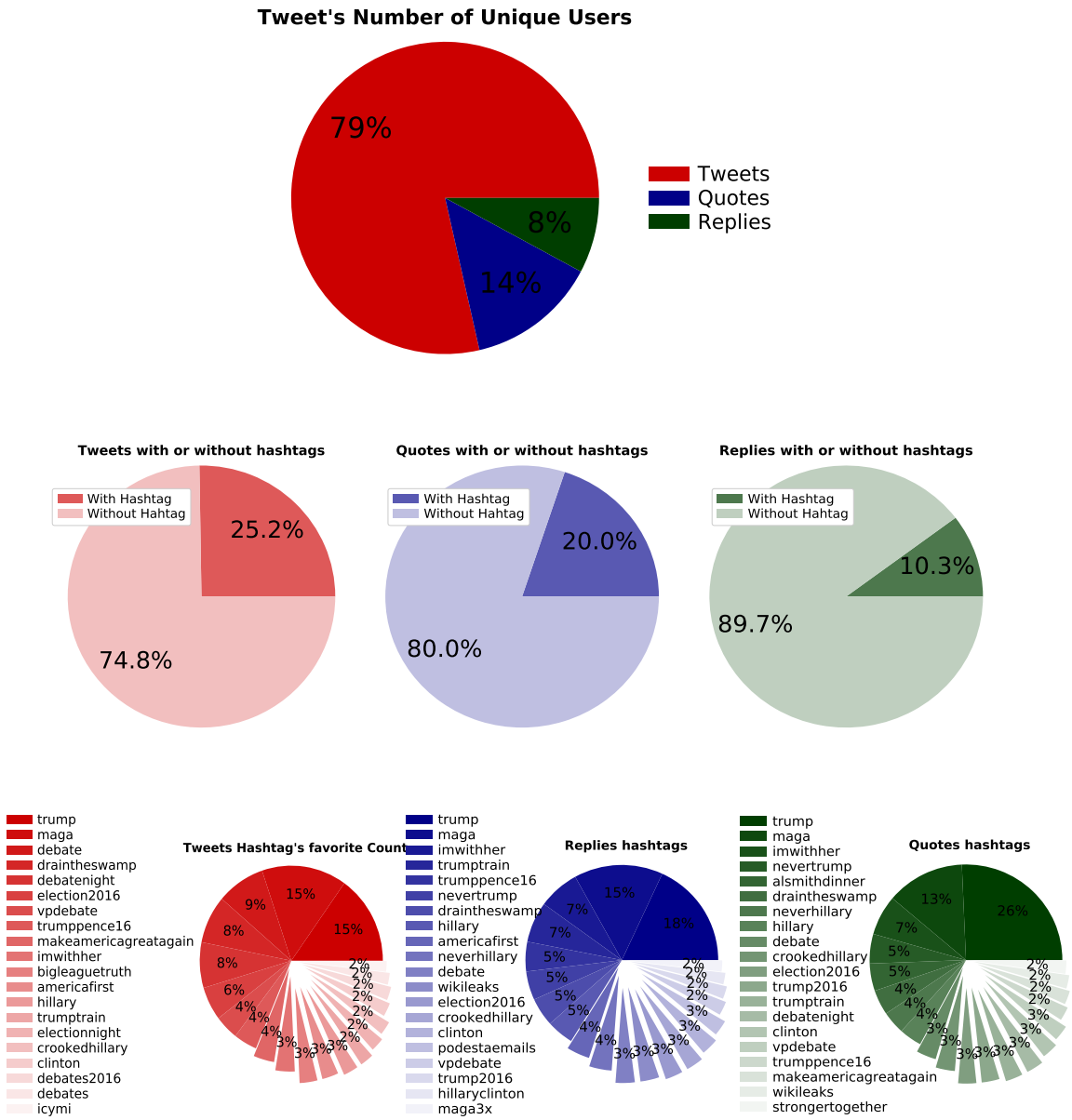


Figure 7.6: Percentage of favourites for different types of tweets and hashtags

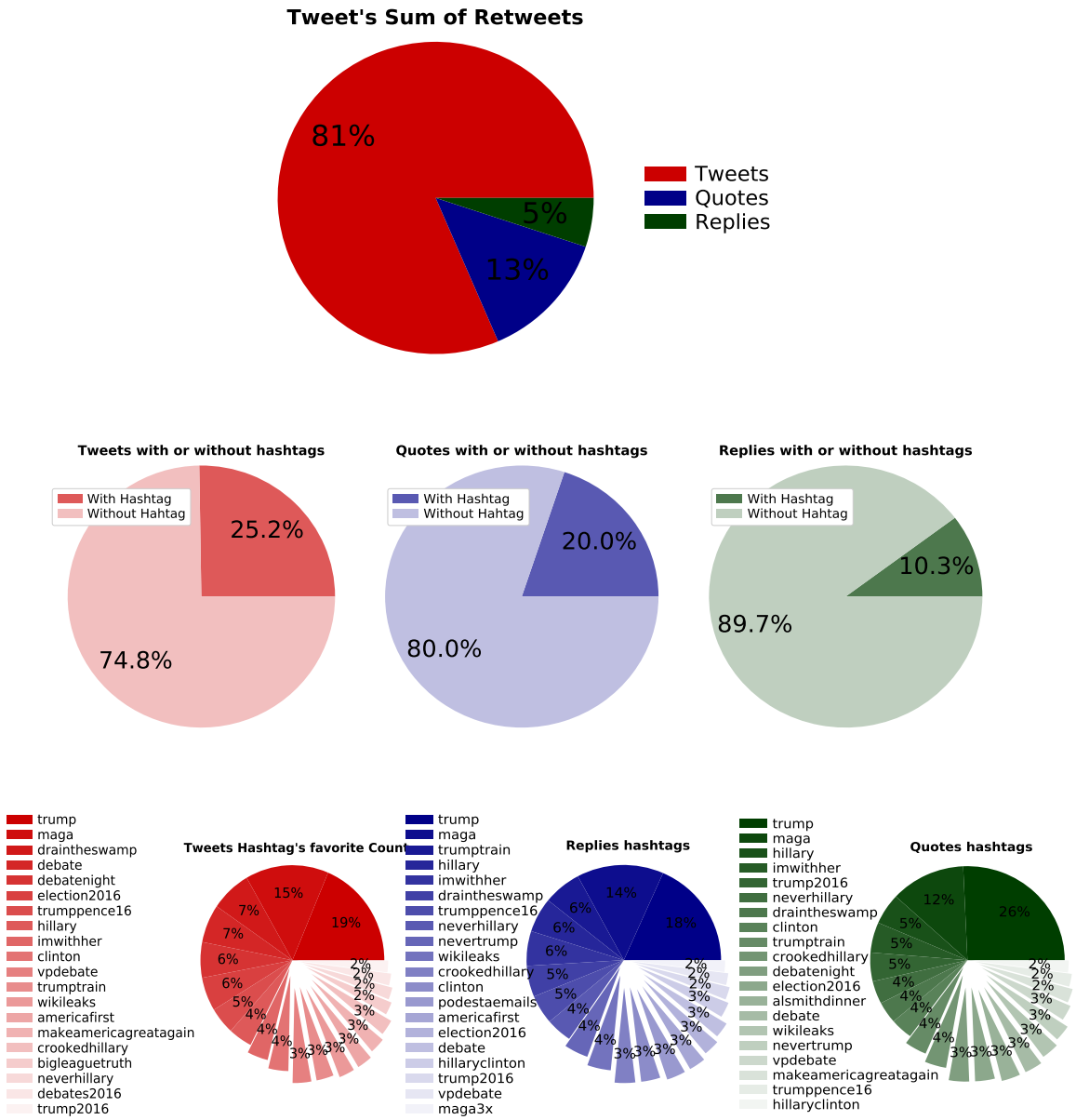


Figure 7.7: Percentage of retweets in different types of tweets and hashtags

user's reviews on mobile phones (reviews dataset), machine learning community on Twitter (community dataset), and 2016 US presidential election (event dataset and bigdata).

The datasets are very unique and have been collected to satisfy the needs of this dissertation. They are enriched with metadatas that are very expensive to collect. For example, the machine learning community dataset contains all the user ids that have retweeted the tweets in our dataset. Collecting retweet list (list of user ids that have retweeted a tweet) from twitter using their API has limitation of 1 tweet per minute. This explains the uniqueness of our datasets and their importance as there is no equivalent dataset in terms of metadata richness. The datasets will be explained more in details in the future chapter when it is needed for readers understanding.

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